Zur Algebra der Funktionaloperationen und Theorie der normalen Operatoren

On the Algebra of Functional Operations and the Theory of Normal Operators

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Contents

Introduction

1. The present paper falls into two essentially independent parts. The first (§§ I–III) is devoted to the investigation of the linear and bounded operators (i.e. matrices) of Hilbert space $\mathfrak{H}$, by considering the algebraic properties of the (noncommutative) ring $\mathfrak{B}$ formed by them. The subject of the second part, on the other hand, is formed by those operators, not necessarily everywhere defined (in $\mathfrak{H}$) and bounded, which admit the so-called Hilbert spectral representation with complex eigenvalues (cf. the more detailed explication of these notions in paragraph 4 of the Introduction). These are the operators to be designated as "normal," which hitherto have been considered only in the bounded case,[1] and for which we shall give a new, more general definition (cf. the passage just cited).

Before we discuss these matters more precisely, let us recall the definition of (complex) Hilbert space $\mathfrak{H}$. It may be realized as the set of all sequences of complex numbers

$$\{x_1, x_2, \dots\} \qquad \text{with} \qquad \sum_{n=1}^{\infty} |x_n|^2 < \infty;$$

we denote its elements by $f, g, \dots, \varphi, \psi, \dots$.[2]

One may calculate with these elements as with vectors, so that the meaning of constructions such as $f \pm g$ and af is immediately clear.[3] There is, moreover, an

[1]Until recently, only in the completely continuous case; cf. the encyclopedia article by Hellinger and Toeplitz, *Encyklopädie der mathematischen Wissenschaften*, II C 13, p. 1562. See also note 25.

[2]By the well-known theorem of Fischer and F. Riesz it may equally well be realized as the set of all functions $f(x)$ on an interval $a < x < b$ for which

$$\int_a^b |f(x)|^2\, dx < \infty$$

($a < b$, and either endpoint may also be infinite); or as the set of all functions $f(P)$, where P ranges over the surface of the unit sphere, for which

$$\iint |f(P)|^2\, d\sigma < \infty$$

($\iint \cdots d\sigma$ denoting integration over the surface of the unit sphere); and so forth. Cf., for example, the paper cited in note 7, especially Chapter I, Appendix I, and the Introduction; for the theorem of Fischer and F. Riesz, see, for instance, F. Riesz, *Göttinger Nachrichten* (1907), pp. 210–273.

[3]When $\mathfrak{H}$ is interpreted as the "sequence space" of the $\{x_1, x_2, \dots\}$ for which $\sum_{n=1}^{\infty} |x_n|^2$ is finite, one defines

$$\{x_1, x_2, \dots\} \pm \{y_1, y_2, \dots\} = \{x_1 \pm y_1, x_2 \pm y_2, \dots\},$$

"inner product" (f, g) as for vectors,[4] which permits the definition of the "absolute value" in Hilbert space,

$$|f| = \sqrt{(f, f)},$$

and of the distance $|f - g|$.[5] In this way Hilbert space $\mathfrak{H}$ becomes a topological (indeed, metric) linear space,[6] in which geometrical-topological expressions such as linear manifold, accumulation point, continuity, and so forth may be used without further explanation.

For a systematic characterization of $\mathfrak{H}$ on this basis, one may compare an earlier paper by the author,[7] whose notation and concepts will also be used here. In this connection compare especially the portions named in note 2; nevertheless, we shall always cite precisely the sections of E. that are used.

$$a\{x_1, x_2, \ldots\} = \{ax_1, ax_2, \ldots\}.$$

In the "function space" of the $f(x)$ for which $\int_a^b |f(x)|^2\, dx$ is finite, the corresponding definitions are $(f \pm g)(x) = f(x) \pm g(x)$ and $(af)(x) = af(x)$; analogously for the other examples. Cf. the reference in note 2.

[4]One defines

$$(\{x_1, x_2, \ldots\}, \{y_1, y_2, \ldots\}) = \sum_{n=1}^{\infty} x_n \overline{y_n}$$

(the series converges absolutely), or, respectively,

$$(f(x), g(x)) = \int_a^b f(x)\overline{g(x)}\, dx,$$

and so forth. Cf. the preceding notes.

[5]Thus

$$|\{x_1, x_2, \ldots\}| = \sqrt{\sum_{n=1}^{\infty} |x_n|^2}, \qquad |f(x)| = \sqrt{\int_a^b |f(x)|^2\, dx}.$$

Unfortunately the notation is ambiguous here: on the left of the second formula stands the absolute value of the function $f(x)$, while on the right stand the absolute values of its values. Such an ambiguity will never arise in our considerations in $\mathfrak{H}$. The other realizations are treated analogously; cf. the preceding notes.

[6]Cf. Hausdorff, "Umgebungsaxiome und topologisch-lineare Räume," in *Grundzüge der Mengenlehre*, 1st ed., Leipzig and Berlin, 1914.

[7] "Zur allgemeinen Eigenwerttheorie Hermitescher Funktionaloperatoren," *Mathematische Annalen* **102**, no. 1 (1929), pp. 49–131. This paper, which will be cited repeatedly below, will always be denoted by E. Several abbreviations from E. that we shall use are important; they are collected here: closed = cl., bounded = bd., extension = ext., Hermitian = H., hypermaximal = hypermax., linear = lin., linear manifold = lin. mfd., maximal = max., normalized = norm., operator = op., orthogonal = orth., projection = P., complete = compl., resolution of the identity = res. id. We shall further abbreviate the term "commutable" (cf. note 16) by comm.

The definitions from E. most important for our purposes are the following (abbreviations as in note 7). An operator is a function whose range and domain are subsets of $\mathfrak{H}$; we denote operators by $A, B, \ldots, R, S, \ldots$, and the value of A at f by Af. (Thus Af need by no means be defined everywhere in $\mathfrak{H}$.) It is linear if, whenever Af and Ag are defined, $A(af)$ and $A(f+g)$ are also defined (i.e. its domain

is a linear manifold),[8] and their values are respectively aAf and $Af + Ag$. It is called closed if it has the following property: whenever all Af_n ($n = 1, 2, \ldots$) are defined and

$$f_n \longrightarrow f, \qquad Af_n \longrightarrow f^* \qquad (n \longrightarrow \infty),$$

then Af is defined and equals f^*.[9]

An operator is continuous if it is so, in the usual sense, throughout its domain: if Af_0 is defined, then for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$|f - f_0| < \delta \quad \Longrightarrow \quad |Af - Af_0| < \epsilon$$

whenever Af is defined. One sees at once that the domain of a closed, continuous operator is necessarily a closed set; if the operator is also linear, its domain is therefore a closed linear manifold.[10]

In the first half of this paper, as already announced, we shall concern ourselves with everywhere-defined, linear, continuous operators, which (in accordance with Hilbert) will be called bounded. (Cf., also for what follows, Appendix I of *E.*) For these everywhere-defined operators the simplest arithmetic operations are defined as follows. If A, B are bounded, then so are aA, $A \pm B$, AB, and

$$(aA)f = a \cdot Af, \qquad (A \pm B)f = Af \pm Bf, \qquad (AB)f = A(Bf).$$

The rules of algebra hold, with two exceptions: there are zero divisors, and multiplication is not commutative.[11] Thus the set $\mathfrak{B}$ of all bounded operators is a ring—with zero divisors and noncommutative. Another important operation is $*$, defined as follows: for every bounded A there is exactly one bounded A^* such that

$$(Af, g) = (f, A^*g), \qquad (f, Ag) = (A^*f, g).$$

[8]That is, together with f, g it contains af, $f + g$. A closed linear manifold must in addition be closed, i.e. contain all its accumulation points.

[9]In compact spaces this continuity-like property is equivalent to continuity, but in $\mathfrak{H}$ it is much weaker; cf. *E.*, note 29.

[10]For linear operators continuity can be reduced to several equivalent formulations; cf. *E.*, Theorem 12. One of these is: there is a fixed c such that $|f| \leq 1$ implies $|Af| \leq c$.

[11]0 and 1 are defined by $0f = 0$ and $1f = f$.

The following rules are easily verified:

$$0^* = 0, \qquad 1^* = 1, \qquad (aA)^* = \overline{a}\, A^*, \qquad (A \pm B)^* = A^* \pm B^*, \qquad (AB)^* = B^* A^*.$$

For the unbounded (more precisely: not necessarily bounded) operators that form the subject of the second half of this paper, we must proceed somewhat differently. For such operators R, S, in defining $aR, R \pm S, RS$, one must first observe that the domains of these operators are thereby restricted, since Rf, Sf need not be defined everywhere. In order that $(R + S)f$ be defined, both Rf and Sf must be defined, and likewise for $(R - S)f$; for RSf, both Sf and $R(Sf)$ must be defined. In the case of R^* it is even uncertain whether such an operator exists. Yet the operation $*$ is of particular importance for us. We shall therefore, when necessary, begin directly with an already specified pair R, R^*, stipulating the following.

Conjugate operator pair. A conjugate operator pair (for short, conj. op. pair) consists of two operators R, R^* having the same domain and satisfying, whenever both sides are defined,

$$(Rf, g) = (f, R^* g), \qquad (f, Rg) = (R^* f, g)$$

(cf. note 12). In addition, in analogy with the H. operators and for the same reasons (cf. *E.*, Introduction V, Definition 6, and note 30), we require that the domain span the closed linear manifold $\mathfrak{H}$, i.e. that it be dense in $\mathfrak{H}$. This definition will prove to be the correct generalization of the operation $*$ in $\mathfrak{B}$; H. operators are included as the special case $R = R^*$.

2. We first make a few remarks about the points of view governing the investigation of $\mathfrak{B}$. We regard this ring as a hypercomplex number system. Indeed, it is probably the simplest system of this kind with infinitely many units—that is, the simplest one that goes beyond the systems usually investigated and characterized essentially

[12] Either relation follows from the other by interchanging f, g and taking complex conjugates. The operation $*$ corresponds, for matrices, to taking the complex-conjugate transpose. More precisely, if $\mathfrak{H}$ is realized as the "sequence space" of the $\{x_1, x_2, \dots\}$ for which $\sum_{n=1}^{\infty} |x_n|^2$ is finite, and A has the matrix $\{a_{\mu\nu}\}$, so that

$$A\{x_1, x_2, \dots\} = \{y_1, y_2, \dots\}, \qquad y_\nu = \sum_{\mu=1}^{\infty} a_{\mu\nu} x_\mu$$

always holds—as in Appendix II of *E.*—then A^* has the matrix $\{\overline{a_{\nu\mu}}\}$. Cf. *E.*, Appendix II.

by the validity of the so-called "chain conditions."[13] Whereas in hypercomplex systems with finitely many units (or with the "chain conditions" that replace this assumption) the topology is so trivial that it plays no noteworthy role, in our infinite system $\mathfrak{B}$ topology plays a decisive role. Thus, for example, in the definition of which subsets $\mathfrak{M}$ of $\mathfrak{B}$ are also rings (in the terminology of the works cited in note 13: "orders"), if we required only that $\mathfrak{M}$ contain $aA, A \pm B, AB$ whenever it contains A, B, we should lose ourselves in an inextricable thicket of pathological constructions. One must also require that $\mathfrak{M}$ be closed in a (still to be formulated) topology of $\mathfrak{B}$.

We shall, moreover, impose one further simplifying requirement on the rings $\mathfrak{M}$: together with A they must also contain A^*. This is a significant restriction, which makes it possible to avoid the complications of elementary-divisor theory.[14] There is good reason to exclude these difficulties at first; compared with the finite-dimensional case,[15] the theory of $\mathfrak{B}$ already furnishes enough new possibilities for complication.

It follows from what has been said that we must first concern ourselves with the topology of $\mathfrak{B}$. A difficulty is that there are several ways of topologizing $\mathfrak{B}$ (i.e. of defining its notion of neighborhood) that are relevant for us. We shall discuss these precisely in Section I; here it is enough to say that rings $\mathfrak{M}$ in $\mathfrak{B}$ will be required to be closed in the so-called weak topology.

Since the intersection of arbitrarily many rings is again a ring, the intersection of all rings containing $\mathfrak{M}$ exists in particular ($\mathfrak{M}$ being any subset of $\mathfrak{B}$). It is plainly characterized uniquely as the smallest ring containing $\mathfrak{M}$; we call it $R(\mathfrak{M})$. Another important construction is the following. Let $\mathfrak{M}$ be a subset of $\mathfrak{B}$. The set of all $A \in \mathfrak{B}$ for which both A and A^* commute with every $B \in \mathfrak{M}$[16] will be denoted by $\mathfrak{M}'$. The operation $'$ can be iterated, thereby producing successively the sets

$$\mathfrak{M}', \ \mathfrak{M}'', \ \mathfrak{M}''', \dots$$

from $\mathfrak{M}$. It has a number of simple properties; among other things we shall show that

$$\mathfrak{M} \subset \mathfrak{M}'', \qquad \mathfrak{M} \subset \mathfrak{N} \Longrightarrow \mathfrak{M}' \supset \mathfrak{N}',$$

[13] Cf. E. Noether, *Mathematische Annalen* **96**, no. 1 (1926), pp. 26–61, especially § 2; also E. Artin, *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* **5**, no. 3 (1927), pp. 251–260.

[14] For matrices in finite-dimensional spaces, E. Fischer was the first to introduce this condition successfully.

[15] Every hypercomplex system with finitely many units can, of course, be represented by matrices (and hence operators) of a finite-dimensional space.

[16] Two elements A, B of $\mathfrak{B}$ are commutable if $AB = BA$.

and, in general,

$$\mathfrak{M}' = \mathfrak{M}''' = \mathfrak{M}^{\mathrm{V}} = \cdots, \qquad \mathfrak{M}'' = \mathfrak{M}^{\mathrm{IV}} = \mathfrak{M}^{\mathrm{VI}} = \cdots.$$

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Together, the two operations $R(\mathfrak{M})$ and $\mathfrak{M}'$ permit a simple characterization of the algebraic relations in $\mathfrak{B}$.

3. An operator A is called normal if A and A^* commute (cf. E., Appendix I). A ring $\mathfrak{M}$ is called Abelian if all its elements commute with one another. One readily shows that a ring $\mathfrak{M}$ is Abelian if and only if all its elements are normal, and that the ring $R(A)$ is Abelian if and only if A is normal. We shall now prove the converse: every Abelian ring $\mathfrak{M}$ can be generated by a normal operator (indeed, by an H. operator); that is, there is such an A with

$$\mathfrak{M} = R(A).$$

There is a further simplification in the case of Abelian rings. Let $\mathfrak{M}$ be a subset of $\mathfrak{B}$, and let $R(\mathfrak{M})$ be the ring generated by $\mathfrak{M}$. One easily shows that $R(\mathfrak{M})$ is obtained by first forming, from the elements of $\mathfrak{M}$, all operators arising through finitely many applications of the operations

$$aA, \qquad A \pm B, \qquad AB, \qquad A^*,$$

and denoting their set by $r(\mathfrak{M})$, and then adjoining to $r(\mathfrak{M})$ all its accumulation points. Here, as mentioned in paragraph 2, accumulation point is to be understood in the sense of the so-called weak topology. If $R(\mathfrak{M})$ is Abelian (this is the case if and only if all elements of $\mathfrak{M}$ commute with one another and with their adjoints), we shall show that it suffices to adjoin to $r(\mathfrak{M})$ a much smaller set of accumulation points—the so-called limits of strongly doubly convergent sequences; cf. the discussion of these notions in Section I. This already produces the whole of $R(\mathfrak{M})$. The result is especially important for the theory of functions of commuting normal (hence, in particular, H. and unitary) operators, although that theory lies beyond the scope of the present paper.

For arbitrary (not necessarily Abelian) rings we shall establish a relation between $R(\mathfrak{M})$ and $\mathfrak{M}''$, where $\mathfrak{M}$ is any subset of $\mathfrak{B}$. In particular, we shall show (our most general result is slightly broader; cf. § II, especially Theorem 5) that if $1 \in \mathfrak{M}$, then

$$R(\mathfrak{M}) = \mathfrak{M}''.$$

[17] $\mathfrak{M} \subset \mathfrak{N}$ or $\mathfrak{N} \supset \mathfrak{M}$ means that $\mathfrak{M}$ is a subset of $\mathfrak{N}$.

To see the significance of this theorem for $\mathfrak{B}$, regarded as a hypercomplex number system, consider what it says when Hilbert space $\mathfrak{H}$ is replaced by a finite-dimensional, say k-dimensional, Euclidean space. Let $\mathfrak{M}$ be, in particular, a finite or continuous group of unitary matrices. It is then immediately clear that $R(\mathfrak{M})$ is the set of all linear combinations of matrices in $\mathfrak{M}$ (among which there can be at most k^2 linearly independent ones).

First suppose that $\mathfrak{M}$, as a set of matrices, is irreducible. As is well known, this means that only the matrices $a1$ commute with all elements of $\mathfrak{M}$;[18] that is, $\mathfrak{M}'$ consists of these alone. Hence $\mathfrak{M}''$ comprises all matrices whatsoever, and by our theorem the same must hold for $R(\mathfrak{M})$. Thus every matrix is a linear combination of matrices from $\mathfrak{M}$, or, equivalently (since there are k^2 linearly independent matrices), there are k^2 linearly independent matrices in $\mathfrak{M}$. Put another way: no fixed linear relation among the k^2 matrix entries can hold for every element of $\mathfrak{M}$. This, however, is Burnside's theorem on irreducible matrix groups.[19] If we do not assume that $\mathfrak{M}$

is irreducible, one sees just as readily that our theorem corresponds to the Frobenius–I. Schur strengthening of Burnside's theorem.[20]

We have therefore proved the analogue of the theorem that, in finitely many dimensions, can serve as the foundation of the theory of unitary representations of groups. We shall not investigate here to what extent an equivalent theory in Hilbert space can be based upon our theorem.

These remarks exhaust the first half of the present paper.

4. The subject of the second half (§§ IV–V) is the conjugate operator pairs mentioned at the end of paragraph 1, independently of any boundedness assumption. The first task is to characterize the normal operators among them. A difficulty now arises: the pair R, R^* ought to be called normal when R and R^* commute; but since R, R^* need not be defined everywhere, the definition of the products RR^*, R^*R, and hence the reduction of commutability to them, is at first uncertain.[21] In

[18]Cf. I. Schur, *Berliner Berichte* (1905), p. 406. The finiteness of $\mathfrak{M}$ always assumed there is, as is well known, immaterial in this case.

[19]Cf. Burnside, *Proceedings of the London Mathematical Society*, ser. 2, **3** (1905), p. 430—though there only for unitary matrices.

[20]Cf. Frobenius and I. Schur, *Berliner Berichte* (1906), p. 209.

[21]The equality $RS = SR$ can by no means characterize the commutability of R, S. Suppose, for example, that Rf is not defined everywhere. The same is then true of $0Rf$, whereas $R0f$ is defined everywhere; hence $0R \neq R0$, i.e. R would fail to commute with 0, which must not occur under any reasonable notion of commutability.

Quite generally, forming AB or $A \pm B$ for operators A, B that are not everywhere defined is a

Appendix III we shall see that the natural matrix definition of commutability likewise fails.

We proceed as follows. If, of two operators R, A, at least A is defined everywhere, we define their commutability by requiring that AR be an extension of RA. That is, whenever Rf is defined, $R(Af)$ must also be defined and must equal $A(Rf)$ (where Af and $A(Rf)$ are in any event defined).[22] We may consequently form $\mathfrak{M}'$ even for a set $\mathfrak{M}$ of entirely arbitrary operators (which need be neither linear nor everywhere defined): it is the set of all $A \in \mathfrak{B}$ for which both A and A^* commute with every $R \in \mathfrak{M}$. Thus $\mathfrak{M}'$ is always a subset of $\mathfrak{B}$. In particular, for every operator R we have the sets

$$(R)',\ (R)'', \ldots \qquad \text{(all subsets of } \mathfrak{B}\text{).}$$

[23]

Now one easily shows (cf. § II.1) that an $A \in \mathfrak{B}$ is normal if and only if $(A)''$ is Abelian. We extend this definition to arbitrary conjugate operator pairs: they are to be called normal if $(R)''$ is Abelian. In Appendix II we shall see that this definition applies to broad classes of operators. In particular, the theory to be developed below is immediately applicable to the (complex, unbounded) Laurent matrices and their generalizations.[24]

We shall then show that these normal operators, and only these, possess one (and indeed only one) Hilbert spectral form with complex eigenvalues.[25] These

questionable matter. There are, for example, hypermaximal H. operators A, B for which Af and Bf are never simultaneously defined (except for $f = 0$), so that $A + B$ cannot be formed at all. Cf. the author's forthcoming paper in *Journal für die reine und angewandte Mathematik*, "Zur Theorie der unbeschränkten Matrizen," where very general classes of counterexamples are given.

[22]The example $R, 0$ in note 21 is precisely what suggests this definition. Moreover, if E is a projection operator, the commutability of R, E says that R is reduced by the closed linear manifold belonging to E (cf. *E.*, Definition 13), exactly as one should reasonably expect.

[23]It will surely cause no misunderstanding that, just as in $R(A)$, we denote the set having the single element R also by R.

[24]Cf. Toeplitz, *Mathematische Annalen* **70** (1911), pp. 351–376.

[25]For a precise formulation of this concept, which generalizes the usual (real) Hilbert spectral form, cf. *E.*, Appendix II.

The concept of normality for finite-dimensional matrices is due to Frobenius [*Journal für die reine und angewandte Mathematik* **84** (1877), pp. 51–54]; he proved that precisely these matrices can be transformed unitarily to diagonal form (with complex diagonal entries). The same result was proved for the completely continuous operators of Hilbert space in the encyclopedia article by Hellinger and Toeplitz (II C 13, pp. 1562–1563).

investigations therefore have an essentially different direction from the preceding sections and are to be regarded rather as a continuation of E.

As one sees, the situation for normal operators is somewhat more favorable than for H. operators: for the latter the Hilbert spectral form was not always attainable,[26] and this is all the more remarkable because, in finite-dimensional spaces and even for bounded operators, H. was a special case of normal ($AA^* = A^*A$ follows from $A = A^*$). The explanation is that we shall prove the following: an H. operator is, in essence, normal if and only if it is hypermaximal, and precisely these H. operators have a Hilbert spectral form (cf. *E.*, Theorem 36). Thus the fact that H. appears as a special case of normal in the situations just named is due simply to the fact that everything there is hypermaximal.

5. The organization of this paper is as follows. In Section I we present and discuss the different ways of topologizing $\mathfrak{B}$. In Section II we introduce the operations $\mathfrak{M}'$ and $R(\mathfrak{M})$ (formation of a ring) and prove various properties of them (including the already announced analogue of the Burnside–Frobenius–Schur theorem).

In Section III we discuss the Abelian rings (cf. what was said above about them). Sections IV and V are almost independent of the preceding material (apart from the definitions); their content is the general theory of normal operators (without an assumption of reality or boundedness) and their spectral representation.

Appendix I treats the bounded H. operators; Appendix II gives an application of the theory of normal operators to Laurent matrices and more general matrices. Appendix III shows the inapplicability, in the unbounded case, of the usual definition of commutability of matrices.

I. Topologizations of $\mathfrak{H}$ and $\mathfrak{B}$

1.

Before considering the topology of $\mathfrak{B}$, we shall describe the somewhat simpler topological relations in $\mathfrak{H}$—less for the sake of their later applications than in order to illustrate the situation in $\mathfrak{B}$.

In *E.*, Appendix I, the author proved that all bounded normal operators can be put into the complex Hilbert spectral form (which is exactly the equivalent of diagonal form). Here the theory is to be completed and extended to unbounded operators. Independently of the author, and using a different method, A. Wintner has since also put the unitary operators (a special case of bounded normal operators) into spectral form. Cf. *Mathematische Zeitschrift* **30**, nos. 1/2 (1929), pp. 228–282.

[26] Cf. *E.*, Introduction VII.

Up to now (as in *E.*) we have considered only the following topology in $\mathfrak{H}$, which is called "strong,"[27] and which, following Hausdorff, we characterize by specifying all neighborhoods of an arbitrary point $f_0 \in \mathfrak{H}$.

Strong topology on $\mathfrak{H}$. Let

$$\mathcal{U}_1(f_0; \epsilon) = \{f \in \mathfrak{H} : |f - f_0| < \epsilon\}.$$

All $\mathcal{U}_1(f_0; \epsilon)$ with $\epsilon > 0$ are the neighborhoods of f_0.

These neighborhoods are therefore the interiors of balls about f_0. Since $|f - f_0|$ is a distance (cf. *E.*, Definition 4 and note 27), it follows at once that the four Hausdorff neighborhood axioms (which are postulated in every topology) are satisfied, namely:[28]

α) A point belongs to each of its neighborhoods.

β) The intersection of two neighborhoods of a point contains a neighborhood of that point.

γ) Every point in a neighborhood itself has a neighborhood that is a subset of the first.

δ) Any two distinct points have disjoint neighborhoods.

We also know already from E. that the operations af, $f \pm g$, (f, g), and $|f|$ are continuous in this sense (and $f \pm g$ and (f, g) are continuous in both variables simultaneously).

In the strong topology the so-called first Hausdorff axiom of countability holds:[29] for every f_0 there is a decreasing sequence of neighborhoods such that every neighborhood of f_0 contains one from this sequence; take

$$\mathcal{U}_1\left(f_0; \frac{1}{n}\right), \qquad n = 1, 2, \ldots .$$

[27] Cf. Weyl, doctoral dissertation, Göttingen, 1908, pp. 8–9.

[28] Cf. Hausdorff, *Grundzüge der Mengenlehre*, 1st ed., Leipzig and Berlin, 1914; the entire topology is developed there from this concept of neighborhood.

We recall how accumulation points and limits are to be defined. A point is an accumulation point of a set if each of its neighborhoods contains points of the set; it is the limit of a sequence if each of its neighborhoods contains all but finitely many terms of that sequence. It is easily seen that being a limit means being an accumulation point of every infinite subset of the sequence.

[29] Cf. the work cited in note 28. Because $\mathfrak{H}$ is separable (*E.*, § I, Axiom C), the second axiom of countability also holds; cf. the same source.

From this one easily infers that, for every accumulation point of a set in $\mathfrak{H}$, there is a sequence of points of the set that has it as its limit.

It is also customary to introduce another topology in $\mathfrak{H}$, the so-called "weak" topology.[30] Ordinarily this means that only weak limits, and not neighborhoods, are defined: $f_1, f_2, \ldots$ converges weakly to f if, for every fixed φ,

$$(f_n, \varphi) \longrightarrow (f, \varphi).$$

It is easy, however, to reduce this to a notion of neighborhood.

Weak topology on $\mathfrak{H}$. Let

$$\mathcal{U}_2(f_0; \varphi_1, \ldots, \varphi_s, \epsilon) = \left\{ f \in \mathfrak{H} : \begin{array}{c} |(f - f_0, \varphi_1)| < \epsilon, \ldots, \\ |(f - f_0, \varphi_s)| < \epsilon \end{array} \right\}.$$

All $\mathcal{U}_2(f_0; \varphi_1, \ldots, \varphi_s, \epsilon)$, with arbitrary $\varphi_1, \ldots, \varphi_s \in \mathfrak{H}$, arbitrary s, and $\epsilon > 0$, are the neighborhoods of f_0.

Again, one easily verifies Hausdorff's axioms α)–δ) and the fact that a limit in the sense of this topology is identical with the weak limit defined above. The continuity of af and $f \pm g$ (the latter in both variables simultaneously) is evident. The continuity of (f, g) in f, with g fixed, follows from the definition; because

$$(f, g) = \overline{(g, f)},$$

it is likewise continuous in g, with f fixed. It is not, however, continuous in both variables simultaneously, for then $|f| = \sqrt{(f, f)}$ would also be continuous, which is false.[31]

Since

$$\mathcal{U}_1(f_0; \epsilon) \subset \mathcal{U}_2(f_0; \varphi_1, \ldots, \varphi_s, \eta)$$

for example when

$$\epsilon = \frac{\eta}{\max(|\varphi_1|, \ldots, |\varphi_s|)},$$

every accumulation point or limit in the strong sense is also one in the weak sense. The converse cannot hold, already because $|f|$ has different continuity properties in the two topologies.

[30] Weyl (cf. note 27) defines it somewhat differently from us; cf. p. 9 of the work cited there.

[31] Indeed, every $\mathcal{U}_2(0; \varphi_1, \ldots, \varphi_s, \epsilon)$ still contains f with $|f| = 1$: it suffices to choose f orthogonal to all $\varphi_1, \ldots, \varphi_s$ and of absolute value 1[T1].

[T1] *Translator's note.* The source prints f_0 in $\mathcal{U}_2(f_0; \ldots)$; the orthogonality argument requires the center 0.

The first Hausdorff axiom of countability does not hold in the weak topology of $\mathfrak{H}$, for its most important consequence fails: there is a set and an accumulation point of it which is not the limit of any sequence of points of that set. Thus, in contrast to what one is otherwise accustomed to, weak closedness cannot be reduced to convergence properties.

Before giving the counterexample, we note one further fact: if $f_1, f_2, \ldots$ converges weakly to f, then, by a remark of E. Schmidt, $|f_1|, |f_2|, \ldots$ are bounded.[32]

The example is the set $\mathfrak{A}$ (we realize $\mathfrak{H}$ as the sequence space of the $\{x_1, x_2, \ldots\}$) of all $\{x_1, x_2, \ldots\}$ for which only two coordinates x_m, x_n are nonzero, with

$$x_m = 1, \qquad x_n = m \qquad (m \neq n;\ m, n \text{ otherwise arbitrary}).$$

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[32]This also follows from a general theorem of St. Banach, *Fundamenta Mathematicae* **3** (1922), p. 157. For completeness we reproduce his elegant proof, specialized to the present case.

Suppose, then, that $f_1, f_2, \ldots$ converges weakly to f; we must prove that $|f_n| \leq c$ for some fixed c. It suffices to prove

$$|(f_n, \varphi)| \leq c|\varphi| \qquad \text{for all } \varphi,$$

for putting $\varphi = f_n$ gives the assertion. The displayed relation certainly always holds if it holds for the φ with $|\varphi| = \epsilon$ ($\epsilon > 0$), i.e. if $|(f_n, \varphi)|$ is bounded for all n and all $|\varphi| = \epsilon$; all the more so if this is true for every $|\varphi| \leq \epsilon$, that is, throughout an arbitrary ball about 0.

If all $|(f_n, \varphi)|$ are bounded while φ varies in a fixed ball $\mathfrak{K}$ given by $|\varphi - \varphi_0| < \delta$, then they are also bounded in a ball about 0 (namely, for $|\varphi| \leq \delta$), since such a φ is certainly the difference of two points of $\mathfrak{K}$ (for example, $\varphi_0 \pm \frac{1}{2}\varphi$). Thus, if the theorem were false, the quantities $|(f_n, \varphi)|$ would be unbounded in every ball $\mathfrak{K}$: for every c there would be a $\varphi_0 \in \mathfrak{K}$ and an n_0 such that $|(f_{n_0}, \varphi_0)| > c$. By continuity, $|(f_{n_0}, \varphi)| > c$ would then hold throughout a suitable ball about φ_0, which may be chosen to lie inside $\mathfrak{K}$.

Now let $\mathfrak{K}$ be any ball; let $\mathfrak{K}'$ be a ball in $\mathfrak{K}$ throughout which $|(f_{n'}, \varphi)| > 1$ (n' fixed), with radius less than 1; let $\mathfrak{K}''$ be a ball in $\mathfrak{K}'$ throughout which $|(f_{n''}, \varphi)| > 2$ (n'' fixed), with radius less than $\frac{1}{2}$; and so forth. Since $\mathfrak{H}$ is complete (cf. *E.*, § I, Axiom E), all the balls $\mathfrak{K}, \mathfrak{K}', \mathfrak{K}'', \ldots$ have a common point $\overline{\varphi}$, and for this point

$$|(f_{n'}, \overline{\varphi})| > 1, \quad |(f_{n''}, \overline{\varphi})| > 2, \quad \ldots;$$

hence

$$\limsup_{n \to \infty} |(f_n, \overline{\varphi})| = \infty,$$

contrary to the hypothesis.

We add the following observation. Let $\varphi_1, \varphi_2, \ldots$ be a complete normalized orthogonal system. If $f_1, f_2, \ldots$ converges weakly to f, then, for every $\nu = 1, 2, \ldots$,

$$\lim_{n \to \infty} (f_n, \varphi_\nu) = (f, \varphi_\nu),$$

and the $|f_n|$ are bounded. This necessary condition is also sufficient: the set of all φ for which

$$\lim_{n \to \infty} (f_n, \varphi) = (f, \varphi)$$

contains $\varphi_1, \varphi_2, \ldots$ and is plainly a linear manifold. Because the $|f_n|$ are bounded (so the (f_n, φ) are equicontinuous in φ), it is closed in the strong sense; hence it comprises all of $\mathfrak{H}$. This proves everything asserted.

The point 0 is a weak accumulation point of $\mathfrak{A}$. Indeed, given

$$\{u_1^1, u_2^1, \ldots\}, \ldots, \{u_1^s, u_2^s, \ldots\}, \epsilon,$$

we can force

$$\left|\sum_{p=1}^{\infty} u_p^1 x_p\right| < \epsilon, \ldots, \left|\sum_{p=1}^{\infty} u_p^s x_p\right| < \epsilon \qquad (\{x_1, x_2, \ldots\} \in \mathfrak{A})$$

as follows. First choose m so that

$$|u_m^1| < \frac{\epsilon}{2}, \ldots, |u_m^s| < \frac{\epsilon}{2}$$

(since $u_p^1 \to 0, \ldots, u_p^s \to 0$), and then choose n so that

$$|u_n^1| < \frac{\epsilon}{2m}, \ldots, |u_n^s| < \frac{\epsilon}{2m}.$$

On the other hand, no sequence of points of $\mathfrak{A}$ converges weakly to 0. In such a sequence the absolute values, namely $\sqrt{1+m^2}$, would be bounded, so only finitely many values of m could occur. Consequently, for some m_0 one would have
$m = m_0$ infinitely often, i.e. $x_{m_0} = 1$, whereas weak convergence would permit $x_{m_0} = 1$ only finitely often.

Finally, attention should be drawn to the following fact. Inside any fixed ball $\mathfrak{K}$, all f are bounded, say $|f| \leq c$. Let $\overline{\varphi}_1, \overline{\varphi}_2, \ldots$ be a complete normalized orthogonal system. Then every

$$\mathcal{U}_2(f_0; \varphi_1, \ldots, \varphi_s, \epsilon)$$

contains a

$$\mathcal{U}_2(f_0; \overline{\varphi}_1, \ldots, \overline{\varphi}_n, \delta),$$

insofar as both are restricted to $\mathfrak{K}$. Indeed, write

$$\varphi_1 = \sum_{\nu=1}^{\infty} a_\nu^1 \overline{\varphi}_\nu, \quad \ldots, \quad \varphi_s = \sum_{\nu=1}^{\infty} a_\nu^s \overline{\varphi}_\nu,$$

choose n so that

$$\sum_{\nu=n+1}^{\infty} |a_\nu^1|^2, \ldots, \sum_{\nu=n+1}^{\infty} |a_\nu^s|^2 < \frac{\epsilon^2}{16c^2},$$

and then choose

$$\delta \leq \frac{\epsilon}{2 \max\left(\sum_{\nu=1}^{n} |a_\nu^1|, \ldots, \sum_{\nu=1}^{n} |a_\nu^s|\right)}.$$

For $r = 1, \dots, s$ and every f in the latter neighborhood, we then have

$$|(f - f_0, \varphi_r)| \leq \sum_{\nu=1}^{n} |a_\nu^r| |(f - f_0, \overline{\varphi}_\nu)| + \left| \left(f - f_0, \sum_{\nu=n+1}^{\infty} a_\nu^r \overline{\varphi}_\nu \right) \right|$$
$$\leq \sum_{\nu=1}^{n} |a_\nu^r| \delta + |f - f_0| \sqrt{\sum_{\nu=n+1}^{\infty} |a_\nu^r|^2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Thus these f belong to the first neighborhood. Since we may plainly take $\delta = 1/p$ $(p = 1, 2, \dots)$, the first Hausdorff axiom of countability is therefore satisfied.[33]

It is well known that the weak, but not the strong, topology is even compact inside $\mathfrak{K}$.[34] Thus the strong topology always satisfies the axiom of countability, but is compact neither in all of $\mathfrak{H}$ nor in $\mathfrak{K}$; the weak topology has neither property in all of $\mathfrak{H}$, but has both in $\mathfrak{K}$.

2.

After this preliminary orientation we turn to the relations in $\mathfrak{B}$, which are our actual concern.

To define convergence of a sequence of operators $A_1, A_2, \dots$ to an operator A, one ordinarily requires either strong convergence of every $A_n f$ to Af, or weak convergence; the latter means

$$(A_n f, g) \longrightarrow (Af, g) \qquad \text{for every } f, g.$$

Thus one requires either

$$|(A_n - A)f| \longrightarrow 0 \quad \text{for every } f,$$

or

$$|((A_n - A)f, g)| \longrightarrow 0 \quad \text{for every } f, g.$$

These are, respectively, strong and weak convergence of operators in $\mathfrak{B}$. As in subsection 1, we reduce these notions of convergence to corresponding topologies in $\mathfrak{B}$.

[33] Since there is a sequence dense everywhere in $\mathfrak{H}$ in the strong, and hence also in the weak, sense, one easily proves the second axiom of countability as well; cf. note 29.

[34] All methods of selection in convergence proofs in Hilbert space amount to this fact.

Strong topology on $\mathfrak{B}$. Let

$$\mathcal{U}_3(A_0; \varphi_1, \dots, \varphi_s, \epsilon) = \left\{ A \in \mathfrak{B} : \begin{array}{c} |(A - A_0)\varphi_1| < \epsilon, \dots, \\ |(A - A_0)\varphi_s| < \epsilon \end{array} \right\}.$$

All $\mathcal{U}_3(A_0; \varphi_1, \dots, \varphi_s, \epsilon)$

with arbitrary $\varphi_1, \dots, \varphi_s \in \mathfrak{H}$, arbitrary s, and $\epsilon > 0$ are the neighborhoods of A_0.

Weak topology on $\mathfrak{B}$. Let

$$\mathcal{U}_4(A_0; \varphi_1, \psi_1, \dots, \varphi_s, \psi_s, \epsilon)$$

be the set of all $A \in \mathfrak{B}$ such that

$$|((A - A_0)\varphi_1, \psi_1)| < \epsilon, \dots, |((A - A_0)\varphi_s, \psi_s)| < \epsilon.$$

All these sets, with arbitrary $\varphi_1, \psi_1, \dots, \varphi_s, \psi_s \in \mathfrak{H}$, arbitrary s, and $\epsilon > 0$, are the neighborhoods of A_0.

One again readily verifies Hausdorff's axioms α)–δ) and that these topologies yield the notions of convergence just described. Furthermore, it is clear that the operations aA and $A + B$ are continuous in both topologies (the latter in both variables simultaneously). The operation A^* is continuous in the weak topology, because

$$((A - A_0)\varphi, \psi) = \overline{((A^* - A_0^*)\psi, \varphi)},$$

but not in the strong topology (we omit the easily supplied counterexamples). The product AB is continuous in either topology in each variable separately when the other is held fixed. Its strong continuity in A is seen by replacing $\varphi_1, \dots, \varphi_s$ with $B\varphi_1, \dots, B\varphi_s$; its strong continuity in B follows from the continuity of the fixed operator A. Its weak continuity in A follows by replacing

$$\varphi_1, \psi_1, \dots, \varphi_s, \psi_s$$

with

$$B\varphi_1, \psi_1, \dots, B\varphi_s, \psi_s;$$

its weak continuity in B follows from the preceding result and

$$AB = (B^* A^*)^*.$$

In neither topology, however, is the product AB continuous in both variables simultaneously (here too the counterexamples are so simple that we omit them).

Since
$$\mathcal{U}_3(A_0; \varphi_1, \dots, \varphi_s, \epsilon) \subset \mathcal{U}_4(A_0; \varphi_1, \psi_1, \dots, \varphi_s, \psi_s, \eta)$$
for example when
$$\epsilon = \frac{\eta}{\max(|\psi_1|, \dots, |\psi_s|)},$$
every accumulation point or limit in the strong sense is also one in the weak sense. The converse cannot hold, already because A^* has different continuity properties in the two topologies.

The first axiom of countability holds in neither of these topologies. In fact, for each there are sets with an accumulation point which is not the limit of any sequence of their points. We shall prove both assertions by exhibiting a set $\mathfrak{A}$ of which 0 is a strong accumulation point, whereas no sequence of its points has 0 even as a weak limit.

First we establish once more that if $A_1, A_2, \dots$ converges weakly (and hence, a fortiori, if it converges strongly) to A, then $A_1, A_2, \dots$ are uniformly bounded: there is a fixed c such that
$$|A_n f| \leq c|f| \qquad \text{for every } n \text{ and } f.$$
It suffices to prove this for weak convergence, where the proof is analogous to that of the corresponding theorem in $\mathfrak{H}$.[35]

It follows incidentally that multiplication AB is continuous in both variables simultaneously with respect to strong *convergence* (not the topology). Indeed, from $A_n \to A$ and $B_n \to B$ it follows that
$$A_n B f \longrightarrow ABf.$$
Moreover, $B_n f - Bf \to 0$, and the A_n are equicontinuous, so
$$A_n B_n f - A_n B f \longrightarrow 0.$$

[35]By E., Theorem 12β, it suffices to prove
$$|(A_n f, g)| \leq C|f||g|.$$
We prove this by St. Banach's method exactly as for the analogous relation of E. Schmidt in note 32, except that the function (f_n, φ) of n, φ is replaced by the function $(A_n f, g)$ of n, f, g, and the balls $\mathfrak{K}, \mathfrak{K}', \mathfrak{K}'', \dots$ for φ by pairs of balls
$$(\mathfrak{K}, \mathfrak{L}),\ (\mathfrak{K}', \mathfrak{L}'),\ (\mathfrak{K}'', \mathfrak{L}''), \dots$$
for f and g, respectively.

Thus $A_n B_n f \longrightarrow ABf$, i.e. $A_n B_n \longrightarrow AB$ (all arrows being in the sense of strong convergence in $\mathfrak{H}$ or $\mathfrak{B}$, as appropriate). We now give the promised counterexample.

Let $\overline{A}_{m,n}$ be the following operator in $\mathfrak{B}$ ($\mathfrak{H}$ again being realized as the sequence space of the $\{x_1, x_2, \ldots\}$):

$$\overline{A}_{m,n}\{x_1, x_2, \ldots\} = \{y_1, y_2, \ldots\},$$

where

$$y_m = x_m, \qquad y_n = m x_n, \qquad y_p = 0 \quad (p \neq m, n), \qquad m \neq n.$$

Let $\mathfrak{A}$ be the set of all $\overline{A}_{m,n}$. The zero operator is a strong accumulation point of $\mathfrak{A}$. Indeed, given

$$\{u_1^1, u_2^1, \ldots\}, \ldots, \{u_1^s, u_2^s, \ldots\}, \epsilon,$$

we can force

$$|\overline{A}_{m,n}\{u_1^1, u_2^1, \ldots\}| < \epsilon, \ldots, |\overline{A}_{m,n}\{u_1^s, u_2^s, \ldots\}| < \epsilon,$$

that is,[T2]

$$\sqrt{|u_m^1|^2 + m^2|u_n^1|^2} < \epsilon, \ldots, \sqrt{|u_m^s|^2 + m^2|u_n^s|^2} < \epsilon,$$

as follows. First choose m so that

$$|u_m^1| < \frac{\epsilon}{\sqrt{2}}, \ldots, |u_m^s| < \frac{\epsilon}{\sqrt{2}}$$

(since $u_p^1 \to 0, \ldots, u_p^s \to 0$), and then choose n so that

$$|u_n^1| < \frac{\epsilon}{\sqrt{2}\, m}, \ldots, |u_n^s| < \frac{\epsilon}{\sqrt{2}\, m}.$$

On the other hand, no sequence of elements of $\mathfrak{A}$ converges weakly to 0. In such a sequence one would have uniformly

$$|\overline{A}_{m,n} f| \leq c|f|,$$

so that m would be bounded and only finitely many values of m could occur. Hence some m_0 would occur infinitely often. If

$$\varphi = \psi = \{x_1, x_2, \ldots\}, \qquad x_{m_0} = 1, \quad x_p = 0\,(p \neq m_0),$$

[T2] *Translator's note.* In the last radical in this estimate the source prints $m^2|u_m^s|^2$. The correction to $m^2|u_n^s|^2$ is required by the subsequent choice of n.

then $(\overline{A}_{m,n}\varphi, \psi) = 1$ infinitely often, whereas weak convergence would permit this equality only finitely often.

The discontinuity of A^* in the strong topology is sometimes inconvenient for our purposes. We therefore introduce *strong double convergence*: it obtains when

$$A_1, A_2, \ldots \longrightarrow A \quad \text{and} \quad A_1^*, A_2^*, \ldots \longrightarrow A^*$$

strongly. It follows from what was said above that, with respect to strong double convergence, the operations $aA, A^*, A+B, AB$ are continuous (the last two in both variables simultaneously).

Finally we show the following.[T3] Suppose that $A_{m,n}$ converges strongly to A_n as $m \to \infty$, with n fixed, and that A_n converges strongly to A as $n \to \infty$. If all $A_{m,n}$ are equicontinuous, i.e. there is a fixed c such that

$$|A_{m,n}f| \leq c|f|,$$

then a suitable subsequence A_{M_p,N_p} converges strongly to A as $p \to \infty$. Let $f_1, f_2, \ldots$ be a sequence everywhere dense in $\mathfrak{H}$ in the strong sense (cf. *E.*, § I, Axiom C). Choose N_p so that

$$|A_{N_p}f_1 - Af_1| < \frac{1}{p}, \ldots, |A_{N_p}f_p - Af_p| < \frac{1}{p}$$

(since $A_n f \to Af$ always), and then choose M_p so that

$$|A_{M_p,N_p}f_1 - A_{N_p}f_1| < \frac{1}{p}, \ldots, |A_{M_p,N_p}f_p - A_{N_p}f_p| < \frac{1}{p}$$

(since $A_{m,n}f \to A_n f$ with n fixed and $m \to \infty$). It follows that

$$A_{M_p,N_p}f \longrightarrow Af$$

for $f = f_1, f_2, \ldots$, i.e. on an everywhere dense subset of $\mathfrak{H}$. But all the A_{M_p,N_p}, as well as A, are equicontinuous; the conclusion therefore holds for every f, as claimed. The same is plainly true for strong double convergence.

3.

There is a third way to topologize $\mathfrak{B}$. Define convergence of $A_1, A_2, \ldots$ to A by requiring $|(A_n - A)f|$ to tend to 0 uniformly, as $n \to \infty$, for all f; that is,

$$|(A_n - A)f| \leq c_n|f|, \qquad c_n \longrightarrow 0.$$

[T3] *Translator's note.* The source interchanges the two indices in this first convergence statement. The form printed here is the one used by the diagonal construction below.

Equivalently, one may require $|((A_n - A)f, g)|$ to tend to 0 uniformly, as $n \to \infty$, for all f, g; that is,

$$|((A_n - A)f, g)| \le c_n |f||g|, \qquad c_n \longrightarrow 0.$$

By E., Theorem 12β, the two conditions are identical. We call this, with the natural terminology, *uniform convergence*; as we see, it is immaterial whether strong or weak convergence is sharpened in this way. The corresponding topology is plainly the following.

Uniform topology on $\mathfrak{B}$**.** Let $\mathcal{U}_5(A_0; \epsilon)$ be the set of all $A \in \mathfrak{B}$ for which there is a fixed $\epsilon' < \epsilon$ such that[T4]

$$|(A - A_0)f| < \epsilon' |f| \qquad \text{for every } f.$$

All $\mathcal{U}_5(A_0; \epsilon)$ with $\epsilon > 0$ are the neighborhoods of A_0.

It is clear that Hausdorff's axioms α)–δ) are satisfied and that the resulting notion of convergence is the one just given. One also sees that the operations $aA, A^*, A + B, AB$ are continuous (the last two in both variables simultaneously).[36]

Every strong neighborhood $\mathcal{U}_3(A_0; \varphi_1, \ldots, \varphi_s, \epsilon)$ contains a sufficiently small uniform neighborhood $\mathcal{U}_5(A_0; \eta)$. [T5]

Consequently every uniform accumulation point or limit is also one in the strong (and hence, a fortiori, in the weak) sense.[37] The converse is false, since the first axiom of countability fails in both the strong and weak topologies, whereas, as we shall now see, it holds in the uniform topology.

Since among the $\mathcal{U}_5(A_0; \epsilon)$ we may restrict ourselves to

$$\mathcal{U}_5\left(A_0; \frac{1}{n}\right), \qquad n = 1, 2, \ldots,$$

[T4]*Translator's note.* The source omits the factor $|f|$ from this displayed inequality; it is restored from the definition of uniform convergence immediately above.

[36]For A^*, this follows because

$$|Af| \le c|f| \text{ for every } f \quad \Longrightarrow \quad |A^* f| \le c|f|;$$

these assertions are equivalent, respectively, to

$$|(Af, g)| \le c|f||g|, \qquad |(A^* f, g)| \le c|f||g|$$

for every f, g (cf. E., Theorem 12β), and $(A^* f, g) = \overline{(Ag, f)}$.

[T5]*Translator's note.* The displayed set inclusion in the source is reversed; the verbal statement here gives the direction required by the two topologies.

[37]Uniform convergence also implies strong double convergence, because $*$ is continuous for the former.

the first axiom of countability holds. This topology, moreover, arises from a metric on the linear space $\mathfrak{B}$: if we define as the modulus of A the upper bound of $|Af|$ for $|f| = 1$, that is, the smallest c for which $|Af| \leq c|f|$ always holds—it shall be denoted by $|A|$—then, first, the following relations are immediate:

$$|aA| = |a|\,|A|, \qquad |A+B| \leq |A| + |B|, \qquad |AB| \leq |A|\,|B|.$$

Thus $|A - B|$ is a distance in the linear space $\mathfrak{B}$ (cf. *E.*, § I, Axiom B, and also Definition 4 and note 27); and, second, $\mathcal{U}_5(A_0; \epsilon)$ is the set of all A for which $|A - A_0| < \epsilon$. In other words, the uniform topology arises from this metric.

As in $\mathfrak{H}$, so also in $\mathfrak{B}$, within every fixed ball $\mathfrak{K}$ (where all A satisfy $|A| \leq c$), the character of the strong and weak topologies changes essentially: there both satisfy the first axiom of countability. Indeed, let $\overline{\varphi}_1, \overline{\varphi}_2, \ldots$ be a complete normalized orthogonal system. Then every $\mathcal{U}_3(A_0; \varphi_1, \ldots, \varphi_s, \epsilon)$, respectively

$$\mathcal{U}_4(A_0; \varphi_1, \psi_1, \ldots, \varphi_s, \psi_s, \epsilon),$$

contains a

$$\mathcal{U}_3(A_0; \overline{\varphi}_1, \ldots, \overline{\varphi}_n, \delta),$$

respectively a[T6]

$$\mathcal{U}_4(A_0; \overline{\varphi}_{k_1}, \overline{\varphi}_{l_1}, \ldots, \overline{\varphi}_{k_m}, \overline{\varphi}_{l_m}, \delta).$$

In the first case put

$$\varphi_1 = \sum_{\nu=1}^{\infty} a_\nu^1 \overline{\varphi}_\nu, \ldots, \qquad \varphi_s = \sum_{\nu=1}^{\infty} a_\nu^s \overline{\varphi}_\nu,$$

and choose n so that

$$\sum_{\nu=n+1}^{\infty} |a_\nu^1|^2 < \frac{\epsilon^2}{16c^2}, \ldots, \qquad \sum_{\nu=n+1}^{\infty} |a_\nu^s|^2 < \frac{\epsilon^2}{16c^2},$$

and choose[T7]

$$\delta \leq \frac{\epsilon}{2\max\left(\sum_{\nu=1}^{n} |a_\nu^1|, \ldots, \sum_{\nu=1}^{n} |a_\nu^s|\right)}.$$

[T6] *Translator's note.* Throughout this argument the source calls the strong and weak neighborhoods $\mathcal{U}_4$ and $\mathcal{U}_5$, respectively. The definitions given earlier require $\mathcal{U}_3$ and $\mathcal{U}_4$, which are used here.

[T7] *Translator's note.* In this displayed choice of δ for the strong topology the source writes the upper summation limits as ∞. They have been corrected to n, as required by the estimate below.

Then, for $r = 1, \ldots, s$ and every A in the latter $\mathcal{U}_3$,

$$\begin{aligned} |(A - A_0)\varphi_r| &\leq \sum_{\nu=1}^{n} |a_\nu^r| |(A - A_0)\overline{\varphi}_\nu| + \left| (A - A_0) \sum_{\nu=n+1}^{\infty} a_\nu^r \overline{\varphi}_\nu \right| \\ &\leq \sum_{\nu=1}^{n} |a_\nu^r| \delta + 2c \sqrt{\sum_{\nu=n+1}^{\infty} |a_\nu^r|^2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Hence these A belong to the first-mentioned $\mathcal{U}_3$.

In the second case put

$$\varphi_1 = \sum_{\nu=1}^{\infty} a_\nu^1 \overline{\varphi}_\nu, \quad \psi_1 = \sum_{\nu=1}^{\infty} b_\nu^1 \overline{\varphi}_\nu, \quad \ldots \qquad \varphi_s = \sum_{\nu=1}^{\infty} a_\nu^s \overline{\varphi}_\nu, \quad \psi_s = \sum_{\nu=1}^{\infty} b_\nu^s \overline{\varphi}_\nu,$$

and choose n so that

$$\sum_{\nu=n+1}^{\infty} |a_\nu^1|^2 < \eta^2, \qquad \sum_{\nu=n+1}^{\infty} |b_\nu^1|^2 < \eta^2, \quad \ldots, \\ \sum_{\nu=n+1}^{\infty} |a_\nu^s|^2 < \eta^2, \qquad \sum_{\nu=n+1}^{\infty} |b_\nu^s|^2 < \eta^2,$$

where $\eta > 0$ has been chosen so that[T8]

$$4c \max(|\varphi_1|, |\psi_1|, \ldots, |\varphi_s|, |\psi_s|)\eta + 2c\eta^2 = \frac{\epsilon}{2}.$$

Choose further[T9]

$$\delta \leq \frac{\epsilon}{2 \max \left(\left(\sum_{\nu=1}^{n} |a_\nu^1| \right) \left(\sum_{\nu=1}^{n} |b_\nu^1| \right), \ldots, \left(\sum_{\nu=1}^{n} |a_\nu^s| \right) \left(\sum_{\nu=1}^{n} |b_\nu^s| \right) \right)}.$$

Finally, let $m = n^2$, and let $(k_1, l_1), \ldots, (k_m, l_m)$ be all pairs (ρ, σ) with $\rho \leq n$, $\sigma \leq n$. Then, for $r = 1, \ldots, s$ and every A in the latter $\mathcal{U}_4$,[T10]

[T8] *Translator's note.* In this choice and in the estimate that follows, the source prints η^2 in place of $2c\eta^2$. Since $|A|, |A_0| \leq c$, one has $|A - A_0| \leq 2c$; the corrected coefficient $2c$ is therefore used here.

[T9] *Translator's note.* In this displayed choice of δ for the weak topology the source prints only the squared sums of the a_ν^r. They have been corrected to the products $(\sum_{\nu=1}^{n} |a_\nu^r|)(\sum_{\nu=1}^{n} |b_\nu^r|)$.

[T10] *Translator's note.* In the second tail term of this estimate the source prints ψ_1. It has been corrected to ψ_r, as required by the subsequent bound.

$$
\begin{aligned}
&\left|((A-A_0)\varphi_r,\psi_r)-\left((A-A_0)\sum_{\nu=1}^{n}a_\nu^r\overline{\varphi}_\nu,\sum_{\nu=1}^{n}b_\nu^r\overline{\varphi}_\nu\right)\right| \\
&=\left|-\left((A-A_0)\varphi_r,\sum_{\nu=n+1}^{\infty}b_\nu^r\overline{\varphi}_\nu\right)-\left((A-A_0)\sum_{\nu=n+1}^{\infty}a_\nu^r\overline{\varphi}_\nu,\psi_r\right)\right. \\
&\qquad\qquad\left.+\left((A-A_0)\sum_{\nu=n+1}^{\infty}a_\nu^r\overline{\varphi}_\nu,\sum_{\nu=n+1}^{\infty}b_\nu^r\overline{\varphi}_\nu\right)\right| \\
&\leq 2c|\varphi_r|\sqrt{\sum_{\nu=n+1}^{\infty}|b_\nu^r|^2}+2c\sqrt{\sum_{\nu=n+1}^{\infty}|a_\nu^r|^2}|\psi_r| \\
&\qquad\qquad+2c\sqrt{\left(\sum_{\nu=n+1}^{\infty}|a_\nu^r|^2\right)\left(\sum_{\nu=n+1}^{\infty}|b_\nu^r|^2\right)} \\
&<4c\max(|\varphi_r|,|\psi_r|)\eta+2c\eta^2\leq\frac{\epsilon}{2},
\end{aligned}
$$

and

$$
\begin{aligned}
\left|\left((A-A_0)\sum_{\nu=1}^{n}a_\nu^r\overline{\varphi}_\nu,\sum_{\nu=1}^{n}b_\nu^r\overline{\varphi}_\nu\right)\right| &\leq\sum_{\mu,\nu=1}^{n}|a_\mu^r|\,|b_\nu^r|\,|((A-A_0)\overline{\varphi}_\mu,\overline{\varphi}_\nu)| \\
&\leq\sum_{\mu,\nu=1}^{n}|a_\mu^r|\,|b_\nu^r|\,\delta\leq\frac{\epsilon}{2}.
\end{aligned}
$$

Consequently

$$
|((A-A_0)\varphi_r,\psi_r)|\leq\frac{\epsilon}{2}+\frac{\epsilon}{2}=\epsilon;
$$

that is, these A belong to the first-mentioned $\mathcal{U}_4$. Since we can plainly choose $\delta = 1/p$ ($p = 1, 2, \ldots$), the assertion above is proved.

4.

In conclusion we note: every subset $\mathfrak{M}$ of $\mathfrak{B}$ has a countable subset $\mathfrak{N}$ such that every element A of $\mathfrak{M}$ is the limit of a strongly doubly convergent sequence from $\mathfrak{N}$.[38]

[38]Thus $\mathfrak{N}$ is everywhere dense in $\mathfrak{M}$, both strongly and weakly; in particular, we may put $\mathfrak{M} = \mathfrak{B}$. This is certainly false for the uniform topology, since there one can readily give a set of continuum many A's whose pairwise distances are 1. If we take $\mathfrak{M}$ to be a ball $\mathfrak{K}$, then from the validity of the first axiom of countability in $\mathfrak{K}$, in both the strong and weak senses, we can also

Let $\mathfrak{M}_c$ be the set of all $A \in \mathfrak{M}$ with $|A| < c$. It is enough to prove the assertion for all $\mathfrak{M}_1, \mathfrak{M}_2, \ldots$ (and take $\mathfrak{N}$ successively equal to $\mathfrak{N}_1, \mathfrak{N}_2, \ldots$), since we can then put $\mathfrak{N} = \mathfrak{N}_1 + \mathfrak{N}_2 + \cdots$. We may therefore assume from the beginning that $|A| < c$ throughout $\mathfrak{M}$.

We realize $\mathfrak{H}$ as sequence space (the $\{x_1, x_2, \ldots\}$ with finite $\sum_{n=1}^{\infty}|x_n|^2$). Then A has a matrix $\{a_{\mu,\nu}\}$, since in general

$$A\{x_1, x_2, \ldots\} = \{y_1, y_2, \ldots\}, \qquad y_\nu = \sum_{\mu=1}^{\infty} a_{\mu,\nu} x_\mu$$

(A is linear and continuous). Let $p = 1, 2, \ldots$, and let $a_{\mu,\nu}^{(p)}$ ($\mu, \nu = 1, \ldots, p$) be arbitrary complex numbers with rational real and imaginary parts (Gaussian rationals) [T11] such that

$$|a_{\mu,\nu}^{(p)} - a_{\mu,\nu}| < \frac{1}{p} \qquad (\mu, \nu = 1, \ldots, p).$$

We define $A^{(p)}$ by

$$A^{(p)}\{x_1, x_2, \ldots\} = \{y_1, y_2, \ldots\}, \qquad y_\nu = \begin{cases} \sum_{\mu=1}^{p} a_{\mu,\nu}^{(p)} x_\mu, & \nu = 1, \ldots, p, \\ 0, & \text{for all other } \nu. \end{cases}$$

readily infer that of the second (cf. note 29 and note 33). We mention further that the weak topology in $\mathfrak{K}$ is compact (cf. the end of subsection 1 and note 34).

[T11] *Translator's note.* The source says only "rational numbers." In the complex Hilbert space considered here this must mean Gaussian rationals; real rational numbers alone cannot approximate arbitrary complex matrix entries.

For $p \geq m$ we then have[T12]

$$\begin{aligned}|A^{(p)}\varphi_m - A\varphi_m|^2 &= \left|\sum_{\mu=1}^{p} a_{m\mu}^{(p)}\varphi_\mu - \sum_{\mu=1}^{\infty} a_{m\mu}\varphi_\mu\right|^2 \\ &= \left|\sum_{\mu=1}^{p}(a_{m\mu}^{(p)} - a_{m\mu})\varphi_\mu - \sum_{\mu=p+1}^{\infty} a_{m\mu}\varphi_\mu\right|^2 \\ &= \sum_{\mu=1}^{p}|a_{m\mu}^{(p)} - a_{m\mu}|^2 + \sum_{\mu=p+1}^{\infty}|a_{m\mu}|^2 < \frac{1}{p} + \sum_{\mu=p+1}^{\infty}|a_{m\mu}|^2.\end{aligned}$$

As $p \to \infty$, this plainly tends to 0,[39] and this holds for every $m = 1, 2, \ldots$. Thus

$$A^{(p)}\varphi_m \longrightarrow A\varphi_m;$$

in precisely the same way one shows

$$A^{(p)*}\varphi_m \longrightarrow A^*\varphi_m.$$

Now consider all operators $B \in \mathfrak{B}$ whose matrix $\{b_{\mu,\nu}\}$ consists entirely of Gaussian rationals, only finitely many of which are nonzero. They plainly form a countable set $B_1, B_2, \ldots$. The $A^{(p)}$ above are such B's. Thus, for every $A \in \mathfrak{B}$, we have a sequence $B_{\nu_1}, B_{\nu_2}, \ldots$ such that

$$B_{\nu_p} f \longrightarrow Af, \qquad B_{\nu_p}^* f \longrightarrow A^* f$$

for all $f = \varphi_1, \varphi_2, \ldots$.

[T12] *Translator's note.* In the middle line of this estimate the source prints $\mu = \nu + 1$ as the lower limit of the final sum. Since the construction is truncated at p, and the following line has the same tail, this has been corrected to $\mu = p + 1$.

[39] Because $\sum_{\mu=1}^{\infty}|a_{m\mu}|^2$ converges. The convergence of all the series

$$\sum_{m=1}^{\infty}|a_{mn}|^2, \qquad \sum_{n=1}^{\infty}|a_{mn}|^2$$

may be seen as follows. The second converges because

$$A\{0, \ldots, 0, \underset{m}{1}, 0, \ldots\} = \{a_{m1}, a_{m2}, \ldots\};$$

the first then does as well if A^* is considered in place of A (where a_{mn} is replaced by $\overline{a}_{nm}$).

We now pass to $\mathfrak{M}$. For each pair $\nu, p = 1, 2, \ldots$, we ask whether there is an $A \in \mathfrak{M}$ such that

$$|B_\nu f - Af| < \frac{1}{p}, \qquad |B_\nu^* f - A^* f| < \frac{1}{p} \quad \text{for } f = \varphi_1, \ldots, \varphi_p.$$

If there is one, choose one such A and call it $A_{\nu,p}$. The $A_{\nu,p}$ form a countable subset $\mathfrak{N}$ of $\mathfrak{M}$; we shall show that it has the required property.

Let A be any element[T13] of $\mathfrak{M}$, and let $p = 1, 2, \ldots$. Form the sequence $B_{\nu_1}, B_{\nu_2}, \ldots$ just described. For this sequence all

$$|B_{\nu_q} f - Af|, \qquad |B_{\nu_q}^* f - A^* f|$$

tend to 0 as $q \to \infty$ ($f = \varphi_1, \varphi_2, \ldots$). Thus, if we choose q sufficiently large and set $n = \nu_q$, then

$$|B_n f - Af| < \frac{1}{p}, \qquad |B_n^* f - A^* f| < \frac{1}{p}$$

for all $f = \varphi_1, \ldots, \varphi_p$. Hence $A_{n,p}$ can be formed (since A, for example, has the required property), and, for all $f = \varphi_1, \ldots, \varphi_p$,

$$|B_n f - A_{n,p} f| < \frac{1}{p}, \qquad |B_n^* f - A_{n,p}^* f| < \frac{1}{p}.$$

Consequently[T14]

$$|A_{n,p} f - Af| < \frac{2}{p}, \qquad |A_{n,p}^* f - A^* f| < \frac{2}{p}.$$

Write $\overline{A}_p$ for $A_{n,p}$ (where n, too, depends on p). It is then plain that

$$\overline{A}_p f \longrightarrow Af, \qquad \overline{A}_p^* f \longrightarrow A^* f$$

for all $f = \varphi_1, \varphi_2, \ldots$, hence also for every linear combination of finitely many of these vectors, that is, on a subset everywhere dense in $\mathfrak{H}$ in the strong sense. Since all the $\overline{A}_p$ belong to $\mathfrak{N}$, hence to $\mathfrak{M}$, and are therefore equicontinuous (uniformly in p), the convergence holds for every f. Thus $\overline{A}_1, \overline{A}_2, \ldots$ converges strongly doubly to A, while it is a sequence from $\mathfrak{N}$.

[T13] *Translator's note.* At this point the source prints $A \in \mathfrak{B}$. The correction to $A \in \mathfrak{M}$ is required both by the assertion being proved and by the sentence below that uses A itself to establish the existence of $A_{n,p}$.

[T14] *Translator's note.* In the second $2/p$ estimate the source prints $|A_{n,p}^* f - Af|$. The second A has been corrected to A^*, as required by the preceding adjoint estimates and by the conclusion $\overline{A}_p^* f \to A^* f$.

II. Properties of $R(\mathfrak{M})$ and $\mathfrak{M}'$

1.

We now turn to the definition of the operations $R(\mathfrak{M})$ and $\mathfrak{M}'$ already mentioned in Introduction 2.

Definition 1. A subset $\mathfrak{M}$ of $\mathfrak{B}$ is called a ring if, first, with A, B it also contains

$$aA, \quad A^*, \quad A+B, \quad AB,$$

and, second, it is weakly closed.[40]

Definition 2. Since the intersection of arbitrarily many rings is again a ring, the intersection of all rings containing a given subset $\mathfrak{M} \subset \mathfrak{B}$ is in particular a ring—the smallest ring containing $\mathfrak{M}$. It shall be denoted by $R(\mathfrak{M})$, the ring generated by $\mathfrak{M}$.

Definition 3. Let $\mathfrak{M}$ be a subset of $\mathfrak{B}$. The set of all $A \in \mathfrak{B}$ for which A and A^* commute with every $B \in \mathfrak{M}$ (cf. note 16) shall be denoted by $\mathfrak{M}'$. Iterates $\mathfrak{M}'', \mathfrak{M}''', \ldots$ may be formed in the same way.

The set $\mathfrak{M}'$ is always a ring. That it contains $aA, A^*, A+B, AB$ along with A, B is clear; but it is also weakly closed. For if A is a weak accumulation point of $\mathfrak{M}'$ and $B \in \mathfrak{M}$, then for every $A' \in \mathfrak{M}'$

$$A'B = BA', \qquad A'^*B = BA'^*,$$

and these four expressions are all continuous in A (weakly, with B fixed). Hence

$$AB = BA, \qquad A^*B = BA^*.$$

If $A \in \mathfrak{M}$ and $B \in \mathfrak{M}'$, then B, B^* commute with A, and therefore A, A^* commute with B; hence $A \in \mathfrak{M}''$. Thus $\mathfrak{M} \subset \mathfrak{M}''$. It is further clear that

$$\mathfrak{M} \subset \mathfrak{N} \quad \Longrightarrow \quad \mathfrak{M}' \supset \mathfrak{N}'.$$

Applying this to $\mathfrak{M} \subset \mathfrak{M}''$ gives $\mathfrak{M}' \supset \mathfrak{M}'''$. But if in the same inclusion $\mathfrak{M}$ is merely replaced by $\mathfrak{M}'$, then $\mathfrak{M}' \subset \mathfrak{M}'''$. Hence $\mathfrak{M}' = \mathfrak{M}'''$. Replacing $\mathfrak{M}$ successively by $\mathfrak{M}', \mathfrak{M}'', \ldots$, we obtain

$$\mathfrak{M}' = \mathfrak{M}''' = \mathfrak{M}^{\mathrm{V}} = \cdots, \qquad \mathfrak{M}'' = \mathfrak{M}^{\mathrm{IV}} = \mathfrak{M}^{\mathrm{VI}} = \cdots.$$

[40]Observe that weak closedness is more than strong closedness.

Since $\mathfrak{M}'$ is always a ring and plainly contains 1, the same is true of $\mathfrak{M}''$. Because $\mathfrak{M} \subset \mathfrak{M}''$, it follows that

$$(\mathfrak{M}, 1)^{41} \subset \mathfrak{M}'', \qquad R(\mathfrak{M}, 1) \subset \mathfrak{M}''$$

(and hence also $R(\mathfrak{M}) \subset \mathfrak{M}''$). We shall soon show (Theorem 5) that in fact

$$R(\mathfrak{M}, 1) = \mathfrak{M}''.$$

We call a set $\mathfrak{M}$ Abelian if all its elements and their adjoints commute with one another. Thus, for sets which contain A^* together with A, for example rings, the commutativity of all the elements suffices. To be Abelian plainly means $\mathfrak{M} \subset \mathfrak{M}'$. Since this implies $\mathfrak{M}'' \subset \mathfrak{M}'''$, $\mathfrak{M}''$ is Abelian whenever $\mathfrak{M}$ is; and because $\mathfrak{M} \subset \mathfrak{M}''$, the converse also holds. From

$$\mathfrak{M} \subset R(\mathfrak{M}) \subset \mathfrak{M}''$$

it follows that $R(\mathfrak{M})$ is Abelian simultaneously with these sets. Thus the Abelian character is equivalent for all three sets $\mathfrak{M}$, $R(\mathfrak{M})$, and $\mathfrak{M}''$.

Let $\mathfrak{M}$ be a ring. If it is Abelian, then every $A \in \mathfrak{M}$ commutes with A^*, that is, A is normal (cf. Introduction 2, at note 16). If it is not Abelian, it contains two noncommuting operators A, B. Put

$$A_1 = \frac{A + A^*}{2}, \qquad A_2 = \frac{A - A^*}{2i}, \qquad B_1 = \frac{B + B^*}{2}, \qquad B_2 = \frac{B - B^*}{2i}.$$

The A_1, A_2, B_1, B_2 are Hermitian operators in $\mathfrak{M}$ and certainly do not all commute (since $A = A_1 + iA_2$, $B = B_1 + iB_2$). We may therefore take A, B themselves to be Hermitian. Then $C = A + iB$ belongs to $\mathfrak{M}$ and fails to commute with $C^* = A - iB$; it is therefore not normal. Consequently: a ring $\mathfrak{M}$ is Abelian if and only if all its elements are normal. An arbitrary $\mathfrak{M}$ has the corresponding property if and only if $R(\mathfrak{M})$ does.

2.

We shall now discuss the relations of the preceding notions to projection operators and unitary operators.

[41] The union of $\mathfrak{M}$ with 1.

Theorem 1. Let $\mathfrak{M}$ be a ring, let A be a bounded Hermitian operator, and let $E(\lambda)$ be its resolution of the identity.[42]

A belongs to $\mathfrak{M}$ if and only if all $E(\lambda)$ with $\lambda < 0$, and all $1 - E(\lambda)$ with $\lambda \geq 0$, belong to $\mathfrak{M}$. If $\mathfrak{M}$ contains 1, we may say more simply: if and only if all $E(\lambda)$ belong to $\mathfrak{M}$.

[42]In E. (Definition 17), a resolution of the identity was defined as a family of projection operators $E(\lambda)$, $-\infty < \lambda < \infty$, with the following properties:

α) If $\lambda \leq \mu$, then $E(\lambda)$ is a part of $E(\mu)$.

β) If $\lambda \geq \lambda_0$ and $\lambda \to \lambda_0$, then $E(\lambda)f \to E(\lambda_0)f$ for every f.

γ) If $\lambda \to -\infty$, respectively $\lambda \to +\infty$, then always $E(\lambda)f \to 0$, respectively $E(\lambda)f \to f$.

All convergences, as always in E., are strong. We would now say in β) that $E(\lambda)$ is a right-continuous function of λ in the strong topology of $\mathfrak{B}$; and in γ), that at $\lambda = \pm\infty$ it can be defined continuously in the strong topology of $\mathfrak{B}$ as 0 and 1, respectively. *The source reverses* 0 *and* 1 *in this bracketed restatement; the order here is required by* γ) *itself.*

The resolution of the identity $E(\lambda)$ belongs to the (not necessarily bounded) Hermitian operator A (cf. E., Theorem 36) if, in addition, the following holds.

Spectral form. The expression Af has meaning if and only if

$$\int_{-\infty}^{\infty} \lambda^2 \, d|E(\lambda)f|^2$$

is finite. This is a Stieltjes integral; since the integrand is always nonnegative and, as can be shown, the expression following d never decreases, the integral is either finite (convergent) or $+\infty$ (properly divergent), and here the former is required. If this is the case, then, for every g,

$$(Af, g) = \int_{-\infty}^{\infty} \lambda \, d(E(\lambda)f, g),$$

the integral being absolutely convergent.

It was established in E. that the so-called hypermaximal Hermitian operators have such resolutions of the identity, and only they do; indeed, all hypermaximal Hermitian operators and all resolutions of the identity correspond one-to-one. This was the generalization of Hilbert's spectral form to the unbounded case (cf. E., Theorem 36).

For a bounded Hermitian operator A, the resolution of the identity always exists. This is the result of Hilbert's theory; cf. also E., where it is shown that all bounded A are hypermaximal (Theorem 48, α, β). The resolution of the identity of such a Hermitian operator is characterized by the fact that condition γ) may be strengthened to say that $E(\lambda)$ is actually constant for all sufficiently small, respectively large, λ, and hence is 0, respectively 1. If this occurs, say, for $\lambda \leq -c$, respectively $\lambda \geq c$, then all $\int_{-\infty}^{\infty}$ above may be replaced by $\int_{-c}^{c}$ (and γ) is automatically satisfied).

This characterization of boundedness is one of the results of Hilbert's theory. For completeness only, we shall give a short proof of it in Appendix I.

Proof. The second assertion follows from the first, so we consider the first.

First, the condition is sufficient. Suppose that all $E(\lambda)$, $\lambda < 0$, and all $1 - E(\lambda)$, $\lambda \geq 0$, belong to $\mathfrak{M}$, and let $-c \leq \lambda \leq c$ be the interval mentioned in note 42, outside which $E(\lambda)$ is 0, respectively 1. Let K_ϵ be a chain of numbers from $-c$ to c:

$$-c = \lambda_0 < \lambda_1 < \cdots < \lambda_{n-1} < \lambda_n = c,$$

with mesh at most ϵ, that is,

$$\lambda_\nu - \lambda_{\nu-1} \leq \epsilon \quad (\nu = 1, 2, \ldots, n;\ \epsilon > 0),$$

and containing 0, say $\lambda_m = 0$. Then

$$\begin{aligned} A^{K_\epsilon} &= \sum_{\nu=1}^{n} \lambda_\nu \big(E(\lambda_\nu) - E(\lambda_{\nu-1})\big) \\ &= \sum_{\nu=1}^{m-1} \lambda_\nu \big(E(\lambda_\nu) - E(\lambda_{\nu-1})\big) \\ &\quad + \sum_{\nu=m+1}^{n} \lambda_\nu \big((1 - E(\lambda_{\nu-1})) - (1 - E(\lambda_\nu))\big) \end{aligned}$$

also belongs to $\mathfrak{M}$, and

$$(A^{K_\epsilon} f, g) = \sum_{\nu=1}^{n} \lambda_\nu \big((E(\lambda_\nu) f, g) - (E(\lambda_{\nu-1}) f, g)\big).$$

As $\epsilon \longrightarrow 0$, this tends, for all f, g, to

$$\int_{-c}^{c} \lambda \, d(E(\lambda) f, g) = (Af, g).$$

Thus A is a weak limit of the A^{K_ϵ},[43] and hence also a weak accumulation point of $\mathfrak{M}$. Consequently $A \in \mathfrak{M}$.

Next, the condition is necessary. Suppose now that $A \in \mathfrak{M}$. Then $A, A^2, A^3, \ldots$, and all their linear combinations, also belong to $\mathfrak{M}$; that is, so does every $p(A)$,

[43] It is even a uniform limit of them. Indeed, if $f_\epsilon(\lambda)$ is defined by

$$f_\epsilon(\lambda) = \lambda_\nu \qquad \text{for } \lambda_{\nu-1} < \lambda \leq \lambda_\nu,$$

where $p(x)$ is any polynomial with $p(0) = 0$. One proves easily[44] that

$$(p(A)f, g) = \int_{-c}^{c} p(\lambda)\, d(E(\lambda)f, g).$$

For $\lambda_0 < 0$, choose $\epsilon > 0$ and a polynomial $p_\epsilon(x)$, with $p_\epsilon(0) = 0$, satisfying

$$\left.\begin{aligned} 1-\epsilon < p_\epsilon(x) &< 1+\epsilon \quad \text{for} \quad -c \le x \le \lambda_0, \\ -\epsilon < p_\epsilon(x) &< 1+\epsilon \quad \text{for} \quad \lambda_0 \le x \le \lambda_0+\epsilon, \\ -\epsilon < p_\epsilon(x) &< \epsilon \quad \text{for} \quad \lambda_0+\epsilon \le x \le c. \end{aligned}\right\}$$

(one sees without difficulty that this is possible). Then, for every f, g,

$$\begin{aligned} &(p_\epsilon(A)f - E(\lambda_0)f, g) \\ &\quad = \int_{-c}^{c} p_\epsilon(\lambda)\, d(E(\lambda)f, g) - \int_{-c}^{\lambda_0} d(E(\lambda)f, g) \\ &\quad = \int_{-c}^{\lambda_0} (p_\epsilon(\lambda) - 1)\, d(E(\lambda)f, g) + \int_{\lambda_0}^{\lambda_0+\epsilon} p_\epsilon(\lambda)\, d(E(\lambda)f, g) \\ &\qquad + \int_{\lambda_0+\epsilon}^{c} p_\epsilon(\lambda)\, d(E(\lambda)f, g). \end{aligned}$$

In these three integrals[T15] the absolute values of the integrands are at most ϵ,

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then

$$\begin{aligned} (A^{K_\epsilon} f, g) &= \int_{-c}^{c} f_\epsilon(\lambda)\, d(E(\lambda)f, g), \\ (Af, g) &= \int_{-c}^{c} \lambda\, d(E(\lambda)f, g), \\ ((A^{K_\epsilon} - A)f, g) &= \int_{-c}^{c} (f_\epsilon(\lambda) - \lambda)\, d(E(\lambda)f, g). \end{aligned}$$

Now $|f_\epsilon(\lambda) - \lambda| \le \epsilon$ always. Moreover,

$$\begin{aligned} &|(E(\mu)f, g) - (E(\lambda)f, g)| \\ &\qquad \le \sqrt{|E(\mu)f|^2 - |E(\lambda)f|^2}\,\sqrt{|E(\mu)g|^2 - |E(\lambda)g|^2} \end{aligned}$$

(cf. the estimate carried out in *E.* in the proof of Theorem 36). Hence

$$|((A^{K_\epsilon} - A)f, g)| \le \epsilon |f|\,|g|,$$

and consequently (cf. § I, 3) $|A^{K_\epsilon} - A| \le \epsilon$. This proves the assertion.

[44] In every detail this calculation is the same as the one carried out for unitary operators in *E.*, Appendix I, in the proof of Theorem 14*. The device goes back to F. Riesz.

[T15] *Translator's note.* The source prints $-c$, rather than c, as the upper limit of the third integral. The interval in question and the polynomial bounds immediately above require c.

$1+\epsilon$, and ϵ, respectively. Using the method of estimation from note 43, we may therefore infer

$$\begin{aligned}
&|(p_\epsilon(A)f - E(\lambda_0)f, g)| \\
&\le \epsilon |E(\lambda_0)f|\,|E(\lambda_0)g| \\
&\quad + (1+\epsilon)\sqrt{|E(\lambda_0+\epsilon)f|^2 - |E(\lambda_0)f|^2}\,\sqrt{|E(\lambda_0+\epsilon)g|^2 - |E(\lambda_0)g|^2} \\
&\quad + \epsilon\sqrt{|f|^2 - |E(\lambda_0+\epsilon)f|^2}\,\sqrt{|g|^2 - |E(\lambda_0+\epsilon)g|^2}\,{}^{[45]} \\
&\le \epsilon|f|\,|g| + \sqrt{|E(\lambda_0+\epsilon)f|^2 - |E(\lambda_0)f|^2}\,\sqrt{|E(\lambda_0+\epsilon)g|^2 - |E(\lambda_0)g|^2} \\
&\le \epsilon|f|\,|g| + \sqrt{|E(\lambda_0+\epsilon)f|^2 - |E(\lambda_0)f|^2}\,|g|.
\end{aligned}$$

Since this holds for every g, it follows that[46]

$$|p_\epsilon(A)f - E(\lambda_0)f| \le \epsilon|f| + \sqrt{|E(\lambda_0+\epsilon)f|^2 - |E(\lambda_0)f|^2}.$$

This tends to 0 with $\epsilon \to 0$. Thus, as $\epsilon \to 0$, for example through the sequence $\epsilon = 1, \frac{1}{2}, \frac{1}{3}, \ldots$,

$$p_\epsilon(A)f \longrightarrow E(\lambda_0)f$$

strongly in $\mathfrak{H}$, for every f; hence $p_\epsilon(A)$ converges strongly in $\mathfrak{B}$ to $E(\lambda_0)$. Therefore $E(\lambda_0) \in \mathfrak{M}$.

For $\lambda_0 \ge 0$, choose polynomials $q_\epsilon(x)$, with $q_\epsilon(0) = 0$, satisfying

$$\left.\begin{array}{rl}
-\epsilon < q_\epsilon(x) < \epsilon & \text{for} \quad -c \le x \le \lambda_0, \\
-\epsilon < q_\epsilon(x) < 1+\epsilon & \text{for} \quad \lambda_0 \le x \le \lambda_0+\epsilon, \\
1-\epsilon < q_\epsilon(x) < 1+\epsilon & \text{for} \quad \lambda_0+\epsilon \le x \le c.
\end{array}\right\}$$

This, too, is plainly possible. Entirely analogous considerations show that, as $\epsilon \to 0$, the $q_\epsilon(A)$ converge strongly in $\mathfrak{B}$ to $1 - E(\lambda_0)$. In this case, therefore, $1 - E(\lambda_0) \in \mathfrak{M}$.

This proves everything. □

[45] As in *E.*, proof of Theorem 36 (cf. note 43 above), the inequality

$$\sqrt{a_1}\sqrt{b_1} + \cdots + \sqrt{a_p}\sqrt{b_p} \le \sqrt{(a_1+\cdots+a_p)(b_1+\cdots+b_p)}$$

is used here.

[46] Put $g = p_\epsilon(A)f - E(\lambda_0)f$.

Theorem 2. Let $\mathfrak{M}$ be a ring, and let $\mathfrak{M}^P, \mathfrak{M}^U$ denote the sets of all projection operators and all unitary operators, respectively, in $\mathfrak{M}$. Then always

$$R(\mathfrak{M}^P) = \mathfrak{M}.$$

The set $\mathfrak{M}^U$ is empty if $1 \notin \mathfrak{M}$; if $1 \in \mathfrak{M}$, then

$$R(\mathfrak{M}^U) = \mathfrak{M}.$$

Proof. By Theorem 1, two rings $\mathfrak{M}, \mathfrak{N}$ are identical if

$$\mathfrak{M}^P = \mathfrak{N}^P.$$

For then they contain the same Hermitian operators, and hence the same operators altogether, since membership of an arbitrary operator A is characterized by that of the two Hermitian operators

$$\frac{A + A^*}{2}, \qquad \frac{A - A^*}{2i}.$$

Now

$$\mathfrak{M}^P \subset \mathfrak{M}, \qquad R(\mathfrak{M}^P) \subset R(\mathfrak{M}) = \mathfrak{M}, \qquad \left(R(\mathfrak{M}^P)\right)^P \subset \mathfrak{M}^P,$$

while, on the other hand,

$$R(\mathfrak{M}^P) \supset \mathfrak{M}^P, \qquad \left(R(\mathfrak{M}^P)\right)^P \supset \mathfrak{M}^P.$$

Thus

$$\left(R(\mathfrak{M}^P)\right)^P = \mathfrak{M}^P,$$

and hence $R(\mathfrak{M}^P) = \mathfrak{M}$.

If U is a unitary element of $\mathfrak{M}$, then U^* and $UU^* = 1$ also belong to $\mathfrak{M}$; hence, if 1 does not belong to $\mathfrak{M}$, then $\mathfrak{M}^U$ is empty. Suppose now that $1 \in \mathfrak{M}$. First,

$$\mathfrak{M}^U \subset \mathfrak{M}, \qquad R(\mathfrak{M}^U) \subset R(\mathfrak{M}) = \mathfrak{M}.$$

Secondly, let $E \in \mathfrak{M}^P$. Then $E, 1$, and hence $2E - 1$, belong to $\mathfrak{M}$; but the last operator is unitary, because E is a projection operator,[47] and therefore belongs to $\mathfrak{M}^U$. Consequently

$$E = \tfrac{1}{2}\left((2E - 1) + 1\right) \in R(\mathfrak{M}^U),$$

[47]Indeed,

$$(2E - 1)^* = 2E - 1, \qquad (2E - 1)^2 = 4E^2 - 4E + 1 = 1.$$

and it follows that

$$\mathfrak{M}^P \subset R(\mathfrak{M}^U), \qquad R(\mathfrak{M}^P) \subset R(\mathfrak{M}^U),$$

so that $R(\mathfrak{M}^U) \supset \mathfrak{M}$. Thus $R(\mathfrak{M}^U) = \mathfrak{M}$, as was to be proved. □

3.

Let $\mathfrak{M}$ be an arbitrary subset of $\mathfrak{B}$. Consider all f for which

$$Af = 0, \qquad A^* f = 0$$

for every $A \in \mathfrak{M}$. These f plainly form a closed linear manifold. Let E_0 be the projection operator of its orthogonal complement (cf. E., Definition 12). Membership in the original manifold is therefore characterized by $E_0 f = 0$. We define:

Definition 4. Let $\mathfrak{M} \subset \mathfrak{B}$ be arbitrary. The projection operator E_0 just constructed, for which $E_0 f = 0$ is equivalent to

$$Af = 0, \qquad A^* f = 0 \quad \text{for every } A \in \mathfrak{M},$$

is called the *principal unit* of $\mathfrak{M}$.

We first show:

Theorem 3. For every $A \in \mathfrak{M}$, and indeed for every $A \in R(\mathfrak{M})$,

$$E_0 A = A E_0 = A.$$

Proof. Let $A \in \mathfrak{M}$. Since identically $E_0(1 - E_0)f = 0$, we have

$$A(1 - E_0)f = 0,$$

and therefore $A(1 - E_0) = 0$, $A = AE_0$. Applying the same argument to A^* gives $A^* = A^* E_0$; taking adjoints then gives $A = E_0 A$.

Now consider all A satisfying $E_0 A = A E_0 = A$. They plainly form a ring, and, as we have just seen, this ring contains $\mathfrak{M}$. It therefore contains $R(\mathfrak{M})$, which proves the assertion. □

Theorem 4. The principal unit E_0 belongs to both $\mathfrak{M}'$ and $\mathfrak{M}''$.

Proof. We have already seen that every $A \in \mathfrak{M}$ commutes with E_0; moreover $E_0^* = E_0$. Hence $E_0 \in \mathfrak{M}'$.

Now let $B \in \mathfrak{M}'$. If $E_0 f = 0$, then $Af = 0$ for every $A \in \mathfrak{M}$, and hence

$$BAf = ABf = 0.$$

Likewise $A^* f = 0$, and therefore

$$BA^* f = A^* Bf = 0.$$

Thus $E_0 Bf = 0$. Since $E_0(1 - E_0)f = 0$ identically, it follows that

$$E_0 B(1 - E_0)f = 0,$$

that is,

$$E_0 B(1 - E_0) = 0, \qquad E_0 B = E_0 B E_0.$$

Take adjoints here and replace B by B^*, which also belongs to $\mathfrak{M}'$. We obtain

$$BE_0 = E_0 B E_0,$$

and consequently $E_0 B = BE_0$. This holds for every $B \in \mathfrak{M}'$, so $E_0 \in \mathfrak{M}''$. □

We now prove the theorem announced in paragraph 3 of the Introduction.

Theorem 5. $R(\mathfrak{M})$ is the set of all $A \in \mathfrak{M}''$ for which

$$E_0 A = AE_0 = A.$$

Proof. We already know that $R(\mathfrak{M})$ is a subset of the indicated set; it remains only to show that every such A actually belongs to $R(\mathfrak{M})$.

Let $\varphi_1, \ldots, \varphi_k$ be arbitrary elements of $\mathfrak{H}$, which we keep fixed for the moment, $k = 1, 2, \ldots$. At the same time, take another Hilbert space $\overline{\mathfrak{H}}$, and choose in it k infinite-dimensional closed linear manifolds

$$\overline{\mathfrak{H}}_1, \ldots, \overline{\mathfrak{H}}_k$$

which are mutually orthogonal and together span $\overline{\mathfrak{H}}$.[48] The spaces $\overline{\mathfrak{H}}_1, \ldots, \overline{\mathfrak{H}}_k$ are themselves Hilbert spaces and hence are isomorphic to $\mathfrak{H}$.[49] Let

$$\Gamma_1, \ldots, \Gamma_k$$

[48]For the terminology, cf., for example, E., §I. The desired construction is as follows. Let $\overline{\varphi}_1, \overline{\varphi}_2, \ldots$ be a complete normalized orthogonal system, and introduce in it a double indexing

$$\overline{\varphi}_{ln}, \qquad l = 1, \ldots, k, \quad n = 1, 2, \ldots.$$

Let $\overline{\mathfrak{H}}_l$ be the closed linear manifold spanned by $\overline{\varphi}_{l1}, \overline{\varphi}_{l2}, \ldots$ $(l = 1, \ldots, k)$. These $\overline{\mathfrak{H}}_1, \ldots, \overline{\mathfrak{H}}_k$ have all the required properties.

[49]Cf. E., Introduction III.

be isomorphic mappings of $\mathfrak{H}$ onto, respectively, $\overline{\mathfrak{H}}_1, \ldots, \overline{\mathfrak{H}}_k$.

To each $A \in \mathfrak{B}$ we now assign the point[50]

$$\Gamma_1(A\varphi_1) + \cdots + \Gamma_k(A\varphi_k) \in \overline{\mathfrak{H}};$$

call it the *image point* of A. Let $\overline{\mathfrak{L}}$ be the set of image points of the operators $A \in R(\mathfrak{M})$, plainly a linear manifold, and let $\overline{\mathfrak{M}}$ be its closed hull (strongly in $\overline{\mathfrak{H}}$), hence a closed linear manifold. Let $\overline{E}$ denote the projection operator onto $\overline{\mathfrak{M}}$.

If A is a bounded operator in $\mathfrak{H}$, there plainly exists one and only one bounded operator $\overline{A}$ in $\overline{\mathfrak{H}}$ such that

$$\overline{A}\big(\Gamma_1 f_1 + \cdots + \Gamma_k f_k\big) = \Gamma_1(A f_1) + \cdots + \Gamma_k(A f_k)$$

(cf. note 50). Plainly

$$\overline{A}^* = \overline{A^*}.$$

Now let $B \in R(\mathfrak{M})$. If $\overline{f} \in \overline{\mathfrak{L}}$, then

$$\overline{f} = \Gamma_1(A\varphi_1) + \cdots + \Gamma_k(A\varphi_k)$$

for some $A \in R(\mathfrak{M})$, and

$$\overline{B}\,\overline{f} = \Gamma_1(BA\varphi_1) + \cdots + \Gamma_k(BA\varphi_k).$$

Since $BA \in R(\mathfrak{M})$, this last point also belongs to $\overline{\mathfrak{L}}$. Thus $\overline{B}$ maps $\overline{\mathfrak{L}}$ into itself, and therefore maps $\overline{\mathfrak{M}}$ into itself. Every $\overline{E f}$ lies in $\overline{\mathfrak{M}}$, hence so does every $\overline{BEf}$; identically,

$$\overline{E}\,\overline{B}\,\overline{E}\,\overline{f} = \overline{B}\,\overline{E}\,\overline{f},$$

and consequently

$$\overline{E}\,\overline{B}\,\overline{E} = \overline{B}\,\overline{E}.$$

Replace B by B^*, which likewise belongs to $R(\mathfrak{M})$, and take adjoints. Since $\overline{B}^* = \overline{B^*}$, this gives

$$\overline{E}\,\overline{B}\,\overline{E} = \overline{E}\,\overline{B}.$$

Therefore

$$\overline{E}\,\overline{B} = \overline{B}\,\overline{E}.$$

[50]The expression $\Gamma_1 f_1 + \cdots + \Gamma_k f_k$ plainly runs through all of $\overline{\mathfrak{H}}$, and does so exactly once, when $f_1, \ldots, f_k$ range independently over $\mathfrak{H}$.

Let us examine $\overline{E}$ more closely. Plainly,

$$\overline{E}\big(\Gamma_1(f_1) + \cdots + \Gamma_k(f_k)\big) = \Gamma_1(g_1) + \cdots + \Gamma_k(g_k),$$

where $g_1, \ldots, g_k$ depend linearly and continuously upon $f_1, \ldots, f_k$. Hence

$$g_l = E_{1l} f_1 + \cdots + E_{kl} f_k \qquad (l = 1, \ldots, k),$$

where all the E_{jl} $(j, l = 1, \ldots, k)$ are bounded operators in $\mathfrak{H}$.[51]

It follows that

$$\begin{aligned}
\overline{E}\,\overline{B}\,\overline{f} &= \overline{E}\,\overline{B}\big(\Gamma_1(f_1) + \cdots + \Gamma_k(f_k)\big) \\
&= \overline{E}\big(\Gamma_1(Bf_1) + \cdots + \Gamma_k(Bf_k)\big) \\
&= \Gamma_1(E_{11}Bf_1 + \cdots + E_{k1}Bf_k) + \cdots + \Gamma_k(E_{1k}Bf_1 + \cdots + E_{kk}Bf_k),
\end{aligned}$$

whereas

$$\begin{aligned}
\overline{B}\,\overline{E}\,\overline{f} &= \overline{B}\,\overline{E}\big(\Gamma_1(f_1) + \cdots + \Gamma_k(f_k)\big) \\
&= \overline{B}\Big(\Gamma_1(E_{11}f_1 + \cdots + E_{k1}f_k) + \cdots + \Gamma_k(E_{1k}f_1 + \cdots + E_{kk}f_k)\Big) \\
&= \Gamma_1(BE_{11}f_1 + \cdots + BE_{k1}f_k) + \cdots + \Gamma_k(BE_{1k}f_1 + \cdots + BE_{kk}f_k).
\end{aligned}$$

Thus we must have

$$E_{11}Bf_1 + \cdots + E_{k1}Bf_k = BE_{11}f_1 + \cdots + BE_{k1}f_k, \quad \ldots,$$

$$E_{1k}Bf_1 + \cdots + E_{kk}Bf_k = BE_{1k}f_1 + \cdots + BE_{kk}f_k.$$

That is, in general,

$$E_{jl}B = BE_{jl}.$$

Since B may be any element of $\mathfrak{M}$, while $B, B^* \in R(\mathfrak{M})$, all the E_{jl} commute with B; therefore every E_{jl} belongs to $\mathfrak{M}'$.

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[51] As is easily calculated[T16],

$$E_{jl} = \Gamma_l^{-1} P_{\overline{\mathfrak{H}}_l}\, \overline{E}\, \Gamma_j,$$

and conversely

$$\overline{E} = \sum_{j,l=1}^{k} \Gamma_l E_{jl} \Gamma_j^{-1} P_{\overline{\mathfrak{H}}_j}.$$

[T16] *Translator's note.* In footnote 51, the source interchanges the indices j and l in both displayed formulas. The indices have been corrected here in accordance with the preceding identity $g_l = E_{1l} f_1 + \cdots + E_{kl} f_k$.

Membership of

$$\overline{f} = \Gamma_1(f_1) + \cdots + \Gamma_k(f_k)$$

in $\overline{\mathfrak{M}}$ is characterized by $\overline{E}\overline{f} = \overline{f}$, that is,

$$\overline{E}\big(\Gamma_1(f_1) + \cdots + \Gamma_k(f_k)\big) = \Gamma_1(f_1) + \cdots + \Gamma_k(f_k),$$

$$\Gamma_1(E_{11}f_1 + \cdots + E_{k1}f_k) + \cdots + \Gamma_k(E_{1k}f_1 + \cdots + E_{kk}f_k) = \Gamma_1(f_1) + \cdots + \Gamma_k(f_k),$$

and hence by

$$(E_{11} - 1)f_1 + \cdots + E_{k1}f_k = 0, \quad \ldots, \quad E_{1k}f_1 + \cdots + (E_{kk} - 1)f_k = 0.$$

If $\overline{f}$ is the image point of A, so that

$$\overline{f} = \Gamma_1(A\varphi_1) + \cdots + \Gamma_k(A\varphi_k),$$

these conditions become

$$(E_{11} - 1)A\varphi_1 + \cdots + E_{k1}A\varphi_k = 0, \quad \ldots, \quad E_{1k}A\varphi_1 + \cdots + (E_{kk} - 1)A\varphi_k = 0.$$

In particular, if $A \in \mathfrak{M}''$, so that it commutes with all E_{jl} (since these belong to $\mathfrak{M}'$), these equations become

$$A\big[(E_{11} - 1)\varphi_1 + \cdots + E_{k1}\varphi_k\big] = 0, \quad \ldots, \quad A\big[E_{1k}\varphi_1 + \cdots + (E_{kk} - 1)\varphi_k\big] = 0.$$

Thus they are conditions of the form

$$A\omega_1 = 0, \quad \ldots, \quad A\omega_k = 0,$$

with fixed $\omega_1, \ldots, \omega_k$.

For every $A \in \mathfrak{M}$, both A and A^* belong to $R(\mathfrak{M})$, and hence their image points belong to $\overline{\mathfrak{M}}$. Moreover $A, A^* \in \mathfrak{M}''$. Consequently

$$A\omega_1 = A^*\omega_1 = \cdots = A\omega_k = A^*\omega_k = 0.$$

By Definition 4, it follows that

$$E_0\omega_1 = \cdots = E_0\omega_k = 0.$$

Therefore, if $A \in \mathfrak{M}''$ and at the same time $E_0A = AE_0 = A$, then

$$A\omega_1 = AE_0\omega_1 = 0, \quad \ldots, \quad A\omega_k = AE_0\omega_k = 0.$$

In other words, the image point of A belongs to $\overline{\mathfrak{M}}$. Hence, for every $\varepsilon > 0$, there is an $A' \in R(\mathfrak{M})$ whose image point (which is the general point of $\overline{\mathfrak{L}}$) has distance less than ε from that of A. But

$$\begin{aligned}
&\left|\Gamma_1(A'\varphi_1) + \cdots + \Gamma_k(A'\varphi_k) - \Gamma_1(A\varphi_1) - \cdots - \Gamma_k(A\varphi_k)\right|^2 \\
&\quad = \left|\Gamma_1((A' - A)\varphi_1) + \cdots + \Gamma_k((A' - A)\varphi_k)\right|^2 \\
&\quad = \left|\Gamma_1((A' - A)\varphi_1)\right|^2 + \cdots + \left|\Gamma_k((A' - A)\varphi_k)\right|^2 \\
&\quad = \left|(A' - A)\varphi_1\right|^2 + \cdots + \left|(A' - A)\varphi_k\right|^2,
\end{aligned}$$

and consequently

$$\left|(A' - A)\varphi_1\right| < \varepsilon, \quad \ldots, \quad \left|(A' - A)\varphi_k\right| < \varepsilon.$$

Thus A' belongs to the strong neighborhood $\mathcal{U}_3(A; \varphi_1, \ldots, \varphi_k; \varepsilon)$ of A.

Since $\varphi_1, \ldots, \varphi_k$ and $\varepsilon > 0$ were wholly arbitrary, A is a strong accumulation point of $R(\mathfrak{M})$. As this ring is even weakly closed, $A \in R(\mathfrak{M})$. This proves the theorem. □

The proof of Theorem 5 also yields the following:

Addendum. Suppose that a set $\mathfrak{M}$ has, of the ring properties α), β) in Definition 1, property α), but in place of β) only the following, less extensive property: if A and A^* are strong accumulation points of $\mathfrak{M}$, then $A \in \mathfrak{M}$. This is weaker than strong closure, and still more so than weak closure. Then $\mathfrak{M}$ is nevertheless a ring.

Proof. In the proof of Theorem 5, only the following properties of $R(\mathfrak{M})$ were used:

$$R(\mathfrak{M}) \supset \mathfrak{M}, \qquad R(\mathfrak{M}) \subset \mathfrak{M}'',$$

every $A \in R(\mathfrak{M})$ satisfies $E_0A = AE_0 = A$, and together with A, B, the operators

$$aA, \qquad A^*, \qquad A + B, \qquad AB$$

also belong to it. Under these assumptions it was shown that every $A \in \mathfrak{M}''$ with $E_0A = AE_0 = A$ is a strong accumulation point of $R(\mathfrak{M})$. The strong closure of $R(\mathfrak{M})$, which was needed only to infer from this that A belongs to it, is not to be invoked now.

For the present $\mathfrak{M}$, we may therefore replace $R(\mathfrak{M})$ by $\mathfrak{M}$ itself. We then obtain: every $A \in \mathfrak{M}''$ with $AE_0 = E_0A = A$ is a strong accumulation point of $\mathfrak{M}$. Since A^* is such a point whenever A is, A^* too is a strong accumulation point of $\mathfrak{M}$. By the second half of the hypothesis, not used until now, $A \in \mathfrak{M}$.

It follows from Theorem 5 that

$$R(\mathfrak{M}) \subset \mathfrak{M},$$

and hence $R(\mathfrak{M}) = \mathfrak{M}$. Thus $\mathfrak{M}$ is a ring, as asserted. □

For sets having ring property α) (Definition 1), strong and weak closure are therefore equivalent. E. Schmidt proved the same thing in $\mathfrak{H}$ for linear manifolds;[52] our present result is its analogue in $\mathfrak{B}$.

4.

We have just characterized $R(\mathfrak{M})$ with the aid of E_0 and $\mathfrak{M}''$, where $\mathfrak{M}$ is again an arbitrary subset of $\mathfrak{B}$. We now wish to reduce $E_0, \mathfrak{M}''$ to $R(\mathfrak{M})$.

Theorem 6. The principal unit E_0 belongs to $R(\mathfrak{M})$; indeed, it is the largest projection operator in $R(\mathfrak{M})$, in the sense that every projection operator in $R(\mathfrak{M})$ is a part of E_0.

Proof. Since $E_0 \in \mathfrak{M}''$ and $E_0 E_0 = E_0$, it belongs to $R(\mathfrak{M})$ by Theorem 5. If E is a projection operator in $R(\mathfrak{M})$, then $EE_0 = E$; that is, E is a part of E_0 (*E.*, Theorem 17). □

Theorem 7. $\mathfrak{M}''$ is the set of all operators

$$A + a \cdot 1,$$

or, equivalently, of all operators

$$A + a(1 - E_0),$$

where $A \in R(\mathfrak{M})$ and a is complex.

Proof. Since $E_0 \in R(\mathfrak{M})$, the two descriptions mean the same thing. Because

$$R(\mathfrak{M}) \subset \mathfrak{M}'', \qquad 1 \in \mathfrak{M}'',$$

the indicated set is plainly contained in $\mathfrak{M}''$. It remains to show that every $B \in \mathfrak{M}''$ has the form

$$A + a(1 - E_0), \qquad A \in R(\mathfrak{M}).$$

[52] Cf. *Rendiconti del Circolo Matematico di Palermo*, **25** (1908), pp. 57–73.

Along with E_0, B, the operator

$$E_0 B = B E_0 = A$$

belongs to $\mathfrak{M}''$, since $E_0 \in \mathfrak{M}'$ and $B \in \mathfrak{M}''$, so they commute. Moreover,

$$E_0 A = E_0^2 B = E_0 B = A, \qquad A E_0 = B E_0^2 = B E_0 = A.$$

By Theorem 5, $A \in R(\mathfrak{M})$. Therefore

$$C = B - A$$

also belongs to $\mathfrak{M}''$, and

$$E_0 C = E_0 B - E_0 A = A - A = 0, \qquad C E_0 = B E_0 - A E_0 = A - A = 0.$$

If we can infer from this that $C = a(1 - E_0)$, the proof is complete.

Let F be a projection operator which is a part of $1 - E_0$. Then F is orthogonal to E_0 (cf. E., Theorem 17), so

$$E_0 F = F E_0 = 0.$$

For every $A \in R(\mathfrak{M})$, therefore,

$$FA = F E_0 A = 0, \qquad AF = A E_0 F = 0.$$

Thus A and F commute. In particular, for every $A \in \mathfrak{M}$, both A and A^* commute with F, so $F \in \mathfrak{M}'$. Hence F and C commute. Together with $E_0 C = C E_0 = 0$, this implies the desired relation[53]

$$C = a(1 - E_0).$$

□

[53]Here is the proof. Let $E_0 f = 0$, $f \neq 0$. As is immediately seen, there corresponds to f, by

$$Fg = \frac{1}{|f|^2}(g, f)f,$$

a projection operator F. From $E_0 f = 0$ it follows that $E_0 F = 0$; hence E_0, F are orthogonal, and F, C commute. Thus

$$Cf = CFf = FCf = \frac{1}{|f|^2}(Cf, f)f = a_f f.$$

We now show that the number a_f is the same for every f. Clearly $a_f = a_{cf}$ for $c \neq 0$, so $a_f = a_g$

Finally we show:

Theorem 8. The rings containing 1, the sets $\mathfrak{M}$ satisfying $\mathfrak{M} = \mathfrak{M}''$, and the sets $\mathfrak{N}'$ are identical classes.

Proof. If a ring $\mathfrak{M}$ contains 1, then its principal unit satisfies

$$E_0 = E_0 1 = 1.$$

For every A we consequently have

$$E_0 A = 1A = A, \qquad AE_0 = A1 = A.$$

By Theorem 5,

$$R(\mathfrak{M}) = \mathfrak{M}'',$$

and, since $\mathfrak{M}$ is a ring, $R(\mathfrak{M}) = \mathfrak{M}$. Thus $\mathfrak{M} = \mathfrak{M}''$.

If a set $\mathfrak{M}$ satisfies $\mathfrak{M} = \mathfrak{M}''$, then $\mathfrak{M} = \mathfrak{N}'$ with $\mathfrak{N} = \mathfrak{M}'$. Every set $\mathfrak{N}'$ is a ring and contains 1 (cf. the beginning of paragraph 1). This proves the equivalence of the three criteria. □

III. The Abelian Rings

1.

We have already seen that, under ring property α) (Definition 1)[T17], weak and strong closure are the same. We now consider Abelian sets and shall show that for them the closure requirements may be weakened still further.

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when f, g are linearly dependent. If they are independent, then

$$C(f+g) = a_{f+g}(f+g), \qquad C(f+g) = Cf + Cg = a_f f + a_g g,$$

and therefore

$$(a_f - a_{f+g})f + (a_g - a_{f+g})g = 0.$$

It follows that $a_f = a_g = a_{f+g}$. Hence $a_f = a$ is indeed constant, and $Cf = af$ whenever $E_0 f = 0$ (the case $f = 0$ is plainly immaterial).

Since $E_0(1 - E_0)f = 0$ identically and $CE_0 = 0$, for every f

$$Cf = Cf - CE_0 f = C(1 - E_0)f = a(1 - E_0)f.$$

Thus $C = a(1 - E_0)$.

[T17] *Translator's note.* The source cites Definition 4 here; Definition 1 is the definition that introduces ring property α).

Let $\mathfrak{M}$ first be an arbitrary subset of $\mathfrak{B}$. Apply, arbitrarily but only finitely often, the operations

$$aA, \qquad A^*, \qquad A+B, \qquad AB$$

to its elements. The resulting set will be denoted by $r(\mathfrak{M})$; plainly

$$\mathfrak{M} \subset r(\mathfrak{M}) \subset R(\mathfrak{M}).$$

If to $r(\mathfrak{M})$ we adjoin all A for which both A and A^* are strong accumulation points of $r(\mathfrak{M})$, we obtain a set $r_1(\mathfrak{M})$. It is contained in $R(\mathfrak{M})$, contains $\mathfrak{M}$, and, by the symmetry of the definition, contains A^* whenever it contains A. Together with A, B, it also contains

$$aA, \qquad A+B, \qquad AB,$$

because $r(\mathfrak{M})$ does so and these operations are continuous in the strong topology. For AB, continuity holds only in each variable separately, but this suffices here, as is easily seen.

If B and B^* are strong accumulation points of $r_1(\mathfrak{M})$, then they are also strong accumulation points of $r(\mathfrak{M})$, since $r_1(\mathfrak{M})$ consists of accumulation points of $r(\mathfrak{M})$. Hence $B, B^* \in r_1(\mathfrak{M})$. The addendum to Theorem 5 is therefore applicable: $r_1(\mathfrak{M})$ is a ring, and consequently

$$r_1(\mathfrak{M}) = R(\mathfrak{M}).$$

Let $r_2(\mathfrak{M})$, respectively $r_3(\mathfrak{M})$, be obtained from $r(\mathfrak{M})$ by adjoining all its strong, respectively all its weak, accumulation points. Then plainly

$$r_1(\mathfrak{M}) \subset r_2(\mathfrak{M}) \subset r_3(\mathfrak{M}) \subset R(\mathfrak{M}),$$

and therefore

$$r_1(\mathfrak{M}) = r_2(\mathfrak{M}) = r_3(\mathfrak{M}) = R(\mathfrak{M}).$$

If now $\mathfrak{M}$ is Abelian, meaning that all elements of $\mathfrak{M}$, together with their adjoints, commute with one another, then a still smaller enlargement of $r(\mathfrak{M})$ suffices to obtain $R(\mathfrak{M})$, namely:

Theorem 9. If $\mathfrak{M}$ is Abelian, then $R(\mathfrak{M})$ is obtained from $r(\mathfrak{M})$ merely by adjoining all limits of strongly doubly convergent sequences of its elements.[54]

[54] This is already a substantial sharpening, since the first axiom of countability holds neither in the strong nor in the weak topology; that is, not every accumulation point is a limit[T18] (cf. § I, 2). It could not be decided whether this theorem also holds independently of the Abelian character of $\mathfrak{M}$.

Proof. Call the resulting set $\overline{r}(\mathfrak{M})$. Plainly

$$\mathfrak{M} \subset \overline{r}(\mathfrak{M}) \subset R(\mathfrak{M}).$$

It is also plain that, together with A, B, this set contains

$$aA, \qquad A^*, \qquad A + B, \qquad AB,$$

because $r(\mathfrak{M})$ has this property and these operations are continuous with respect to strong double convergence. Hence, if $\overline{r}(\mathfrak{M})$ is strongly closed, it is a ring by the addendum to Theorem 5, and therefore equals $R(\mathfrak{M})$. It remains only to prove strong closure.[55]

Let A be a strong accumulation point of $\overline{r}(\mathfrak{M})$. Then $A \in R(\mathfrak{M})$, and is therefore normal (§ I, 1). Given arbitrary $\varphi_1, \ldots, \varphi_k$ and $\varepsilon > 0$, there is an $A' \in \overline{r}(\mathfrak{M})$ in

$$\mathcal{U}_3(A; \varphi_1, \ldots, \varphi_k; \varepsilon);$$

that is,

$$|(A' - A)\varphi_1| < \varepsilon, \quad \ldots, \quad |(A' - A)\varphi_k| < \varepsilon.$$

Since $A' \in \overline{r}(\mathfrak{M})$, it too is normal. Moreover, A, A^*, A', A'^*, as elements of the Abelian ring $R(\mathfrak{M})$, all commute. Put

$$\frac{1}{2}(A + A^*) = B, \qquad \frac{1}{2i}(A - A^*) = C,$$

$$\frac{1}{2}(A' + A'^*) = B', \qquad \frac{1}{2i}(A' - A'^*) = C'.$$

These are all commuting bounded Hermitian operators, with $B', C' \in \overline{r}(\mathfrak{M})$; furthermore,

$$A' - A = (B' - B) + i(C' - C).$$

If B_1, C_1 are two commuting bounded Hermitian operators, then

$$\begin{aligned} |(B_1 + iC_1)f|^2 &= |B_1 f|^2 + (B_1 f, iC_1 f) + (iC_1 f, B_1 f) + |iC_1 f|^2 \\ &= |B_1 f|^2 + (-iC_1 B_1 f, f) + (iB_1 C_1 f, f) + |C_1 f|^2 \\ &= |B_1 f|^2 + |C_1 f|^2. \end{aligned}$$

[T18]*Translator's note.* In note 54 the source omits the word "not" before "every accumulation point is a limit." The first axiom of countability fails here, so the negation is required.

[55]Observe that even the closure of $\overline{r}(\mathfrak{M})$ with respect to strong double convergence is not self-evident.

The preceding inequalities therefore imply

$$|(B' - B)\varphi_1| < \varepsilon, \quad \ldots, \quad |(B' - B)\varphi_k| < \varepsilon,$$

and

$$|(C' - C)\varphi_1| < \varepsilon, \quad \ldots, \quad |(C' - C)\varphi_k| < \varepsilon.$$

Write

$$c = \max(|B|, |C|), \qquad c' = \max(|B'|, |C'|).$$

Thus, for every f,

$$|Bf| \leq c|f|, \quad |Cf| \leq c|f|, \quad |B'f| \leq c'|f|, \quad |C'f| \leq c'|f|.$$

While c is fixed, c' depends upon $\varphi_1, \ldots, \varphi_k, \varepsilon$. By increasing both if necessary, we assume

$$c' \geq c > 0.$$

We now examine the relation between B and B' more closely. Let $p(x)$ be a polynomial with the following properties:

$$\left.\begin{array}{rll} -c < p(x) < \min\{c, -c + \delta\} & \text{for } -c' \leq x \leq -c, \\ \max\{x - \delta, -c\} < p(x) < \min\{x + \delta, c\} & \text{for } -c \leq x \leq c, \\ \max\{-c, c - \delta\} < p(x) < c & \text{for } c \leq x \leq c', \end{array}\right\}$$

where $\delta > 0$ is still at our disposal. We see that, for $|x| \leq c$,

$$|p(x) - x| < \delta;$$

for $|x| \leq c'$,

$$|p(x)| < c;$$

and, for $|x|, |y| \leq c'$,

$$|p(x) - p(y)| < |x - y| + 2\delta,$$

hence

$$\big(p(x) - p(y)\big)^2 - (x - y)^2 < 8c'\delta + 4\delta^2.$$

We now choose δ so that in the first inequality $|p(x) - x| < \varepsilon$, and in the last one

$$\big(p(x) - p(y)\big)^2 - (x - y)^2 < \varepsilon^2$$

holds. (The polynomial $p(x)$ depends upon c', δ, and ε, and hence, directly and indirectly, upon $\varphi_1, \ldots, \varphi_k, \varepsilon$.)

A theorem of F. Riesz (cf. *E.*, Appendix II, Theorems 3^*, 4^*) gives, for every f,

$$|p(B)f - Bf| \leq \varepsilon|f|, \qquad |p(B')f| \leq c|f|.$$

Furthermore, the polynomial

$$\varepsilon^2 + (x-y)^2 - \big(p(x) - p(y)\big)^2 = q(x,y)$$

is everywhere nonnegative in the square $|x|, |y| \leq c'$. From this, from the commutativity of B, B', and from $|B|, |B'| \leq c'$, one may infer, by an analogue of the theorem just used—to which we shall return presently—that the Hermitian operator $q(B, B')$ is definite. That is,

$$\big(q(B,B')f, f\big) \geq 0,$$

$$\big((p(B') - p(B))^2 f, f\big) \leq \big((B' - B)^2 f, f\big) + \varepsilon^2(f, f),$$

and therefore

$$|(p(B') - p(B))f|^2 \leq |(B' - B)f|^2 + \varepsilon^2|f|^2.$$

For $f = \varphi_1, \ldots, \varphi_k$ we consequently have

$$|(p(B') - p(B))\varphi_i| \leq \varepsilon\sqrt{1 + |\varphi_i|^2}.$$

Together with the preceding inequality this gives

$$|(p(B') - B)\varphi_i| \leq \varepsilon\Big(|\varphi_i| + \sqrt{1 + |\varphi_i|^2}\Big).$$

Thus, if we put

$$\varepsilon = \frac{\eta}{\max_{i=1,\ldots,k}\big(|\varphi_i| + \sqrt{1 + |\varphi_i|^2}\big)}, \qquad \eta > 0 \text{ arbitrary},$$

then $p(B')$ lies in $\mathcal{U}_3(B; \varphi_1, \ldots, \varphi_k; \eta)$. At the same time, as we have seen, $|p(B')| \leq c$. Thus B is a strong accumulation point of that part of $r(\mathfrak{M})$ which lies in the ball $|B^*| \leq c$, and, more precisely, of the Hermitian operators in it. By §I, 3, it is therefore also the limit of a strongly convergent sequence in that set; and, since only Hermitian operators are involved, this sequence is even doubly convergent. In other words,

$$B \in \bar{r}(\mathfrak{M}).$$

In exactly the same way one shows that C belongs to it as well, and hence so does $A = B + iC$. This proves everything.

It remains to supply the proof, omitted above, of the generalization of F. Riesz's theorem. Let B, B' be two commuting Hermitian operators with $|B|, |B'| \leq c'$. By *E.*, Appendix II, Theorem 11* (put $A = B + iB'$), there are two resolutions of the identity $E(\lambda), F(\lambda)$ such that

$$(Bf, g) = \int_{-c'}^{c'} \lambda\, d(E(\lambda)f, g), \qquad (B'f, g) = \int_{-c'}^{c'} \lambda\, d(F(\lambda)f, g),$$

and all the $E(\lambda)$ and $F(\mu)$ commute. As there, we put

$$\Delta(\lambda, \mu) = E(\lambda)F(\mu) = F(\mu)E(\lambda)$$

and note that

$$\Delta(\lambda', \mu')\Delta(\lambda'', \mu'') = \Delta\big(\min(\lambda', \lambda''), \min(\mu', \mu'')\big).$$

It follows from the preceding formulas (all double integrals are to be extended over $\int_{-c'}^{c'}\int_{-c'}^{c'}$) that

$$(Bf, g) = \iint \lambda\, d(\Delta(\lambda, \mu)f, g), \qquad (B'f, g) = \iint \mu\, d(\Delta(\lambda, \mu)f, g).$$

Exactly as in the proof of Theorem 1 (cf. the reference in note 44), one then has

$$\big(q(B, B')f, g\big) = \iint q(\lambda, \mu)\, d(\Delta(\lambda, \mu)f, g),$$

$$\big(q(B, B')f, f\big) = \iint q(\lambda, \mu)\, d(\Delta(\lambda, \mu)f, f) = \iint q(\lambda, \mu)\, d|\Delta(\lambda, \mu)f|^2.$$

Now $q(\lambda, \mu) \geq 0$ throughout the region of integration, and

$$\iint_{\mathfrak{R}} d|\Delta(\lambda, \mu)f|^2 \geq 0$$

for every rectangle $\mathfrak{R}$ (cf. *E.*, Appendix II, §5); hence the last integral on the right is nonnegative. Since $\big(q(B, B')f, f\big) \geq 0$ for every f, the operator $q(B, B')$ is definite, as asserted.[56] □

[56] Observe that in this proof it was necessary to call upon the resolutions of the identity of given Hermitian operators, whereas, conversely, their existence can be proved from F. Riesz's theorem. For the Stieltjes–Radon double integrals used here in *E.*, Appendix II, see *Sitzungsberichte der Wiener Akademie*, **122**, IIa (1913), pp. 1295–1438, especially §§I–II.

2.

A ring $\mathfrak{M} = R(A)$ with one generator A is Abelian if and only if it is the set (A), that is, if and only if A, A^* commute: hence this is so for normal A. We shall now prove the converse announced in Introduction 3: every Abelian ring (that is, every ring having only normal elements) can be put in the form $R(A)$, where the normal A may even be chosen Hermitian.

Theorem 10. As A ranges over all normal operators, $R(A)$ ranges over all Abelian rings, and no others. Moreover, for a given $\mathfrak{M} = R(A)$, A can always be chosen to be a Hermitian operator.

Proof. In view of what was just said, it is enough to show that for every Abelian ring $\mathfrak{M}$ there exists a Hermitian operator A with $\mathfrak{M} = R(A)$.

First,

$$\mathfrak{M} = R(\mathfrak{M}^P) \qquad \text{(Theorem 2)}.$$

By § I, 4 there is in $\mathfrak{M}^P$ a countable set $\mathfrak{N} \subset \mathfrak{M}^P$ which is strongly dense everywhere. Consequently

$$\mathfrak{M}^P \subset R(\mathfrak{N}), \qquad \mathfrak{M} = R(\mathfrak{M}^P) \subset R(\mathfrak{N}),$$

while, on the other hand,

$$R(\mathfrak{N}) \subset R(\mathfrak{M}^P) = \mathfrak{M}.$$

Thus $\mathfrak{M} = R(\mathfrak{N})$. Here $\mathfrak{N}$ is a sequence of projection operators $E_1, E_2, \ldots$ belonging to $\mathfrak{M}$; these therefore commute.

In what follows we write $E \leq F$, more briefly, to say that E is a part of F. Two projection operators E, F are *comparable*[T19] if $E \leq F$ or $F \leq E$. We have $\mathfrak{M} = R(E_1, E_2, \ldots)$, where all the E_n commute. We shall now alter them, while preserving the preceding property, so that they are even comparable.

More precisely, we shall give a sequence $F_1, F_2, \ldots$ of projection operators with the following properties:

$$F_1 = E_1; \qquad F_2 \leq F_1 \leq F_3,$$

and F_1, F_2, F_3 are obtainable from E_1, E_2, and conversely, by the operations $+, -, \cdot$; further,

$$\cdots, \quad F_{2^{n-1}} \leq F'_1 \leq F_{2^{n-1}+1} \leq F'_2 \leq \cdots \leq F'_{2^{n-1}-2} \leq F_{2^n-2} \leq F'_{2^{n-1}-1} \leq F_{2^n-1}.$$

[T19]*Translator's note.* In the second alternative the source prints $F \geq E$, which merely repeats $E \leq F$. It has been corrected to $F \leq E$, the reverse ordering required by comparability.

Here $F'_1, \ldots, F'_{2^{n-1}-1}$ are $F_1, \ldots, F_{2^{n-1}-1}$, arranged in the order of magnitude determined by $\leq$; and $F_1, \ldots, F_{2^n-1}$ are obtainable from $E_1, \ldots, E_n$, and conversely, by the operations $+, -, \cdot$; and so on. If we have such F_p, it is clear that each F_p is formed by $+, -, \cdot$ from finitely many E_n, and conversely. Hence

$$R(F_1, F_2, \ldots) = R(E_1, E_2, \ldots) = \mathfrak{M},$$

and all the F_p are manifestly comparable. We shall now construct them.

The construction is inductive. For $n = 1$ it is trivial. Suppose it has already been carried out for $n = m - 1$; we carry it out for $n = m$. Let $F'_1, \ldots, F'_{2^{m-1}-1}$ again be $F_1, \ldots, F_{2^{m-1}-1}$ in the $\leq$-ordering, and put

$$\begin{aligned} F_{2^{m-1}} &= E_m F'_1, \\ F_{2^{m-1}+1} &= F'_1 + E_m(F'_2 - F'_1), \\ F_{2^{m-1}+2} &= F'_2 + E_m(F'_3 - F'_2), \quad \ldots, \\ F_{2^m-2} &= F'_{2^{m-1}-2} + E_m(F'_{2^{m-1}-1} - F'_{2^{m-1}-2}), \\ F_{2^m-1} &= F'_{2^{m-1}-1} + E_m(1 - F'_{2^{m-1}-1}). \end{aligned}$$

The preceding formulas express $F_{2^{m-1}}, \ldots, F_{2^m-1}$ in terms of $F'_1, \ldots, F'_{2^{m-1}-1}, E_m$, that is, in terms of $F_1, \ldots, F_{2^{m-1}-1}, E_m$, and hence, by the induction hypothesis, in terms of $E_1, \ldots, E_{m-1}, E_m$. Conversely, among the E_n only E_m remains to be considered, and for it we have

$$E_m = F_{2^{m-1}} + (F_{2^{m-1}+1} - F'_1) + \cdots + (F_{2^m-1} - F'_{2^{m-1}-1}).$$

Finally, it is also easy to verify that

$$F_{2^{m-1}} \leq F'_1 \leq F_{2^{m-1}+1} \leq F'_2 \leq \cdots \leq F'_{2^{m-1}-2} \leq F_{2^m-2} \leq F'_{2^{m-1}-1} \leq F_{2^m-1}.$$

Let us now consider $F_1, F_2, \ldots$. Their order is

$$F_2 \leq F_1 \leq F_3,$$

$$F_4 \leq F_2 \leq F_5 \leq F_1 \leq F_6 \leq F_3 \leq F_7,$$

$$\begin{aligned} F_8 \leq F_4 \leq F_9 \leq F_2 & \\ &\leq F_{10} \leq F_5 \leq F_{11} \leq F_1 \\ &\leq F_{12} \leq F_6 \leq F_{13} \leq F_3 \leq F_{14} \leq F_7 \leq F_{15}, \ldots. \end{aligned}$$

This order can be characterized as follows. We renumber the F_p by writing F_ρ for

$$F_p, \qquad p = 2^{l_1} + 2^{l_2} + \cdots + 2^{l_m}, \qquad l_1 > l_2 > \cdots > l_m \geq 0$$

(the dyadic expansion), where

$$\rho = \frac{1}{2 \cdot 3^{l_1}} + \frac{2}{3^{l_1 - l_2}} + \cdots + \frac{2}{3^{l_1 - l_m}}.$$

They then stand in their order of magnitude: $\rho < \sigma$ implies $F_\rho \leq F_\sigma$. As is readily seen, the index ρ ranges over the set of all midpoints of the intervals omitted from the familiar Cantor ternary set[57]

—that is, over a countable set lying in the interval $0 < x < 1$ and containing none of its own accumulation points.

We now construct a resolution of the identity $G(\lambda)$ having the following properties:

$$G(\lambda) = 0 \quad (\lambda < -1), \qquad G(\lambda) = 1 \quad (\lambda \geq 0);$$

and, for $-1 \leq \lambda < 0$, every $G(\lambda)$ is a strong accumulation point of the family F_ρ, with, in particular,

$$G(\rho - 1) = F_\rho.$$

Let $\mathfrak{N}$ be the set of all $G(\lambda)$ with $\lambda < 0$ and all $1 - G(\lambda)$ with $\lambda \geq 0$. Then $\mathfrak{N} \subset \mathfrak{M}^P$, and hence

$$R(\mathfrak{N}) \subset R(\mathfrak{M}) = \mathfrak{M}.$$

On the other hand, $\mathfrak{N}$ contains all the F_ρ, that is, $F_1, F_2, \ldots$, and therefore

$$R(\mathfrak{N}) \supset R(F_1, F_2, \ldots) = \mathfrak{M}.$$

Thus $R(\mathfrak{N}) = \mathfrak{M}$. If A is the bounded Hermitian operator having the resolution of the identity $G(\lambda)$ just described (cf. Appendix I), then Theorem 1 gives

$$R(A) = R(\mathfrak{N}) = \mathfrak{M},$$

and everything is proved. It remains, therefore, to construct $G(\lambda)$, and it is enough to define it for $-1 \leq \lambda < 0$.

In each open interval omitted from the Cantor set (cf. note 57), let

$$\overline{G}(\lambda - 1) = F_\rho$$

[57] The set in question comprises all numbers $\sum_{s=1}^{\infty} \alpha_s / 3^s$, where every α_s is 0 or 2. It is well known to be perfect and nowhere dense, and it omits the intervals

$$\sum_{s=1}^{k} \frac{\alpha_s}{3^s} + \frac{1}{3^{k+1}} < x < \sum_{s=1}^{k} \frac{\alpha_s}{3^s} + \frac{2}{3^{k+1}}.$$

Their midpoints are $\sum_{s=1}^{k} \alpha_s / 3^s + 1/(2 \cdot 3^k)$, another way of writing our ρ.

when ρ is the midpoint of that interval. Thus $\overline{G}(\lambda)$ is defined on an open set dense everywhere in $-1 \le \lambda < 0$; moreover,

$$\lambda < \mu \quad \Longrightarrow \quad \overline{G}(\lambda) \le \overline{G}(\mu),$$

and

$$\lambda \longrightarrow \lambda_0 \quad \Longrightarrow \quad \overline{G}(\lambda) f \longrightarrow \overline{G}(\lambda_0) f$$

whenever both sides are defined ($\overline{G}(\lambda)$ is strongly semicontinuous in λ). We may therefore extend its definition to the whole of $-1 \le \lambda < 0$ as follows. Let $\lambda_1 > \lambda_2 > \cdots$, $\lambda_n \to \lambda$, be a sequence for which $\overline{G}(\lambda_1), \overline{G}(\lambda_2), \ldots$ are already defined. Since

$$\overline{G}(\lambda_1) \ge \overline{G}(\lambda_2) \ge \cdots,$$

the $\overline{G}(\lambda_n)$ have a projection operator G as their strong limit (cf. E., Theorem 19). The operator G depends only upon λ. Indeed, from two such sequences

$$\lambda_1' > \lambda_2' > \cdots, \qquad \lambda_1'' > \lambda_2'' > \cdots$$

we can combine subsequences into a new sequence

$$\lambda_{m_1}' > \lambda_{n_1}'' > \lambda_{m_2}' > \lambda_{n_2}'' > \cdots,$$

which must also converge; hence the two limits are the same. If $\overline{G}(\lambda)$ was already defined, then the preceding remarks show that $G = \overline{G}(\lambda)$. In general we denote G by $G(\lambda)$.

For $\mu < \lambda$, choose

$$\lambda_1 > \lambda_2 > \cdots, \quad \lambda_n \to \lambda, \qquad \mu_1 > \mu_2 > \cdots, \quad \mu_n \to \mu,$$

with $\mu_n < \lambda_n$. Then $\overline{G}(\mu_n) \le \overline{G}(\lambda_n)$, and consequently, by continuity, $G(\mu) \le G(\lambda)$. Furthermore, for a given f and a suitable λ_n,

$$|\overline{G}(\lambda_n) f|^2 \le |G(\lambda) f|^2 + \varepsilon.$$

Hence, for $\lambda \le \lambda' \le \lambda_n$ ($\lambda_n > \lambda$), all the more

$$|G(\lambda') f|^2 \le |\overline{G}(\lambda_n) f|^2 \le |G(\lambda) f|^2 + \varepsilon,$$

$$|G(\lambda') f - G(\lambda) f|^2 = |G(\lambda') f|^2 - |G(\lambda) f|^2 \le \varepsilon$$

(cf. E., Theorem 16). Thus, for $\lambda' \ge \lambda$, $\lambda' \to \lambda$, $G(\lambda') f \to G(\lambda) f$. Consequently $G(\lambda)$ is the desired resolution of the identity, and we have reached our goal. $\square$

If $\mathfrak{M}$ is an Abelian ring, then by Theorem 10 just proved it equals $R(A)$, where A is a Hermitian operator; and by Theorem 9, $R(A)$ is the set of strong double limits of $r(A)$. In the present case, however, $r(A)$, because $A = A^*$, is the set of all $p(A)$, where $p(x)$ ranges over all polynomials satisfying $p(0) = 0$. Thus $\mathfrak{M}$ is the set of the limits of all strongly doubly convergent sequences

$$p_1(A), p_2(A), \ldots$$

($p_1(x), p_2(x), \ldots$ being polynomials vanishing at $x = 0$). With a more general interpretation of the notion $f(A)$ than the one we have used hitherto (namely, by extending it to discontinuous $f(x)$), one could show that $\mathfrak{M}$ is the set of all $f(A)$, where $f(x)$ is any function vanishing at $x = 0$ for which the extended definition applies. We shall not enter into these matters here.

IV. General Properties of (Unbounded) Normal Operators

1.

We pass to our second subject, the study of unbounded operators and the definition of normality for them. We must therefore begin by considering arbitrary operators (not necessarily everywhere defined or continuous; for the present we do not even require them to be linear and closed), which we shall denote by $R, S, \ldots$, in contrast with the $A, B, \ldots$ of $\mathfrak{B}$.

Definition 5. The operator aR is defined for those f for which Rf is defined, and its value is then $a \cdot Rf$. The operator $R \pm S$ is defined for those f for which Rf and Sf are defined, and its value is then $Rf \pm Sf$. The operator RS is defined for those f for which Sf and $R(Sf)$ are defined, and its value is then $R(Sf)$.[58]

The operators R, A *commute* (it is essential that $A \in \mathfrak{B}$) if RA is an extension of AR; that is, if, whenever Rf is defined, $R(Af)$ is also defined—Af and $A(Rf)$ are defined in any case—and is equal to $A(Rf)$.[59]

[58] It is clear that, with this definition, the commutative and associative laws for addition and the associative law for multiplication hold. Of the distributive laws, one always has $(R \pm S)T = RT \pm ST$, whereas $R(S \pm T) = RS \pm RT$ only when R is linear and everywhere defined (if R is merely linear, the left-hand side is an extension of the right-hand side). We have $R + 0 = R$, $1R = R1 = R$, and $R0 = 0$ (provided Rf vanishes for $f = 0$); on the other hand, $0R$ is defined only where R is. The relation between $+$ and $-$ is analogous. Thus, in the unbounded case, the operations $+, -, \cdot$ are no longer as transparent as in the bounded case.

[59] According to *E.*, Introduction V, R is an extension of S when Rf is defined wherever Sf is, and

We can now extend Definition 3 to sets $\mathfrak{M}$ of arbitrary operators.

Definition $3'$**.** If $\mathfrak{M}$ is a set of arbitrary operators (no longer necessarily a subset of $\mathfrak{B}$), let $\mathfrak{M}'$ be the set of all $A \in \mathfrak{B}$ for which A and A^* commute with every $R \in \mathfrak{M}$.

Thus in every case $\mathfrak{M}' \subset \mathfrak{B}$, and $\mathfrak{M}'', \mathfrak{M}''', \ldots$ consequently always fall under the old definition. Which of the properties of $\mathfrak{M}'$ are still retained?

It is immediately seen that 1 always belongs to $\mathfrak{M}'$, and, together with A, B, so do A^* and AB. If, in addition, all the elements of $\mathfrak{M}$ are linear, then aA and $A + B$ also belong to it whenever A, B do. Now suppose that all the elements of $\mathfrak{M}$ are closed, that A is a strong accumulation point of $\mathfrak{M}'$, and that $R \in \mathfrak{M}$. If Rf is defined, there is an $A'_n \in \mathfrak{M}'$ in

$$\mathcal{U}_3\left(A; f, Rf; \frac{1}{n}\right);$$

that is,

$$A'_n f \longrightarrow Af, \qquad RA'_n f = A'_n Rf \longrightarrow ARf.$$

Since R is closed, $R(Af)$ is therefore defined and equals ARf. Thus A and R commute.

Consequently, if A and A^* are strong accumulation points of $\mathfrak{M}'$, then $A \in \mathfrak{M}'$. Let us now assume, once and for all, that every element of $\mathfrak{M}$ is closed and linear. Then, by the foregoing, all the hypotheses of the supplement to Theorem 5 are satisfied, and $\mathfrak{M}'$ is a ring. Since 1 plainly belongs to it,

$$\mathfrak{M}' = \mathfrak{M}''' \qquad \text{(Theorem 8)}.$$

For arbitrary $\mathfrak{M}$, we may apply the results from the beginning of § II, 1 to $\mathfrak{M}' \subset \mathfrak{B}$:

$$\mathfrak{M}' \subset \mathfrak{M}''' = \mathfrak{M}^{\mathrm{V}} = \mathfrak{M}^{\mathrm{VII}} = \cdots, \qquad \mathfrak{M}'' = \mathfrak{M}^{\mathrm{IV}} = \mathfrak{M}^{\mathrm{VI}} = \cdots.$$

We cannot, however, conclude that $\mathfrak{M}' = \mathfrak{M}'''$, since the relation $\mathfrak{M} \subset \mathfrak{M}''$ is missing ($\mathfrak{M}' \subset \mathfrak{B}$, but not necessarily $\mathfrak{M} \subset \mathfrak{B}$). As we saw, the equality does hold when $\mathfrak{M}$ consists entirely of closed linear operators.

Finally, let us record which elements occur in $\mathfrak{M}'^P$ and $\mathfrak{M}'^U$. Every projection operator E commuting with all $R \in \mathfrak{M}$ belongs to $\mathfrak{M}'^P$ (since $E = E^*$); that is, every

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the two are equal there. If R too is everywhere defined (for example, if $R \in \mathfrak{B}$), the present definition of commutativity says $RA = AR$, that is, it is ordinary commutativity. For arbitrary R, it would scarcely be possible to require $RA = AR$, since then R would not commute with 0 (cf. note 21 and note 58). The asymmetry of our definition (between R and A) shows that a direct definition of commutativity for arbitrary R, S must encounter difficulties.

E which reduces all the $R \in \mathfrak{M}$ (cf. E., Definition 13). Every unitary U for which both U and $U^* = U^{-1}$ commute with every $R \in \mathfrak{M}$ belongs to $\mathfrak{M}'^U$. In other words, RU is an extension of UR, and RU^{-1} is an extension of $U^{-1}R$. It is immediately seen that this means

$$RU = UR$$

(with identical domains as well), or, equivalently,

$$U^{-1}RU = R.$$

2.

We now introduce the notion of a conjugate pair (briefly, a *conjugate pair*) of operators.

Definition 5. Two operators R, R^* form a conjugate pair if they have the same domain, if that domain spans the closed linear manifold $\mathfrak{H}$, and if throughout it

$$(Rf, g) = (f, R^*g), \qquad (f, Rg) = (R^*f, g)$$

holds (cf. the beginning of note 12).

It is immediately seen that, together with R, R^*, the operators R^*, R also form a conjugate pair. One R cannot form conjugate pairs with two different operators R_1^*, R_2^*: the domains of R_1^*, R_2^* are prescribed to be the same, as are $(R_1^* f, g)$ and $(R_2^* f, g)$ for a set of g's spanning the closed linear manifold $\mathfrak{H}$, and hence $R_1^* f$ and $R_2^* f$ themselves. Nor can two different R_1, R_2 have the same R^*, by the first observation. The symbol * is justified by noting that, for $A \in \mathfrak{B}$, precisely A, A^* form a conjugate pair. The operator R is Hermitian if and only if R, R form a conjugate pair.

A conjugate pair S, S^* is an extension of another pair R, R^* if the common domain of S, S^* contains that of R, R^*, and if on the latter[60]

$$Rf = Sf, \qquad R^*f = S^*f.$$

If the two pairs are not identical, we speak, exactly as for Hermitian operators (cf. E., §II), of a *proper extension*. A conjugate pair without a proper extension is *maximal*.

[60]Each of the two following relations follows from the other: the restrictions of S, S^* to the domain of R, R^* still form a conjugate pair.

We also note that every extension of a Hermitian operator is Hermitian: if R, R^* has the extension S, S^*, then

$$R = R^* \quad \Longrightarrow \quad S = S^*.$$

Indeed, if Sf and Rg are defined, then

$$(Sf, g) = (f, S^*g) = (f, R^*g) = (f, Rg) = (f, Sg) = (S^*f, g).$$

Since these g's span the closed linear manifold $\mathfrak{H}$, it follows that $Sf = S^*f$, and hence $S = S^*$.

It is quite easy to see (exactly as for Hermitian operators; cf. *E.*, §II, Theorem 9) that every conjugate pair can be extended to one consisting of two linear operators. It is not, however, immediately clear whether they can also be made closed in this way.[61] We shall not pursue this question here.

3.

An $A \in \mathfrak{B}$ is normal if A, A^* commute, that is, if (A) is Abelian, or, equivalently (cf. § II, 1), if $(A)''$ is Abelian. We transfer this definition to arbitrary R (since always $(R)'' \subset \mathfrak{B}$).

Definition 6. A conjugate pair R, R^* is *normal* if $(R)''$ (a subset of $\mathfrak{B}$) is Abelian.

By what was said above, it is clear that for operators in $\mathfrak{B}$ this is the old definition.

Theorem 11. The pair R, R^* is normal if and only if R^*, R is normal.

Proof. We prove

$$(R)' = (R^*)'.$$

It then follows that $(R)'' = (R^*)''$, and hence the assertion. By symmetry, it is enough

to prove $(R)' \subset (R^*)'$; that is, if A, A^* commute with R, they must also commute with R^*.

[61] The methods in question yield only a kind of common closure of the two operators R, R^* of the pair: from $f_n \to f$, $Rf_n \to f^*$, and $R^* f_n \to f^{**}$, one infers that Rf, R^*f are defined and equal to f^*, f^{**}, respectively[T20].

[T20] *Translator's note.* At the end of note 61 the source identifies Rf, R^*f with f^{**}, f^{***}. The convergence assumptions immediately preceding instead give f^*, f^{**}, as printed here.

Suppose, then, that $R^* f$ is defined. Then Rf, RAf, $RA^* f$ are defined as well, and hence so are R^*Af, R^*A^*f. If R^*g is also defined, we have

$$\begin{aligned}(R^*Af, g) &= (Af, Rg) = (f, A^*Rg) = (f, RA^*g) \\ &= (R^*f, A^*g) = (AR^*f, g),\end{aligned}$$

$$\begin{aligned}(R^*A^*f, g) &= (A^*f, Rg) = (f, ARg) = (f, RAg) \\ &= (R^*f, Ag) = (A^*R^*f, g).\end{aligned}$$

Since these g's span the closed linear manifold $\mathfrak{H}$,

$$R^*Af = AR^*f, \qquad R^*A^*f = A^*R^*f.$$

Thus A, A^* commute with R^*, as asserted. □

Theorem 12. If R, R (R a Hermitian operator) is assumed linear and closed, then it is normal if and only if R is hypermaximal.[62]

Proof. We begin with some general observations about closed linear Hermitian operators. Form the Cayley transform U of R. Let $\mathfrak{E}$, $\mathfrak{F}$ be its domain and range, respectively (cf. *E.*, §V, Theorem 2); both are closed linear manifolds. Let E be the projection operator onto $\mathfrak{E}$ (cf. *E.*, §III, Theorem 13).

Suppose A commutes with R. Every φ for which $U\varphi$ is defined has the form $Rf + if$; therefore RAf is defined, and

$$A\varphi = ARf + iAf = RAf + iAf.$$

Thus $U(A\varphi)$ is also defined, and

$$U(A\varphi) = RAf - iAf = ARf - iAf = A(U\varphi);$$

that is, U, A commute. Conversely, suppose A commutes with U. Every f for which Rf is defined has the form $\varphi - U\varphi$; hence $UA\varphi$ is defined and

$$Af = A\varphi - AU\varphi = A\varphi - UA\varphi.$$

[62] "Hypermaximal" is abbreviated *hypermax.*; cf. *E.*, Definition 9. If R is not linear and closed, consider its closure $\widetilde{R}$ (*E.*, §II, Theorem 9). Since it is Hermitian, $\widetilde{R}, \widetilde{R}$ is a conjugate pair; and since everything commuting with R also commutes with $\widetilde{R}$, one has $(R)' \subset (\widetilde{R})'$ and $(R)'' \supset (\widetilde{R})''$. Thus $(\widetilde{R})''$ is Abelian as well. Hence $\widetilde{R}, \widetilde{R}$ is normal, and the present theorem applies to the closed linear operator $\widetilde{R}$.

Thus $R(Af)$ is defined as well, and

$$R(Af) = i(A\varphi + UA\varphi) = i(A\varphi + AU\varphi) = A(Rf);$$

that is, R, A commute. It follows that

$$(R)' = (U)', \qquad (R)'' = (U)''.$$

The sufficiency of the condition is now immediate. If R is hypermaximal, then U is unitary (cf. E., Theorem 35), and therefore belongs to $\mathfrak{B}$. Moreover, it commutes with $U^* = U^{-1}$; hence U, U^* is normal, and, because $(R)'' = (U)''$, so is R, R.

It remains to prove necessity. Let R, R be normal, and let $A \in (R)'$. Then $A \in (U)'$. Along with Uf, the vector UAf is defined; that is, together with f, Af belongs to $\mathfrak{E}$. The vector Ef always belongs to $\mathfrak{E}$, and hence so does AEf. Therefore

$$EAEf = AEf, \qquad \text{or} \qquad EAE = AE.$$

Taking adjoints and replacing A by A^* (which also belongs to $(R)'$) gives $EAE = EA$. Hence $AE = EA$: A, E commute. Thus A commutes with both U and E, and therefore with

$$UE = V.$$

This is important because V is everywhere defined and hence belongs to $\mathfrak{B}$. Since this holds for every $A \in (R)'$ (which contains A^* together with A), it follows that $V \in (R)''$.

Thus $V^* \in (R)''$ as well; and, since $(R)''$ is Abelian, V, V^* commute. For all f, g,

$$(V^*Vf, g) = (Vf, Vg) = (U(Ef), U(Eg)) = (Ef, Eg) = (Ef, g).$$

Consequently

$$V^*V = VV^* = E,$$

and this operator too must belong to $(R)''$. Hence V, E commute, with the consequence

$$EV = VE = UE \cdot E = UE = V.$$

If $\varphi \in \mathfrak{E}$, then

$$\varphi = E\varphi, \qquad U\varphi = UE\varphi = V\varphi = EV\varphi.$$

Thus $U\varphi \in \mathfrak{E}$, and therefore $\varphi - U\varphi \in \mathfrak{E}$. But the vectors $\varphi - U\varphi$ must be dense everywhere (E., Theorem 24); hence $\mathfrak{E}$ is an everywhere dense closed linear manifold. Thus

$$\mathfrak{E} = \mathfrak{H}, \qquad E = 1.$$

It follows that

$$V = U \cdot 1 = U, \qquad V^*V = VV^* = 1,$$

and hence

$$U^*U = UU^* = 1.$$

Thus U is unitary, and R is hypermaximal. □

We emphasize once more the fact found incidentally in the proof of this theorem: for a hypermaximal Hermitian operator R and its unitary Cayley transform U,

$$(R)' = (U)',$$

and consequently

$$(R)'' = (U)'', \quad (R)''' = (U)''', \ldots.$$

Now $(U)''$ is Abelian, so

$$(U)'' \subset (U)''' = (U)', \qquad (R)'' \subset (R)'.$$

Moreover,

$$(R)' = (R)''' = \cdots, \qquad (R)'' = (R)^{\mathrm{IV}} = \cdots,$$

because this is so for U, or by § IV, 1.

Theorem 13. Let R, as above, be a hypermaximal Hermitian operator, U its Cayley transform, $E(\lambda)$ its resolution of the identity, and $\mathcal{E}$ the set of all $E(\lambda)$. Then

$$(R)' = (U)' = \mathcal{E}',$$

and hence

$$(R)'' = (U)'' = \mathcal{E}'', \qquad (R)''' = (U)''' = \mathcal{E}''', \ldots.$$

Proof. Everything follows from the first equality; and since $(R)' = (U)'$ has already been established, only $(U)' = \mathcal{E}'$ remains to be proved. This follows, in any case, from[63]

$$R(U) = R(\mathcal{E}).$$

[63]In general,

$$\mathfrak{M} \subset R(\mathfrak{M}) \subset \mathfrak{M}'', \qquad \mathfrak{M}' \supset (R(\mathfrak{M}))' \supset \mathfrak{M}''' = \mathfrak{M}';$$

hence $(R(\mathfrak{M}))' = \mathfrak{M}'$: the ring $R(\mathfrak{M})$ determines $\mathfrak{M}'$.

It is therefore enough to show that $U \in R(\mathcal{E})$, and that every $E(x)$[64] belongs to $R(U)$.

From the relation

$$(Uf,g) = \int_0^1 e^{2\pi i x}\, d(E(x)f,g),$$

one proves literally as in the corresponding part of the first half of the proof of Theorem 1, for the bounded Hermitian operator A used there, that $U \in R(\mathcal{E})$. To see, conversely, that $E(x) \in R(U)$, recall how these $E(x)$ were constructed in E., Appendix II. The $E(x)$ were differences

of strong limits of certain $F(\rho)$, $0 < \rho < 1$ (cf. §6 there); it is therefore enough that these belong to $R(U)$. The $F(\rho)$, however, were formed by addition, subtraction, and multiplication from the $E(U,a,b)$, $-\infty < a,b < \infty$ (cf. §§4 and 5 there), so only the latter matter. But $E(U,a,b)$ was a product of projection operators from the resolutions of the identity[65] of

$$\frac{U+U^*}{2} \qquad \text{and} \qquad \frac{U-U^*}{2i};$$

and, by our Theorem 2, these belong to $R(U)$ provided the last two operators do. This is clear: $R(U)$ contains U, U^*, and hence also the two operators just displayed. □

4.

After these preparations we return to our proper subject. Let R, R^* again be a conjugate pair.

Theorem 14. Let R, R^* be normal. Form the two Hermitian operators

$$S_1 = \frac{R+R^*}{2}, \qquad S_2 = \frac{R-R^*}{2i},$$

[64] We write $E(x)$, $0 < x < 1$, in place of $E(\lambda)$, $-\infty < \lambda < \infty$, with the correspondence $\lambda = -\cot \pi x$, $x = -\frac{1}{\pi}\operatorname{arccot}\lambda$ (cf. E., proof of Theorem 36).

[65] $E(U,a,b)$ (called U_A in the passage cited) is defined as

$$P_{\mathfrak{M}}(R,a)\,P_{\mathfrak{N}}(S,b), \qquad R = \frac{U+U^*}{2}, \quad S = \frac{U-U^*}{2i},$$

and only the properties of these projection operators stated in Theorems $8^*, 9^*$ there are used; these properties plainly belong to the corresponding resolutions of the identity.

and then $\widetilde{S}_1, \widetilde{S}_2$ (cf. the beginning of note 62); the latter are hypermaximal. The operators

$$\overline{R} = \widetilde{S}_1 + i\widetilde{S}_2, \qquad \overline{R}^* = \widetilde{S}_1 - i\widetilde{S}_2$$

(both defined on the intersection of the domains of $\widetilde{S}_1, \widetilde{S}_2$) form a conjugate pair, and in fact a normal one. The pair $\overline{R}, \overline{R}^*$ is an extension of R, R^*.

Proof. Form S_1, S_2 as prescribed; their Hermitian character is evident; then form $\widetilde{S}_1, \widetilde{S}_2$. Let $A \in \mathfrak{B}$. If A commutes with R, R^*, it also commutes with S_1, S_2, and hence with $\widetilde{S}_1, \widetilde{S}_2$. It follows that

$$(R, R^*)' \subset (\widetilde{S}_1)', \qquad (R, R^*)' \subset (\widetilde{S}_2)'.$$

By $(R)' = (R^*)'$ (cf. the proof of Theorem 11), $(R)' = (R, R^*)'$; hence

$$(R)' \subset (\widetilde{S}_1)', \qquad (R)' \subset (\widetilde{S}_2)', \qquad (R)'' \supset (\widetilde{S}_1)'', \qquad (R)'' \supset (\widetilde{S}_2)''.$$

Along with $(R)''$, therefore, $(\widetilde{S}_1)''$ and $(\widetilde{S}_2)''$ are Abelian. Thus the conjugate pairs $\widetilde{S}_1, \widetilde{S}_1$ and $\widetilde{S}_2, \widetilde{S}_2$ are normal, and by Theorem 12 the operators $\widetilde{S}_1, \widetilde{S}_2$ are hypermaximal.

Since $\widetilde{S}_1, \widetilde{S}_2$ are extensions of S_1, S_2, respectively, the pair $\overline{R}, \overline{R}^*$ is an extension of R, R^*. It is clear, since $\widetilde{S}_1, \widetilde{S}_2$ are Hermitian, that $\overline{R}, \overline{R}^*$ is a conjugate pair. Its normality is seen as follows. We have

$$(\overline{R})' = (\overline{R}^*)',$$

both being equal to $(\overline{R}, \overline{R}^*)'$; and this set plainly contains $(\widetilde{S}_1, \widetilde{S}_2)'$, which in turn, as we already know, contains $(R)'$. Thus

$$(\overline{R})' \supset (R)', \qquad (\overline{R})'' \subset (R)''.$$

Since $(R)''$ is Abelian, so is $(\overline{R})''$; hence $\overline{R}, \overline{R}^*$ is normal. □

Theorem 15. The vectors $\overline{R}f$ and $\overline{R}^*f$ are defined if and only if there exist two vectors f^*, f^{**} such that, for every g in the common domain of R and R^*,

$$(f, Rg) = (f^{**}, g), \qquad (f, R^*g) = (f^*, g).$$

In that case[T21]

$$\overline{R}f = f^*, \qquad \overline{R}^*f = f^{**}.$$

[T21] *Translator's note.* In the second formula of the conclusion the source prints $\overline{R}^* f = f^{***}$. The defining identity immediately above requires f^{**}, as printed here.

Proof. Necessity is clear:

$$(f, Rg) = (f, \overline{R}g) = (\overline{R}^* f, g), \qquad (f, R^*g) = (f, \overline{R}^* g) = (\overline{R} f, g).$$

For sufficiency, observe that

$$\frac{f^* + f^{**}}{2}, \qquad \frac{f^* - f^{**}}{2i}$$

are, by definition, the vectors assigned to f by S_1, S_2, respectively (cf. E., Definitions 8 and 9), and hence also by $\widetilde{S}_1, \widetilde{S}_2$. Since the latter are hypermaximal, $\widetilde{S}_1 f, \widetilde{S}_2 f$ are defined and equal, respectively, to

$$\frac{f^* + f^{**}}{2}, \qquad \frac{f^* - f^{**}}{2i}.$$

Consequently $\overline{R} f, \overline{R}^* f$ are defined as well, and are equal to f^*, f^{**}, respectively. □

Theorem 16. Every extension T, T^* of R, R^* (a conjugate pair, not necessarily a normal one) has $\overline{R}, \overline{R}^*$ as an extension. The pair $\overline{R}, \overline{R}^*$, even regarded merely as a conjugate pair, is maximal; indeed, it is the unique maximal extension of R, R^*.

Proof. The last two assertions follow from the first, so we need consider only that one. Suppose Tf, T^*f are defined, and put

$$Tf = f^*, \qquad T^*f = f^{**}.$$

Then the hypotheses of Theorem 15 are satisfied. Indeed, when Rg, R^*g are defined,

$$(f, Rg) = (f, Tg) = (T^* f, g), \qquad (f, R^*g) = (f, T^*g) = (Tf, g).$$

Thus $\overline{R} f, \overline{R}^* f$ are also defined, and equal to Tf, T^*f, respectively. Hence $\overline{R}, \overline{R}^*$ is an extension of T, T^*. □

For normal conjugate pairs of operators, therefore, the situation is much simpler than for Hermitian operators (cf. E.): a maximal extension is always possible here in one and only one way. In what follows we shall consequently restrict ourselves to maximal pairs R, R^*; by Theorem 16 these are characterized by

$$\overline{R} = R, \qquad \overline{R}^* = R^*.$$

For such pairs we shall establish the analogue of Hilbert's spectral representation.

V. Spectral Form of Normal Operators

1.

As announced, we investigate normal maximal conjugate pairs R, R^*.

Theorem 17. Let R, R^* be a normal maximal conjugate pair. There then exists a family of projection operators $\Delta(z)$, indexed by all complex numbers z, with the following properties.

α)

$$\Delta(z')\Delta(z'') = \Delta(z'')\Delta(z') = \Delta(z),$$

where

$$\operatorname{Re} z = \min(\operatorname{Re} z', \operatorname{Re} z''), \qquad \operatorname{Im} z = \min(\operatorname{Im} z', \operatorname{Im} z'').$$

β) For every f,

$$\Delta(z) f \longrightarrow \Delta(z_0) f$$

as $z \to z_0$, while $\operatorname{Re} z \geq \operatorname{Re} z_0$ and $\operatorname{Im} z \geq \operatorname{Im} z_0$. The convergence is uniform both when $f, \operatorname{Re} z$ are fixed and $\operatorname{Im} z$ is arbitrary, and when $f, \operatorname{Im} z$ are fixed and $\operatorname{Re} z$ is arbitrary.

γ) For every f,

$$\Delta(z) f \longrightarrow 0$$

when $\operatorname{Re} z \to -\infty$ or $\operatorname{Im} z \to -\infty$, whereas

$$\Delta(z) f \longrightarrow f$$

when $\operatorname{Re} z \to +\infty$ and $\operatorname{Im} z \to +\infty$.

(A family of projection operators satisfying α)–γ) is called a *complex resolution of the identity*.)

δ) The vector Rf (and $R^* f$) is defined if and only if

$$\iint_{\mathbb{C}} |z|^2 \, d|\Delta(z) f|^2$$

is finite. (The double integral extends over the whole complex plane. For every rectangle $\mathfrak{R}$[66],

$$\iint_{\mathfrak{R}} d|\Delta(z) f|^2 \geq 0,$$

and the integrand $|z|^2$ is always nonnegative. Thus the integral displayed in δ) is, by its nature, either convergent and finite or properly divergent to $+\infty$; the latter possibility is to be excluded.)

ε) In the case just described, for every g,

$$(Rf, g) = \iint_{\mathbb{C}} z\, d(\Delta(z)f, g), \qquad (R^* f, g) = \iint_{\mathbb{C}} \overline{z}\, d(\Delta(z)f, g).$$

For such f, and arbitrary g, these integrals converge absolutely.[67]

(On the basis of δ)–ε), R, R^* and $\Delta(z)$ are said to *belong together*.)

ζ) If the Hermitian operators $\widetilde{S}_1, \widetilde{S}_2$ are formed as in Theorem 14, then $\widetilde{S}_1 f$,

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[66] If the rectangle $\mathfrak{R}$ has the vertices

$$a' + b'i, \quad a' + b''i, \quad a'' + b''i, \quad a'' + b'i \qquad (a' \le a'',\ b' \le b''),$$

then

$$\iint_{\mathfrak{R}} d|\Delta(z)f|^2 = |\Delta(a' + b'i)f|^2 - |\Delta(a' + b''i)f|^2 \\ + |\Delta(a'' + b''i)f|^2 - |\Delta(a'' + b'i)f|^2.$$

Since $|Ef|^2 = (f, Ef)$ for a projection operator E, this equals

$$\left|\{\Delta(a' + b'i) - \Delta(a' + b''i) + \Delta(a'' + b''i) - \Delta(a'' + b'i)\}\, f\right|^2 \ge 0,$$

provided the operator in braces is a projection operator. By α), it is indeed so, for

$$\Delta(a' + b'i) - \Delta(a' + b''i) + \Delta(a'' + b''i) - \Delta(a'' + b'i) \\ = \Delta(a'' + b''i)\big(1 - \Delta(a' + b''i)\big)\big(1 - \Delta(a'' + b'i)\big),$$

which makes the assertion evident.

[67] All these arguments are closely analogous to those of E., Theorem 36, with which they may be compared. For completeness we give here a proof of the absolute convergence of $\iint z\, d(\Delta(z)f, g)$; the proof for $\iint \overline{z}\, d(\Delta(z)f, g)$ is the same.

Let $\mathfrak{R}_1, \dots, \mathfrak{R}_k$ be any rectangles with no common interior points, together filling a finite part of the plane, and let $\zeta_1, \dots, \zeta_k$ be arbitrary points in them. If E_h is the projection operator calculated for the rectangle $\mathfrak{R}_h$ as in note 66, then, for $h = 1, \dots, k$,

$$\left|\iint_{\mathfrak{R}_h} d(\Delta(z)f, g)\right| = |(E_h f, g)| = |(E_h f, E_h g)| \\ \le |E_h f|\,|E_h g| \\ = \sqrt{\iint_{\mathfrak{R}_h} d|\Delta(z)f|^2 \iint_{\mathfrak{R}_h} d|\Delta(z)g|^2}.$$

respectively $\widetilde{S}_2 f$, is defined if and only if

$$\iint_{\mathbb{C}} (\operatorname{Re} z)^2 \, d|\Delta(z) f|^2, \qquad \text{respectively} \qquad \iint_{\mathbb{C}} (\operatorname{Im} z)^2 \, d|\Delta(z) f|^2$$

is finite. (For the nature of these integrals see p. 411 and note 66.)

η) In that case, for every g,

$$(\widetilde{S}_1 f, g) = \iint_{\mathbb{C}} \operatorname{Re} z \, d(\Delta(z) f, g),$$

$$(\widetilde{S}_2 f, g) = \iint_{\mathbb{C}} \operatorname{Im} z \, d(\Delta(z) f, g).$$

The absolute convergence of these integrals is assured; cf. ε) and note 67.

Proof. Since $\widetilde{S}_1, \widetilde{S}_2$ are hypermaximal, there belong to them two resolutions of the identity $E(\lambda), F(\lambda)$ (cf. *E.*, Theorem 36, and our note 42). By definition, $\widetilde{S}_1 f$, respectively $\widetilde{S}_2 f$, is defined if

$$\int_{-\infty}^{\infty} \lambda^2 \, d|E(\lambda) f|^2, \qquad \text{respectively} \qquad \int_{-\infty}^{\infty} \lambda^2 \, d|F(\lambda) f|^2,$$

is finite; and then, for every g,

$$(\widetilde{S}_1 f, g) = \int_{-\infty}^{\infty} \lambda \, d(E(\lambda) f, g),$$

$$(\widetilde{S}_2 f, g) = \int_{-\infty}^{\infty} \lambda \, d(F(\lambda) f, g).$$

It follows (cf. the estimates in *E.*, proof of Theorem 36, and our note 45) that

$$\left| \sum_{h=1}^{k} \zeta_h \iint_{\mathfrak{R}_h} d(\Delta(z) f, g) \right| \leq \sum_{h=1}^{k} |\zeta_h| \sqrt{\iint_{\mathfrak{R}_h} d|\Delta(z) f|^2 \iint_{\mathfrak{R}_h} d|\Delta(z) g|^2}$$
$$\leq \sqrt{\sum_{h=1}^{k} |\zeta_h|^2 \iint_{\mathfrak{R}_h} d|\Delta(z) f|^2} \sqrt{\sum_{h=1}^{k} \iint_{\mathfrak{R}_h} d|\Delta(z) g|^2}.$$

But

$$\iint_{\mathbb{C}} |z|^2 \, d|\Delta(z) f|^2 \quad \text{and} \quad \iint_{\mathbb{C}} d|\Delta(z) g|^2 = |g|^2$$

are finite and hence converge absolutely (cf. the observation following δ)). The same therefore follows for $\iint z \, d(\Delta(z) f, g)$.

Let the set of all $E(\lambda)$, respectively all $F(\lambda)$, be $\mathcal{E}$, respectively $\mathcal{F}$. Every $E(\lambda)$ belongs to $\mathcal{E}$, hence to $\mathcal{E}''$, and therefore by Theorem 13 to $(\widetilde{S}_1)''$. But already in the proof of Theorem 14 we saw that $(\widetilde{S}_1)'' \subset (R)''$; hence $\mathcal{E} \subset (R)''$. Similarly, $\mathcal{F} \subset (R)''$. Since $(R)''$ is Abelian, the elements of $\mathcal{E}$ commute with those of $\mathcal{F}$: every $E(a)$ commutes with every $F(b)$. We may therefore define a family of projection operators $\Delta(z)$ by

$$\Delta(a+ib) = E(a)F(b) = F(b)E(a).$$

It is readily verified that α)–γ) hold; that is, this is a complex resolution of the identity.

Furthermore, the integrals

$$\int_{-\infty}^{\infty} \lambda^2 \, d|E(\lambda)f|^2, \qquad \int_{-\infty}^{\infty} \lambda^2 \, d|F(\lambda)f|^2,$$

$$\int_{-\infty}^{\infty} \lambda \, d(E(\lambda)f, g), \qquad \int_{-\infty}^{\infty} \lambda \, d(F(\lambda)f, g)$$

pass respectively into

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} a^2 \, d|\Delta(a+ib)f|^2, \qquad \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} b^2 \, d|\Delta(a+ib)f|^2,$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} a \, d(\Delta(a+ib)f, g), \qquad \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} b \, d(\Delta(a+ib)f, g).$$

Thus, on putting $z = a + ib$, they become precisely the integrals in ζ)–η). Those assertions are therefore established.

Since $R = \overline{R}$ and $R^* = \overline{R}^*$, the vector Rf (and R^*f) is defined if and only if $\widetilde{S}_1 f, \widetilde{S}_2 f$ are both defined; that is, if and only if both integrals in ζ) are finite. In view of what was said there about their essentially nonnegative nature, this means that their sum is finite. Since

$$|z|^2 = (\operatorname{Re} z)^2 + (\operatorname{Im} z)^2,$$

this is exactly assertion δ). Assertion ε) follows immediately from η). □

Theorem 18. The family $\Delta(z)$ is already determined uniquely by conditions α)–ε). Thus, to every maximal conjugate pair R, R^* there belongs one and only one complex resolution of the identity $\Delta(z)$.

Proof. Let $\Delta(z)$ be the family constructed in the proof of Theorem 17, and let $\Delta'(z)$ be another family satisfying α)–ε). We must prove

$$\Delta(z) = \Delta'(z)$$

for every z.

First, η) follows from δ), at least for those f for which Rf is defined, that is, for which both $\widetilde{S}_1 f, \widetilde{S}_2 f$ are defined. For these f, it therefore holds with $\Delta'(z)$ as well. But $\widetilde{S}_1$, respectively $\widetilde{S}_2$, restricted to the domain of R, is S_1, respectively S_2 (both have the same domain). We are therefore dealing with the f's for which $S_1 f, S_2 f$ are defined.

As b increases, with a fixed, $\Delta'(a+ib)$ never decreases in the ordering of projection operators (cf. *E.*, §III, Theorem 14). Hence, as $b \to \infty$, a strong limit exists (*E.*, §III, Theorem 19):

$$\Delta'(a+ib) \longrightarrow E'(a).$$

Similarly, with b fixed, a strong limit exists as $a \to \infty$:

$$\Delta'(a+ib) \longrightarrow F'(b).$$

The operators $E'(a), F'(b)$ are projection operators which, like $\Delta'(a+ib)$, never decrease as a, b increase. For $a' \geq a,\ b' \geq b$, condition α) gives

$$\Delta'(a'+ib)\Delta'(a+ib') = \Delta'(a+ib')\Delta'(a'+ib) = \Delta'(a+ib).$$

On passing to the limit $a', b' \to +\infty$, this yields

$$F'(b)E'(a) = E'(a)F'(b) = \Delta'(a+ib).$$

From

$$\Delta'(a+ib)f \longrightarrow \begin{cases} E'(a)f, & b \to +\infty, \\ 0, & b \to -\infty, \end{cases} \qquad \Delta'(a+ib)f \longrightarrow \begin{cases} F'(b)f, & a \to +\infty, \\ 0, & a \to -\infty, \end{cases}$$

one immediately obtains

$$\begin{aligned} \iint_{\mathbb{C}} \operatorname{Re} z \, d(\Delta'(z)f, g) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} a \, d(E'(a)F'(b)f, g) \\ &= \int_{-\infty}^{\infty} a \, d(E'(a)f, g), \end{aligned}$$

$$\iint_{\mathbb{C}} \operatorname{Im} z \, d(\Delta'(z)f, g) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} b \, d(E'(a)F'(b)f, g)$$
$$= \int_{-\infty}^{\infty} b \, d(F'(b)f, g),$$

in the sense that each right-hand side converges whenever the corresponding left-hand side does. The two equalities hold independently of one another.

It follows without difficulty from α)–γ) that $E'(\lambda), F'(\lambda)$ are resolutions of the identity. Let the Hermitian operators belonging to them be T_1, T_2, respectively (cf. E., Theorem 36). By what has already been shown, whenever $T_1 f$ and $S_1 g$ are both defined,

$$(f, S_1 g) = \int_{-\infty}^{\infty} \lambda \, d(f, E'(\lambda)g) = \int_{-\infty}^{\infty} \lambda \, d(E'(\lambda)f, g) = (T_1 f, g).$$

Thus f is an extension element of S_1, with assigned value $T_1 f$, and hence also an extension element of $\widetilde{S}_1$. But $\widetilde{S}_1$ is hypermaximal; consequently $\widetilde{S}_1 f$ is defined and equals $T_1 f$. Thus $\widetilde{S}_1$ is an extension of T_1; and, since T_1 too is hypermaximal,

$$\widetilde{S}_1 = T_1.$$

Likewise one proves

$$\widetilde{S}_2 = T_2.$$

Since the resolutions of the identity of $\widetilde{S}_1, \widetilde{S}_2$ are $E(\lambda), F(\lambda)$, respectively, it follows that

$$E'(\lambda) = E(\lambda), \qquad F'(\lambda) = F(\lambda).$$

Therefore

$$\Delta'(a + ib) = E'(a)F'(b) = E(a)F(b) = \Delta(a + ib),$$

that is, $\Delta'(z) = \Delta(z)$, as was to be proved. □

Theorem 19. To every complex resolution of the identity, that is, to every family $\Delta(z)$ satisfying α)–γ), there belongs one and only one conjugate pair R, R^*, according to δ)–ε); and this pair is maximal and normal.

Proof. If such a pair exists, it is unique: condition δ) determines its domain, and ε) determines $(Rf, g), (R^*f, g)$, hence Rf, R^*f themselves. As regards existence, it remains to say this: if we have any pair of operators R, R^* defined by δ), ε) (and if the domain is dense everywhere), then those two conditions themselves make them a conjugate pair; only maximality and normality then remain to be checked.

The existence of R, R^* according to δ), ε) is established once we know the following. If

$$\iint_{\mathbb{C}} |z|^2 \, d|\Delta(z) f|^2$$

is finite, then there exist two vectors f^*, f^{**} such that, for every g,

$$\iint_{\mathbb{C}} z \, d(\Delta(z) f, g) = (f^*, g), \qquad \iint_{\mathbb{C}} \overline{z} \, d(\Delta(z) f, g) = (f^{**}, g).$$

These are then Rf and $R^* f$, respectively. For the moment call the two integrals on the left $L^*(g), L^{**}(g)$, with f fixed. Plainly,

$$L^*(a_1 g_1 + \cdots + a_n g_n) = \overline{a}_1 L^*(g_1) + \cdots + \overline{a}_n L^*(g_n),$$

$$L^{**}(a_1 g_1 + \cdots + a_n g_n) = \overline{a}_1 L^{**}(g_1) + \cdots + \overline{a}_n L^{**}(g_n).$$

By F. Riesz's theorem (cf. *E.*, note 52) our goal is reached as soon as we find a constant c such that

$$|L^*(g)| \leq c|g|, \qquad |L^{**}(g)| \leq c|g|.$$

But the estimate in note 67 gives

$$|L^*(g)| \leq \sqrt{\iint |z|^2 \, d|\Delta(z) f|^2} \sqrt{\iint d|\Delta(z) g|^2} = \sqrt{\iint |z|^2 \, d|\Delta(z) f|^2} \, |g|,$$

and likewise

$$|L^{**}(g)| \leq \sqrt{\iint |z|^2 \, d|\Delta(z) f|^2} \, |g|,$$

which proves everything required.

The domain, that is, the set of f for which $\iint |z|^2 \, d|\Delta(z) f|^2$ is finite, is everywhere dense. To see this, proceed as follows. Just as, in the proof of Theorem 18, the resolutions of the identity $E'(\lambda)$ and $F'(\lambda)$ were formed from $\Delta'(z)$, we now form from $\Delta(z)$ two resolutions of the identity $E(\lambda)$ and $F(\lambda)$. The integral $\iint |z|^2 \, d|\Delta(z) f|^2$ is finite whenever

$$\iint (\Re z)^2 \, d|\Delta(z) f|^2, \qquad \iint (\Im z)^2 \, d|\Delta(z) f|^2$$

are finite, that is, whenever

$$\int_{-\infty}^{\infty} \lambda^2 \, d|E(\lambda) f|^2, \qquad \int_{-\infty}^{\infty} \lambda^2 \, d|F(\lambda) f|^2$$

are finite (cf. the proof cited above). This is certainly so for every f for which

$$E(\lambda)f = F(\lambda)f = \begin{cases} f, & \lambda \geq c, \\ 0, & \lambda \leq -c \end{cases} \qquad (c \text{ arbitrary but fixed})$$

holds,[68] and hence for every

$$f = (E(c) - E(-c))(F(c) - F(-c))g.$$

As this converges to g when $c \to \infty$, while g is arbitrary, the set in question is indeed everywhere dense.

It remains to show that R, R^* is maximal and normal. First, normality. Let $\mathcal{D}$ be the set of all $\Delta(z)$. Since these operators are Hermitian and mutually commuting, $\mathcal{D}$ is Abelian, and consequently so is $\mathcal{D}''$. It therefore suffices to prove

$$(R)'' \subset \mathcal{D}'',$$

and a fortiori it suffices to prove $(R)' \supset \mathcal{D}'$. This will be established if we show that every A which commutes with all $\Delta(z)$ also commutes with R.

Suppose, then, that Rf is defined. We assert that $R(Af)$ is defined and equals $A(Rf)$. First we must infer the finiteness of

$$\iint |z|^2 \, d|\Delta(z)Af|^2$$

from that of

$$\iint |z|^2 \, d|\Delta(z)f|^2,$$

and then prove, for example,

$$(R(Af), g) = (A(Rf), g) \qquad \text{for every } g.$$

Equivalently, we must prove

$$(Af, R^*g) = (Rf, A^*g), \qquad \text{or} \qquad \iint z \, d(Af, \Delta(z)g) = \iint z \, d(\Delta(z)f, A^*g).$$

The operator A is bounded, so that $|Af| \leq c|f|$ always. Hence, for every rectangle $\mathfrak{R}$, if $E_{\mathfrak{R}}$ denotes the projection operator constructed from it as in note 66,

$$\iint_{\mathfrak{R}} d|\Delta(z)Af|^2 = |E_{\mathfrak{R}}Af|^2 = |AE_{\mathfrak{R}}f|^2 \leq c^2|E_{\mathfrak{R}}f|^2 = c^2 \iint_{\mathfrak{R}} d|\Delta(z)f|^2.$$

[68]For then the integrals from $-\infty$ to ∞ may be replaced by integrals from $-c$ to c, and the integrand λ^2 is bounded.

Since $|z|^2 \geq 0$, it follows that[T22]

$$\iint |z|^2\, d|\Delta(z)Af|^2 \leq c^2 \iint |z|^2\, d|\Delta(z)f|^2,$$

which proves the first assertion. The second is immediate from

$$(Af, \Delta(z)g) = (\Delta(z)Af, g) = (A\Delta(z)f, g) = (\Delta(z)f, A^*g).$$

We now consider maximality. It states that

$$\overline{R} = R, \qquad \overline{R}^* = R^*.$$

Since the intersection of the domains of $\widetilde{S}_1$ and $\widetilde{S}_2$ (cf. Theorem 14) is the domain of $\overline{R}, \overline{R}^*$, we must require that R, and hence also R^*, be defined everywhere on this intersection. We again use the resolutions of the identity $E(\lambda)$ and $F(\lambda)$ above, and let the corresponding (hypermaximal) Hermitian operators be T_1, T_2. Evidently T_1 is an extension of S_1, hence, being closed and linear, also of $\widetilde{S}_1$; and since the latter is hypermaximal, $T_1 = \widetilde{S}_1$. In precisely the same way, $T_2 = \widetilde{S}_2$. Thus, in the intersection of the domains of $\widetilde{S}_1$ and $\widetilde{S}_2$, both $T_1 f$ and $T_2 f$ are defined; that is, the integrals

$$\int_{-\infty}^{\infty} \lambda^2\, d|E(\lambda)f|^2 = \iint (\Re z)^2\, d|\Delta(z)f|^2,$$

$$\int_{-\infty}^{\infty} \lambda^2\, d|F(\lambda)f|^2 = \iint (\Im z)^2\, d|\Delta(z)f|^2$$

are finite. Their sum $\iint |z|^2\, d|\Delta(z)f|^2$ is therefore finite as well. Hence Rf is defined, and the proof is complete. ☐

Theorem 17–Theorem 19 show that maximal normal conjugate pairs of operators correspond one-to-one to complex resolutions of the identity, just as hypermaximal Hermitian operators correspond to real resolutions of the identity (by *E.*, Theorem 36). Because maximal extensions are unique in general, and because there is no analogue here of the hypermaximality condition, the situation is in fact somewhat simpler.

[T22] *Translator's note.* In this sentence and in both integrals immediately below, the source prints λ^2. The integration is over the complex z-plane and the preceding estimates require $|z|^2$; that expression is therefore used here.

2. Some elementary properties of maximal normal conjugate pairs

Theorem 20. Let R, R^* be a maximal normal conjugate pair, let $\Delta(z)$ be its complex resolution of the identity, and let $\mathcal{D}$ be the set of all $\Delta(z)$. Then

$$(R)' = \mathcal{D}', \qquad (R)'' = \mathcal{D}'', \qquad (R)''' = \mathcal{D}''', \quad \ldots.$$

Proof. It plainly suffices to prove the first equality. In the preceding proof we already saw that $(R)' \supset \mathcal{D}'$; it remains only to show $(R)' \subset \mathcal{D}'$. In the proof of Theorem 14 we obtained $(R)' \subset (\widetilde{S}_1)'$ and $(R)' \subset (\widetilde{S}_2)'$, hence

$$(R)' \subset (\widetilde{S}_1, \widetilde{S}_2)'.$$

Thus it is enough to see that

$$(\widetilde{S}_1, \widetilde{S}_2)' \subset \mathcal{D}'.$$

Let the resolutions of the identity of $\widetilde{S}_1$ and $\widetilde{S}_2$ be $E(\lambda)$ and $F(\lambda)$, respectively, and let $\mathcal{E}$ and $\mathcal{F}$ denote the sets of all $E(\lambda)$ and $F(\lambda)$. By Theorem 13,

$$(\widetilde{S}_1)' = \mathcal{E}', \qquad (\widetilde{S}_2)' = \mathcal{F}',$$

and therefore

$$(\widetilde{S}_1, \widetilde{S}_2)' = (\mathcal{E}, \mathcal{F})'.$$

If $A \in \mathfrak{B}$ commutes with every element of both $\mathcal{E}$ and $\mathcal{F}$, then it commutes with every

$$\Delta(a + ib) = E(a)F(b),$$

and hence with every element of $\mathcal{D}$. Thus $(\mathcal{E}, \mathcal{F})' \subset \mathcal{D}'$, as required. □

Theorem 21. If R, R^* is a normal conjugate pair, then in general

$$|Rf| = |R^* f|.$$

If R, R^* is also maximal, then both R and R^* are closed linear operators.

Proof. Since R, R^* has at most one maximal extension (Theorem 16), it is enough to prove the first assertion for maximal R, R^*. Let $\Delta(z)$ be its complex resolution of the identity.

Whenever Rf (and R^*f) is defined, we have

$$\begin{aligned}(Rf, \Delta(a+ib)g) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a'+ib')\,d\big(\Delta(a'+ib')f, \Delta(a+ib)g\big)\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a'+ib')\,d\big(\Delta(a+ib)\Delta(a'+ib')f, g\big)\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a'+ib')\,d\big(\Delta(\min(a,a')+i\min(b,b'))f, g\big)\\ &= \int_{-\infty}^{a}\int_{-\infty}^{b}(a'+ib')\,d\big(\Delta(a'+ib')f, g\big).\end{aligned}$$

In particular,[T23]

$$\begin{aligned}(\Delta(a+ib)f, Rf) &= \int_{-\infty}^{a}\int_{-\infty}^{b}(a'-ib')\,d\big(f, \Delta(a'+ib')f\big)\\ &= \int_{-\infty}^{a}\int_{-\infty}^{b}(a'-ib')\,d|\Delta(a'+ib')f|^2.\end{aligned}$$

It follows that[69]

$$\begin{aligned}|Rf|^2 &= (Rf, Rf)\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a+ib)\,d\big(\Delta(a+ib)f, Rf\big)\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a+ib)\,d\left[\int_{-\infty}^{a}\int_{-\infty}^{b}(a'-ib')\,d|\Delta(a'+ib')f|^2\right]\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a+ib)(a-ib)\,d|\Delta(a+ib)f|^2\\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a^2+b^2)\,d|\Delta(a+ib)f|^2.\end{aligned}$$

In the same way,

$$|R^*f|^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a^2+b^2)\,d|\Delta(a+ib)f|^2.$$

This proves $|Rf| = |R^*f|$.

[T23] *Translator's note.* In the last Stieltjes differentials above the source prints the unprimed $\Delta(a+ib)$. The primed integration variables shown here are required by the surrounding integrals.

[69] For this transformation of the integral, cf. *E.*, note 55.

We turn to the second assertion. The linearity of R, as of R^*, follows, for example, from Theorem 15 (maximality gives $R = \overline{R}$ and $R^* = \overline{R}^*$). We now prove that R is closed; the proof for R^* is identical. Suppose that $f_n \to f$, that every Rf_n, and hence every $R^* f_n$, is defined, and that $Rf_n \to f^*$. The Rf_n satisfy the Cauchy condition,

$$|R(f_m - f_n)| \longrightarrow 0 \qquad (m, n \to \infty),$$

and therefore so do the $R^* f_n$, because

$$|R(f_m - f_n)| = |R^*(f_m - f_n)|.$$

Consequently there exists an f^{**} such that $R^* f_n \to f^{**}$. It follows at once that f, f^*, f^{**} satisfy the hypotheses of Theorem 15. Thus Rf is defined and equals f^*. Hence R is closed, as asserted. □

Appendix I

For completeness, we give here the familiar Hilbertian characterization of the resolution of the identity of a bounded Hermitian operator. Let R be hypermaximal and let $E(\lambda)$ be a resolution of the identity.

By E., § II, Theorem 12γ, the boundedness of R is characterized by

$$|(Rf, f)| \leq c|f|^2 \qquad (c \text{ fixed}),$$

and

$$(Rf, f) = \int_{-\infty}^{\infty} \lambda \, d(E(\lambda) f, f) = \int_{-\infty}^{\infty} \lambda \, d|E(\lambda) f|^2.$$

If

$$E(\lambda) = \begin{cases} 1, & \lambda \geq c, \\ 0, & \lambda \leq -c, \end{cases}$$

then the integral from $-\infty$ to ∞ may be replaced by the integral from $-c$ to c. The integrand is then at most c in absolute value, while $|E(\lambda) f|^2$ never decreases; hence the integral is, in absolute value, at most

$$c \int_{-\infty}^{\infty} d|E(\lambda) f|^2 = c|f|^2.$$

Thus R is bounded.

Conversely, suppose that $E(\lambda) \neq 0$ for arbitrarily small λ, respectively $E(\lambda) \neq 1$ for arbitrarily large λ; in other words, that $E(\lambda)$ is not constant for arbitrarily large $|\lambda|$. (Since $E(\lambda)f \to 0$, respectively f, as $\lambda \to -\infty$, respectively $+\infty$, no constant value other than 0, respectively 1, can occur.) Then, for every D, there are λ, μ such that

$$E(\lambda) \neq E(\mu), \qquad \lambda < \mu < -D \quad \text{or} \quad D < \lambda < \mu.$$

Thus $E(\mu) - E(\lambda)$ is a nonzero projection operator. Let $f \neq 0$ be a point of its closed linear manifold, so that

$$(E(\mu) - E(\lambda))f = f.$$

Then[70]

$$E(\lambda')f = \begin{cases} f, & \lambda' \geq \mu, \\ 0, & \lambda' \leq \lambda. \end{cases}$$

Consequently,

$$\begin{aligned} |(Rf, f)| &= \left| \int_{-\infty}^{\infty} \lambda' \, d|E(\lambda')f|^2 \right| \\ &= \left| \int_{\lambda}^{\mu} \lambda' \, d|E(\lambda')f|^2 \right| \\ &\geq D \int_{\lambda}^{\mu} d|E(\lambda')f|^2 \\ &= D \int_{-\infty}^{\infty} d|E(\lambda')f|^2 = D|f|^2. \end{aligned}$$

For $D > c$, this is greater than $c|f|^2$; hence R is unbounded.

In precisely the same way one can read off the boundedness of a maximal normal conjugate pair R, R^* from its complex resolution of the identity $\Delta(z)$. It is defined by

$$|Rf| \leq c|f| \qquad (c \text{ fixed}),$$

and it is convenient to use the following relation (Theorem 21):

$$|Rf|^2 = \iint |z|^2 \, d|\Delta(z)f|^2.$$

[70]Indeed,

$$|E(\mu)f|^2 - |E(\lambda)f|^2 = |(E(\mu) - E(\lambda))f|^2 = |f|^2,$$

while $|E(\mu)f|^2 \leq |f|^2$ and $|E(\lambda)f|^2 \geq 0$ (cf. E., § III). Therefore $|E(\mu)f| = |f|$ and $|E(\lambda)f| = 0$. A fortiori, for $\lambda' \geq \mu$, respectively $\lambda' \leq \lambda$, we have $|E(\lambda')f| = |f|$, respectively 0; that is, $E(\lambda')f = f$, respectively 0 (cf. the passage cited), as asserted.

Appendix II

1. Operators of a prescribed class

We shall apply the results of § IV–V, in particular to Laurent matrices and their generalizations.

Let $\mathcal{P}$ be a subset of $\mathfrak{B}$ for which $\mathcal{P}'$ is Abelian. If R, R^* is a conjugate pair, we say that it is *of class* $\mathcal{P}$ when

$$(R)' \supset \mathcal{P},$$

that is, when R commutes with every A, A^* as A ranges over all of $\mathcal{P}$.

Then $(R)'' \subset \mathcal{P}'$, so $(R)''$ is Abelian; that is, R, R^* is normal. By Theorem 16 it therefore has a unique maximal extension $\overline{R}, \overline{R}^*$, which is likewise normal. In the proof of Theorem 14 we saw that

$$(\overline{R})' \supset (R)',$$

and hence $(\overline{R})' \supset \mathcal{P}$; thus $\overline{R}$ is also of class $\mathcal{P}$. It is therefore enough to consider maximal normal pairs R, R^*. We now assume R, R^* to be such a pair, write $\Delta(z)$ for its complex resolution of the identity, and let $\mathcal{D}$ be the set of all $\Delta(z)$. Since $(R)' = \mathcal{D}'$ by Theorem 20, the pair R, R^* is of class $\mathcal{P}$ if and only if

$$\mathcal{D}' \supset \mathcal{P}.$$

Because $(\Delta(z))' \supset \mathcal{D}' \supset \mathcal{P}$, every $\Delta(z)$ then belongs to class $\mathcal{P}$; and since $\mathcal{D}'$ is the intersection of all $(\Delta(z))'$, this condition is also sufficient. Moreover,

$$\mathcal{D}' \supset \mathcal{P} \implies \mathcal{P}' \supset \mathcal{D}'' \supset \mathcal{D},$$

whereas

$$\mathcal{P}' \supset \mathcal{D} \implies \mathcal{D}' \supset \mathcal{P}'' \supset \mathcal{P}.$$

Consequently $\mathcal{D} \subset \mathcal{P}'$ as well; in other words, it is necessary and sufficient that every $\Delta(z)$ belong to $\mathcal{P}'$.

We now construct a set $\mathcal{P}$ of the kind just described. Let G be a countably infinite Abelian group (the method can also be generalized to continuous groups G, but we shall not discuss this here), with elements $\alpha, \beta, \ldots$. Let $\varphi_\alpha, \varphi_\beta, \ldots$ be a complete normalized orthogonal system in $\mathfrak{H}$, indexed not by the integers $1, 2, \ldots$, but by the elements $\alpha, \beta, \ldots$ of G.

It is immediate that the operator U_α defined by

$$U_\alpha\left(\sum_{\beta\in G} x_\beta\varphi_\beta\right) = \sum_{\beta\in G} x_{\alpha\beta}\varphi_\beta, \qquad \left(\sum_{\beta\in G}|x_\beta|^2 < \infty\right),$$

is unitary. We have

$$U_\alpha U_\beta = U_{\alpha\beta}, \qquad U_\alpha^* = U_\alpha^{-1} = U_{\alpha^{-1}}.$$

Let $\mathcal{P}$ be the set of all U_α. When does an $A \in \mathfrak{B}$ belong to $\mathcal{P}'$? It must commute with every $U_\alpha \in \mathcal{P}$ (the relations $U_\alpha^* = U_{\alpha^{-1}}$ add nothing new), so that

$$AU_\alpha = U_\alpha A.$$

This means

$$(AU_\alpha f, g) = (U_\alpha A f, g)$$

for all f, g. Since both sides are linear and continuous in f and in g, it is enough to require this on a complete normalized orthogonal system. Taking $f = \varphi_\beta$ and $g = \varphi_\gamma$, we obtain

$$(AU_\alpha\varphi_\beta, \varphi_\gamma) = (A\varphi_{\alpha^{-1}\beta}, \varphi_\gamma),$$

and

$$(U_\alpha A\varphi_\beta, \varphi_\gamma) = (A\varphi_\beta, U_\alpha^*\varphi_\gamma) = (A\varphi_\beta, U_{\alpha^{-1}}\varphi_\gamma) = (A\varphi_\beta, \varphi_{\alpha\gamma}).$$

Thus

$$(A\varphi_{\alpha^{-1}\beta}, \varphi_\gamma) = (A\varphi_\beta, \varphi_{\alpha\gamma});$$

equivalently, $(A\varphi_\beta, \varphi_\gamma)$ depends only on $\beta^{-1}\gamma$.

If $A, B \in \mathcal{P}'$, write

$$(A\varphi_\varepsilon, \varphi_\eta) = a(\varepsilon^{-1}\eta), \qquad (B\varphi_\varepsilon, \varphi_\eta) = b(\varepsilon^{-1}\eta).$$

Then

$$\begin{aligned}
(AB\varphi_\beta, \varphi_\gamma) &= (B\varphi_\beta, A^*\varphi_\gamma) \\
&= \sum_{\delta\in G}(B\varphi_\beta, \varphi_\delta)(\varphi_\delta, A^*\varphi_\gamma) \\
&= \sum_{\delta\in G}(B\varphi_\beta, \varphi_\delta)(A\varphi_\delta, \varphi_\gamma) \\
&= \sum_{\delta\in G} a(\delta^{-1}\gamma)b(\beta^{-1}\delta) \\
&= \sum_{\substack{\varepsilon,\eta\in G \\ \varepsilon\eta=\beta^{-1}\gamma}} a(\varepsilon)b(\eta).
\end{aligned}$$

Exactly the same expression results for

$$(BA\varphi_\beta, \varphi_\gamma) = \sum_{\substack{\varepsilon,\eta \in G \\ \varepsilon\eta = \beta^{-1}\gamma}} a(\varepsilon)b(\eta).$$

Thus $(ABf, g) = (BAf, g)$ at least on the complete normalized orthogonal system $\{\varphi_\alpha\}$, and therefore, since both sides are linear and continuous in f and in g, for all f, g. Hence $AB = BA$: the operators A and B commute. We have thereby shown that $\mathcal{P}'$, which contains A^* whenever it contains A, is Abelian.

2. Generalized Laurent matrices

We now apply to the set $\mathcal{P}$ just constructed what was said at the beginning of § 1; that is, we investigate the pairs R, R^* of this class.

Let a_α be a sequence of numbers, likewise indexed by the elements $\alpha \in G$, subject only to

$$\sum_{\alpha \in G} |a_\alpha|^2 < \infty.$$

We define two operators R, R^*, on the vectors φ_α and only on these, by

$$R\varphi_\alpha = \sum_{\beta \in G} a_{\alpha^{-1}\beta}\varphi_\beta, \qquad R^*\varphi_\alpha = \sum_{\beta \in G} \overline{a_{\alpha\beta^{-1}}}\varphi_\beta.$$

Indeed,

$$\sum_{\beta \in G} |a_{\alpha^{-1}\beta}|^2 = \sum_{\beta \in G} |a_{\alpha\beta^{-1}}|^2 = \sum_{\beta \in G} |a_\beta|^2 < \infty.$$

It is immediate that R, R^* form a conjugate pair in the sense of Definition 5, and also that R commutes with every U_α, and hence with every $U_\alpha^* = U_{\alpha^{-1}}$. Thus $(R)' \supset \mathcal{P}$. The pair R, R^* is therefore of class $\mathcal{P}$; it is normal, and we may form $\overline{R}, \overline{R}^*$. This pair is maximal normal and also belongs to class $\mathcal{P}$.

Using Theorem 15, we now determine the domain and range of $\overline{R}$ and $\overline{R}^*$. According to that theorem, $\overline{R}f$ and $\overline{R}^*f$ are defined if and only if there exist two vectors f^*, f^{**} such that, whenever Rg and R^*g are defined,

$$(f, Rg) = (f^{**}, g), \qquad (f, R^*g) = (f^*, g).$$

Since g must be one of the φ_α, writing

$$f = \sum_{\beta \in G} x_\beta \varphi_\beta, \qquad \sum_{\beta \in G} |x_\beta|^2 < \infty,$$

this says

$$\sum_{\beta \in G} x_\beta \overline{a_{\alpha^{-1}\beta}} = (f^{**}, \varphi_\alpha), \qquad \sum_{\beta \in G} x_\beta a_{\alpha\beta^{-1}} = (f^*, \varphi_\alpha).$$

Put

$$y_\alpha = \sum_{\beta \in G} a_{\alpha\beta^{-1}} x_\beta, \qquad z_\alpha = \sum_{\beta \in G} \overline{a_{\alpha^{-1}\beta}} x_\beta.$$

Then we must have

$$(f^*, \varphi_\alpha) = y_\alpha, \qquad (f^{**}, \varphi_\alpha) = z_\alpha.$$

By E., § I, Theorems 5 and 7, this is possible if and only if

$$\sum_{\alpha \in G} |y_\alpha|^2 < \infty, \qquad \sum_{\alpha \in G} |z_\alpha|^2 < \infty,$$

and in that case

$$f^* = \sum_{\alpha \in G} y_\alpha \varphi_\alpha, \qquad f^{**} = \sum_{\alpha \in G} z_\alpha \varphi_\alpha.$$

Our result is therefore the following. For

$$f = \sum_{\alpha \in G} x_\alpha \varphi_\alpha, \qquad \sum_{\alpha \in G} |x_\alpha|^2 < \infty,$$

the vectors $\overline{R} f$ and $\overline{R}^* f$ are defined if and only if the two sums

$$\sum_{\alpha \in G} |y_\alpha|^2, \qquad \sum_{\alpha \in G} |z_\alpha|^2$$

are finite, where y_α and z_α are as defined above. In that case

$$\overline{R} f = \sum_{\alpha \in G} y_\alpha \varphi_\alpha, \qquad \overline{R}^* f = \sum_{\alpha \in G} z_\alpha \varphi_\alpha.$$

On the other hand, by Theorem 17 and Theorem 18 the pair $\overline{R}, \overline{R}^*$ has exactly one resolution of the identity $\Delta(z)$; as established in § 1, each $\Delta(z)$ belongs to $\mathcal{P}'$. In the present case this means, as we saw there, that

$$(\Delta(z)\varphi_\alpha, \varphi_\beta)$$

depends only on $\alpha^{-1}\beta$. This is the general element of the matrix of the projection operator $\Delta(z)$ in the complete normalized orthogonal system $\varphi_\alpha, \varphi_\beta, \ldots$.

If, for example, we choose for G the infinite cyclic group (the integers $0, \pm 1, \pm 2, \dots$, with addition as the composition law), we obtain a general eigenvalue theory of Laurent matrices,[71] including non-real and unbounded ones. Other choices of G lead to analogous but more general classes of matrices, which are now equally under our control.

Appendix III

1. A matrix criterion for commutativity

We shall analyze somewhat more closely the notion of commutativity for unbounded operators. We defined the normality of R, that is, the commutativity of R, R^*, by requiring $(R)''$ to be Abelian. Thus, if we form the Hermitian operators

$$S_1 = \frac{R + R^*}{2}, \qquad S_2 = \frac{R - R^*}{2i},$$

their commutativity is expressed by the Abelian character of $(S_1, S_2)''$.[72] It is natural, however, to try to use for this purpose the usual definition of commutativity of matrices.

Let, for example, R, S be two Hermitian operators and let $\varphi_1, \varphi_2, \dots$ be a complete normalized orthogonal system in which both possess matrices, say $\{a_{\mu\nu}\}$ and $\{b_{\mu\nu}\}$.[73][T24] Since all row and column sums of squares of absolute values

$$\sum_{\mu=1}^{\infty} |a_{\mu\nu}|^2 = \sum_{\mu=1}^{\infty} |a_{\nu\mu}|^2, \qquad \sum_{\mu=1}^{\infty} |b_{\mu\nu}|^2 = \sum_{\mu=1}^{\infty} |b_{\nu\mu}|^2$$

are finite, the following series converge absolutely as well:

$$\sum_{\rho=1}^{\infty} a_{\mu\rho} b_{\rho\nu}, \qquad \sum_{\rho=1}^{\infty} b_{\mu\rho} a_{\rho\nu},$$

[71] These and similar classes of matrices were related by Toeplitz to questions in the theory of functions. Cf. also the work cited above, note 24.

[72] In the proof of Theorem 14 we saw that $(R)' = (R, R^*)' = (S_1, S_2)'$, and therefore $(R)'' = (S_1, S_2)''$.

[73] Cf. *E.*, Appendix III, and also the author's paper "On the Theory of Unbounded Matrices," forthcoming in the *Journal für die reine und angewandte Mathematik*.

[T24] *Translator's note.* The cited paper subsequently appeared as J. von Neumann, "Zur Theorie der unbeschränkten Matrizen," *Journal für die reine und angewandte Mathematik* **161** (1929), 208–236, doi:10.1515/crll.1929.161.208.

and one might try to define commutativity by

$$\sum_{\rho=1}^{\infty} a_{\mu\rho}b_{\rho\nu} = \sum_{\rho=1}^{\infty} b_{\mu\rho}a_{\rho\nu} \qquad (\mu, \nu = 1, 2, \ldots).$$

Since

$$a_{\mu\nu} = (R\varphi_\mu, \varphi_\nu), \qquad b_{\mu\nu} = (S\varphi_\mu, \varphi_\nu)$$

(cf., for example, *E.*, Appendix III), we have

$$\begin{aligned}\sum_{\rho=1}^{\infty} a_{\mu\rho}b_{\rho\nu} &= \sum_{\rho=1}^{\infty}(R\varphi_\mu, \varphi_\rho)(S\varphi_\rho, \varphi_\nu)\\ &= \sum_{\rho=1}^{\infty}(R\varphi_\mu, \varphi_\rho)(\varphi_\rho, S\varphi_\nu)\\ &= (R\varphi_\mu, S\varphi_\nu),\end{aligned}$$

and similarly

$$\begin{aligned}\sum_{\rho=1}^{\infty} b_{\mu\rho}a_{\rho\nu} &= \sum_{\rho=1}^{\infty}(S\varphi_\mu, \varphi_\rho)(R\varphi_\rho, \varphi_\nu)\\ &= \sum_{\rho=1}^{\infty}(S\varphi_\mu, \varphi_\rho)(\varphi_\rho, R\varphi_\nu)\\ &= (S\varphi_\mu, R\varphi_\nu).\end{aligned}$$

Thus the commutativity condition is simply

$$(R\varphi_\mu, S\varphi_\nu) = (S\varphi_\mu, R\varphi_\nu).$$

If all $R(S\varphi_\mu)$ and $S(R\varphi_\mu)$ are defined, this may be written

$$(SR\varphi_\mu, \varphi_\nu) = (RS\varphi_\mu, \varphi_\nu), \qquad ((RS - SR)\varphi_\mu, \varphi_\nu) = 0.$$

Since the φ_ν form a complete normalized orthogonal system, $(RS - SR)\varphi_\mu = 0$. If any closed linear operator T is an extension of $i(RS - SR)$, then $Tf = 0$ for every $f = \varphi_\mu$, and hence for every finite linear aggregate of these vectors, that is, on an everywhere dense set. Since T is closed, it follows that Tf is defined for every f, and equals 0; thus $T = 0$. On the other hand, $i(RS - SR)$ is plainly Hermitian: it is defined for every φ_μ, so its domain spans the closed linear manifold $\mathfrak{H}$. We may therefore form

$$T = i(\widetilde{RS - SR}),$$

and this operator must be 0.

In this case we may therefore, with some justification, speak of commutativity of R and S. In particular, this applies to $R, S \in \mathfrak{B}$, which are bounded. For then $i(RS - SR)$ is defined everywhere and is itself closed and linear, so

$$i(RS - SR) = 0, \qquad RS = SR.$$

But what of arbitrary R, S, for which the vectors $R(S\varphi_\mu)$ and $S(R\varphi_\mu)$ need not all be defined? To what extent does

$$(R\varphi_\mu, S\varphi_\nu) = (S\varphi_\mu, R\varphi_\nu) \qquad (\mu, \nu = 1, 2, \ldots)$$

express a reasonable commutativity of R, S, when R, S have matrices in the complete normalized orthogonal system $\varphi_1, \varphi_2, \ldots$ (cf. note 73)?

We shall show by a counterexample that, in this generality, the displayed relation can hardly have any significance.

First, however, let us mention the notion of normality (for conjugate pairs R, R^*) to which this notion of commutativity (for Hermitian operators) leads. We must form

$$S_1 = \frac{R + R^*}{2}, \qquad S_2 = \frac{R - R^*}{2i},$$

and require

$$(S_1\varphi_\mu, S_2\varphi_\nu) = (S_2\varphi_\mu, S_1\varphi_\nu),$$

or, equivalently,

$$(R\varphi_\mu, R\varphi_\nu) = (R^*\varphi_\mu, R^*\varphi_\nu).$$

This plainly means

$$|Rf| = |R^* f|$$

for every finite linear aggregate of the φ_μ. We know from Theorem 21 that this is necessary for normality.

2. A counterexample

Let $\varphi_1, \varphi_2, \ldots$ be a complete normalized orthogonal system. Define two matrices $\{a_{\mu\nu}\}$ and $\{b_{\mu\nu}\}$ as follows. Unless both μ and ν are equal to either $2n - 1$ or $2n$, for the same $n = 1, 2, \ldots$, set

$$a_{\mu\nu} = b_{\mu\nu} = 0.$$

Within each such 2×2 block, set

$$a_{2n-1,2n-1} = a_{2n,2n-1} = a_{2n-1,2n} = a_{2n,2n} = n,$$

$$b_{2n-1,2n-1} = b_{2n,2n} = n, \qquad b_{2n-1,2n} = -b_{2n,2n-1} = in.$$

Let the closed linear Hermitian operators associated with the (plainly Hermitian-square-summable) matrices $\{a_{\mu\nu}\}$ and $\{b_{\mu\nu}\}$, relative to $\{\varphi_1, \varphi_2, \ldots\}$, be R and S, respectively (cf. note 73).

The formulas[T25]

$$\psi_{2n-1} = \frac{1}{\sqrt{2}}\varphi_{2n-1} + \frac{1}{\sqrt{2}}\varphi_{2n}, \qquad \psi_{2n} = \frac{1}{\sqrt{2}}\varphi_{2n-1} - \frac{1}{\sqrt{2}}\varphi_{2n},$$

$$\chi_{2n-1} = \frac{1}{\sqrt{2}}\varphi_{2n-1} + \frac{i}{\sqrt{2}}\varphi_{2n}, \qquad \chi_{2n} = \frac{1}{\sqrt{2}}\varphi_{2n-1} - \frac{i}{\sqrt{2}}\varphi_{2n}$$

define two further complete normalized orthogonal systems $\psi_1, \psi_2, \ldots$ and $\chi_1, \chi_2, \ldots$, and

$$R\psi_{2n-1} = 2n\,\psi_{2n-1}, \qquad R\psi_{2n} = 0,$$

$$S\chi_{2n-1} = 2n\,\chi_{2n-1}, \qquad S\chi_{2n} = 0.$$

The operators R, S are closed and linear. In the complete normalized orthogonal systems above, R and S, respectively, therefore have diagonal matrices; hence both are hypermaximal.[74]

In particular, for

$$f = \sum_{\nu=1}^{\infty} y_\nu \psi_\nu, \qquad \sum_{\nu=1}^{\infty} |y_\nu|^2 < \infty,$$

the vector Rf is defined if and only if

$$\sum_{n=1}^{\infty} 4n^2 |y_{2n-1}|^2 < \infty.$$

[T25] *Translator's note.* In the displayed definition of χ_{2n}, the source prints φ_{2n} in the first term. The correction to φ_{2n-1} is forced both by orthonormality and by the immediately following diagonalization.

[74] More generally, if

$$T\xi_n = \alpha_n \xi_n \qquad (n = 1, 2, \ldots),$$

where T is a closed linear operator and $\xi_1, \xi_2, \ldots$ is a complete normalized orthogonal system, then T is hypermaximal. Indeed, the closed linear manifolds $\mathfrak{E}, \mathfrak{F}$ (cf. *E.*, Chapter V) contain all

$$(T \pm i1)\xi_n = (\alpha_n \pm i)\xi_n,$$

and hence all ξ_n. Both manifolds are therefore equal to $\mathfrak{H}$.

If instead

$$f = \sum_{\nu=1}^{\infty} x_\nu \varphi_\nu, \qquad \sum_{\nu=1}^{\infty} |x_\nu|^2 < \infty,$$

then

$$y_{2n-1} = \frac{1}{\sqrt{2}} x_{2n-1} + \frac{1}{\sqrt{2}} x_{2n},$$

so the characteristic condition is

$$\sum_{n=1}^{\infty} n^2 |x_{2n-1} + x_{2n}|^2 < \infty.$$

Similarly, for

$$f = \sum_{\nu=1}^{\infty} z_\nu \chi_\nu, \qquad \sum_{\nu=1}^{\infty} |z_\nu|^2 < \infty,$$

the vector Sf is defined precisely when

$$\sum_{n=1}^{\infty} 4n^2 |z_{2n-1}|^2 < \infty.$$

For $f = \sum_{\nu=1}^{\infty} x_\nu \varphi_\nu$, where

$$z_{2n-1} = \frac{1}{\sqrt{2}} x_{2n-1} - \frac{i}{\sqrt{2}} x_{2n},$$

this is characterized by

$$\sum_{n=1}^{\infty} n^2 |x_{2n-1} - i x_{2n}|^2 < \infty.$$

The finiteness of both sums is, as one readily sees, equivalent to the finiteness of

$$\sum_{n=1}^{\infty} n^2 \left(|x_{2n-1}|^2 + |x_{2n}|^2 \right).$$

If, in addition, $R(Sf)$ and $S(Rf)$ are to be defined, write

$$Rf = \sum_{\nu=1}^{\infty} u_\nu \varphi_\nu, \qquad Sf = \sum_{\nu=1}^{\infty} v_\nu \varphi_\nu,$$

where

$$u_{2n-1} = n x_{2n-1} + n x_{2n}, \qquad u_{2n} = n x_{2n-1} + n x_{2n},$$

$$v_{2n-1} = nx_{2n-1} - inx_{2n}, \qquad v_{2n} = inx_{2n-1} + nx_{2n}.$$

We must then require the finiteness of

$$\sum_{n=1}^{\infty} n^2|v_{2n-1} + v_{2n}|^2, \qquad \sum_{n=1}^{\infty} n^2|u_{2n-1} - iu_{2n}|^2,$$

that is, of

$$\sum_{n=1}^{\infty} n^4|x_{2n-1} + x_{2n}|^2, \qquad \sum_{n=1}^{\infty} n^4|x_{2n-1} - ix_{2n}|^2.$$

The finiteness of these two sums is readily seen to be equivalent to that of

$$\sum_{n=1}^{\infty} n^4\big(|x_{2n-1}|^2 + |x_{2n}|^2\big).$$

For such f, we now calculate

$$i(RS - SR)f = \sum_{\nu=1}^{\infty} w_\nu \varphi_\nu.$$

The result is[T26]

$$w_{2n-1} = -2n^2 x_{2n-1}, \qquad w_{2n} = 2n^2 x_{2n}.$$

Thus $i(RS - SR)$ is a closed linear Hermitian operator and has a diagonal matrix in the system $\varphi_1, \varphi_2, \ldots$; it is therefore hypermaximal (cf. note 74). Plainly it is not 0; indeed,

$$f \neq 0 \quad \Longrightarrow \quad i(RS - SR)f \neq 0.$$

There can consequently be no question of commutativity of R and S. Nevertheless, we shall find a complete normalized orthogonal system $\omega_1, \omega_2, \ldots$ in which both R and S have matrices and the commutativity relations of § 1 are satisfied.

3. Construction of the common matrix system

For R and S to possess matrices $\{a_{\mu\nu}\}$ and $\{b_{\mu\nu}\}$, respectively, in a complete normalized orthogonal system $\omega_1, \omega_2, \ldots$, means the following (cf. note 73). Necessarily

$$a_{\mu\nu} = (R\omega_\mu, \omega_\nu), \qquad b_{\mu\nu} = (S\omega_\mu, \omega_\nu).$$

[T26] *Translator's note.* After defining the coefficients w_ν, the source prints z_{2n-1} and z_{2n} in these two formulas. They have been corrected to w_{2n-1} and w_{2n}.

The elementary operators of these matrices, R' and S', are the restrictions of R and S, respectively, to the set $\omega_1, \omega_2, \ldots$, and their associated closed linear operators are $\widetilde{R'}$ and $\widetilde{S'}$. One requires

$$\widetilde{R'} = R, \qquad \widetilde{S'} = S.$$

That is, for every f for which Rf, respectively Sf, is defined, there must be a sequence $f_1, f_2, \ldots$ in the linear manifold—not merely the closed linear manifold—spanned by the $\omega_1, \omega_2, \ldots$, such that

$$f_n \to f, \qquad Rf_n \to Rf,$$

respectively

$$f_n \to f, \qquad Sf_n \to Sf.$$

If both Rf and Sf are defined, the two approximating sequences may be different.

Suppose that $\omega_1, \omega_2, \ldots$ is obtained from an everywhere dense sequence $g_1, g_2, \ldots$ by E. Schmidt's orthogonalization procedure.[75] The required condition certainly holds if the sequences $f_1, f_2, \ldots$ just mentioned, with

$$f_n \to f, \qquad Rf_n \to Rf,$$

respectively

$$f_n \to f, \qquad Sf_n \to Sf,$$

can be chosen as subsequences of $g_1, g_2, \ldots$. Moreover, if

$$(Rg_p, Sg_q) = (Sg_p, Rg_q)$$

always holds, then it holds also for the $\omega_1, \omega_2, \ldots$, which are linear aggregates of the g's:

$$(R\omega_\mu, S\omega_\nu) = (S\omega_\mu, R\omega_\nu).$$

Thus R, S satisfy the commutativity relation of § 1. Notice that we have assumed Rg_p and Sg_p to be defined, but not $R(Sg_p)$ and $S(Rg_p)$.

Let $\overline{g}_1, \overline{g}_2, \ldots$ be the set of all linear aggregates of finitely many $\varphi_1, \varphi_2, \ldots$, with rational coefficients, written as a sequence. Since R, S have matrices in the system $\varphi_1, \varphi_2, \ldots$, it is immediate that the sequences $f_1, f_2, \ldots$ mentioned above can be selected from $\overline{g}_1, \overline{g}_2, \ldots$; the rationality condition on the coefficients causes no

[75] Cf. E. Schmidt at the place cited in note 52; see also *E.*, § I, Theorem 8.

difficulty. We now choose $h_1, h_2, \ldots$ so that all Rh_n, Sh_n are defined and, with strong convergence as always in $\mathfrak{H}$,

$$h_n \longrightarrow 0, \qquad Rh_{2n-1} \longrightarrow 0, \qquad Sh_{2n} \longrightarrow 0 \qquad (n \to \infty).$$

The sequences $f_1, f_2, \ldots$ above can then always be selected as subsequences of

$$\overline{g}_1 + h_1,\ \overline{g}_2 + h_3, \ldots, \qquad \overline{g}_1 + h_2,\ \overline{g}_2 + h_4, \ldots.$$

For brevity, write

$$\overline{g}_1, \overline{g}_1, \overline{g}_2, \overline{g}_2, \ldots$$

as $\widetilde{g}_1, \widetilde{g}_2, \ldots$, and choose

$$g_1, g_2, \ldots = \widetilde{g}_1 + h_1, \widetilde{g}_2 + h_2, \ldots.$$

Everything will therefore be accomplished once

$$(Rg_p, Sg_q) = (Sg_p, Rg_q)$$

has been enforced by a suitable choice of $h_1, h_2, \ldots$. There is still some freedom in the choice of the h's, whereas the $\widetilde{g}_1, \widetilde{g}_2, \ldots$ have already been fixed.

We reindex the sequence $\varphi_1, \varphi_2, \ldots$ as follows. Divide it into pairs $\varphi_{2n-1}, \varphi_{2n}$, replace the single index $n = 1, 2, \ldots$ by the triple p, q, μ, where each index ranges over $1, 2, \ldots$, and denote $\varphi_{2n-1}, \varphi_{2n}$ by

$$\varphi'_{pq\mu}, \qquad \varphi''_{pq\mu},$$

respectively. Denote the integer n belonging to p, q, μ by $n(pq\mu)$, and arrange the indexing so that

$$n(pq\mu) \geq \mu^2$$

always. Define

$$h_p = \sum_{q,\mu=1}^{\infty} \frac{1}{pq\,[n(pq\mu)]^{3/2}}\, \varphi'_{pq\mu} + \sum_{r=1}^{p-1} x_{rp} v_{rp},$$

where[T27]

$$v_{rp} = \begin{cases} \varphi'_{r,p,\rho(rp)} - \varphi''_{r,p,\rho(rp)}, & p \text{ odd}, \\ \varphi'_{r,p,\rho(rp)} - i\varphi''_{r,p,\rho(rp)}, & p \text{ even}. \end{cases}$$

[T27] *Translator's note.* In the even-p branch the source prints $\varphi'_{r,q,\rho(rp)}$ in the first term. The free index q has been corrected to p, as required by the definition of v_{rp}.

The quantities x_{rp} and $\rho(rp)$, for $r < p$, will be chosen later.

First consider

$$\xi_p = \sum_{q,\mu=1}^{\infty} \frac{1}{pq\,[n(pq\mu)]^{3/2}}\,\varphi'_{pq\mu}.$$

These sums converge, since the sums of the squares of the absolute values of their coefficients satisfy

$$\sum_{q,\mu=1}^{\infty} \frac{1}{p^2q^2[n(pq\mu)]^3} \leq \sum_{q,\mu=1}^{\infty} \frac{1}{p^2q^2\mu^6}$$
$$= \frac{1}{p^2}\left(\sum_{q=1}^{\infty}\frac{1}{q^2}\right)\left(\sum_{\mu=1}^{\infty}\frac{1}{\mu^6}\right).$$

Thus $\xi_p \longrightarrow 0$ as $p \longrightarrow \infty$. Moreover,

$$\sum_{q,\mu=1}^{\infty} \frac{[n(pq\mu)]^2}{p^2q^2[n(pq\mu)]^3} = \sum_{q,\mu=1}^{\infty} \frac{1}{p^2q^2n(pq\mu)}$$
$$\leq \sum_{q,\mu=1}^{\infty} \frac{1}{p^2q^2\mu^2}$$
$$= \frac{1}{p^2}\left(\sum_{q=1}^{\infty}\frac{1}{q^2}\right)\left(\sum_{\mu=1}^{\infty}\frac{1}{\mu^2}\right)$$

is finite, and even tends to 0 with $p \longrightarrow \infty$. Hence $R\xi_p$ and $S\xi_p$ are defined.

The sum

$$\sum_{r=1}^{p-1} x_{rp}v_{rp}$$

is finite, and R, S are therefore applicable to it. This sum tends to 0 as $p \longrightarrow \infty$ whenever

$$\sum_{r=1}^{p-1} |x_{rp}|^2 \longrightarrow 0,$$

which certainly holds, for example, if

$$|x_{rp}| \leq \frac{1}{rp}.$$

Furthermore, for odd p we have $Rv_{rp} = 0$, and for even p we have $Sv_{rp} = 0$. Thus applying R, respectively S, to $\sum_{r=1}^{p-1} x_{rp}v_{rp}$ gives 0. In summary:[T28]

$$Rh_p \text{ and } Sh_p \text{ are always defined,} \qquad h_p \to 0,$$

$$Rh_p \to 0 \quad \text{for odd } p, \qquad Sh_p \to 0 \quad \text{for even } p.$$

It remains to ensure, by a suitable choice of x_{rp} and $\rho(rp)$, that

$$(Rg_p, Sg_q) - (Sg_p, Rg_q) = 0.$$

It is plainly enough to consider $p < q$, as we shall do, while retaining $|x_{rp}| \leq 1/(rp)$. Substitute

$$g_p = \widetilde{g}_p + \sum_{t,\mu=1}^{\infty} \frac{1}{pt\,[n(pt\mu)]^{3/2}} \varphi'_{pt\mu} + \sum_{r=1}^{p-1} x_{rp}v_{rp},$$

$$g_q = \widetilde{g}_q + \sum_{\mu,\nu=1}^{\infty} \frac{1}{q\mu\,[n(q\mu\nu)]^{3/2}} \varphi'_{q\mu\nu} + \sum_{s=1}^{q-1} x_{sq}v_{sq}$$

in the formula above and expand. In view of the existing orthogonalities, and since $p < q$, the result is

$$\begin{aligned}
&(\{SR - RS\}\widetilde{g}_p, \widetilde{g}_q) + (\{SR - RS\}\widetilde{g}_p, \xi_q) + (\xi_p, \{RS - SR\}\widetilde{g}_q) \\
&\quad + \sum_{s=1}^{q-1} \overline{x}_{sq}(\{SR - RS\}\widetilde{g}_p, v_{sq}) \\
&\quad + \sum_{r=1}^{p-1} x_{rp}(v_{rp}, \{RS - SR\}\widetilde{g}_q) \\
&\quad + \frac{\overline{x}_{pq}}{pq\,[n(p,q,\rho(pq))]^{3/2}} \left(\{SR - RS\}\varphi'_{p,q,\rho(pq)}, v_{pq}\right) = 0.
\end{aligned}$$

For brevity put

$$i(RS - SR) = T.$$

We also introduce the following notation. The vector $\widetilde{g}_l$ is a linear aggregate of finitely many φ_n, and hence of finitely many $\varphi'_{pq\mu}$ and $\varphi''_{pq\mu}$; let $\mu(l)$ denote the largest μ occurring in it. Require henceforth that

$$\rho(rt) > \max\bigl(\mu(1), \ldots, \mu(t-1)\bigr)$$

[T28] *Translator's note.* At this point the source prints $h_p \to \infty$; the construction and the estimate immediately above require $h_p \to 0$.

always. The fourth term in the preceding expression then vanishes identically, leaving

$$(T\widetilde{g}_p, \widetilde{g}_q) + (T\widetilde{g}_p, \xi_q) + (\xi_p, T\widetilde{g}_q) + \sum_{r=1}^{p-1} x_{rp}(v_{rp}, T\widetilde{g}_q) \\ + \frac{\overline{x}_{pq}}{pq\,[n(p,q,\rho(pq))]^{3/2}}(T\varphi'_{p,q,\rho(pq)}, v_{pq}) = 0.$$

Since

$$(T\varphi'_{p,q,\rho(pq)}, v_{pq}) = -2[n(p,q,\rho(pq))]^2,$$

we obtain[T29]

$$\overline{x}_{pq} = \frac{pq}{2[n(p,q,\rho(pq))]^{1/2}} \begin{bmatrix} (T\widetilde{g}_p, \widetilde{g}_q) + (T\widetilde{g}_p, \xi_q) + (\xi_p, T\widetilde{g}_q) \\ + \sum_{r=1}^{p-1} x_{rp}(v_{rp}, T\widetilde{g}_q) \end{bmatrix}.$$

Here x_{pq} is expressed in terms of the x_{rp} and v_{rp}, and hence of the $\rho(rp)$, for $r = 1, \dots, p-1$, together with $\rho(pq)$. The latter remains arbitrary. If the x_{rs} and $\rho(rs)$ are ordered by increasing $r+s$, the displayed equation is therefore a recursion for the x_{rs}, while the $\rho(rs)$ may be chosen freely. It remains only to maintain

$$|x_{pq}| \le \frac{1}{pq}, \qquad \rho(pq) \ge \max\bigl(\mu(1), \dots, \mu(q-1)\bigr).$$

Choose the $\rho(rs)$ in the same order. If C denotes the absolute value of the expression in brackets in the formula for $\overline{x}_{pq}$, then C depends only on the x_{rs} and $\rho(rs)$ for which $r+s < p+q$, and not on $\rho(pq)$. Hence

$$|x_{pq}| = \frac{pq\,C}{2[n(p,q,\rho(pq))]^{1/2}} \le \frac{pq\,C}{2\rho(pq)}.$$

It is therefore enough to choose

$$\rho(pq) \ge \frac{1}{2}Cp^2q^2, \qquad \rho(pq) \ge \max\bigl(\mu(1), \dots, \mu(q-1)\bigr),$$

and our objective is attained.

[T29] *Translator's note.* The source prints a minus sign and omits the factor pq in this recursion. Solving the preceding equation gives the corrected formula displayed here; the ensuing estimate and choice of $\rho(pq)$ have been corrected consistently.

We have now constructed the sequences

$$h_1, h_2, \ldots, \qquad g_1, g_2, \ldots,$$

and the system

$$\omega_1, \omega_2, \ldots$$

which we sought. The desired construction is therefore complete.

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